USDA FOREST SERVICE
LAKE WEDINGTON
WATERLINE DISTRIBUTION PROJECT
SCHEDULE OF ITEMS-SCHEDULE B

DATE:
BY:

ITEM NO.	DESCRIPTION	METHOD OF MEASUREMENT	PAY UNIT	EST. QTY.	UNIT PRICE	AMOUNT
1	Mobilization	LSQ	LPSM	1		
2	6" HDPE Waterline	CQ	FT	5,293		
3	4" HDPE Waterline	CQ	FT	5,303		
4	3" HDPE Waterline	CQ	FT	597		
5	2" HDPE Waterline	CQ	FT	2,877		
6	1-1/2" HDPE Waterline	CQ	FT	580		
7	1" HDPE Waterline	CQ	FT	1,286		
8	Excavation, Pipe Installation, Tracer Wire, Bedding, Incidental Accessories, Testing, and Backfill	CQ	FT	15,936		
9	HDPE Tees with MJ's & Accessories (see schedule)	CQ	EA	50		
10	Reducers	CQ	EA	16		
11	Mainguard #77 Blow Off Assembly	CQ	EA	15		
12	Frost Free Hydrant with Concrete Pad	CQ	EA	11		
13	6" Gate Valves with MJ's and all required Incidental Accessories for connection	CQ	EA	8		
14	4" Gate Valves with MJ's and all required Incidental Accessories for connection	CQ	EA	5		
15	3" Gate Valves with MJ's and all required Incidental Accessories for connection	CQ	EA	1		
16	2" Gate Valves with MJ's and all required Incidental Accessories for connection	CQ	EA	27		
17	1-1/2" Gate Valves with MJ's and all required Incidental Accessories for connection	CQ	EA	1		

18	1” Gate Valves with MJ’s and all required Incidental Accessories for connection	CQ	EA	25		
19	3 Way Valves w/ Accessories	CQ	EA	5		
20	Bypass Meters	CQ	EA	3		
21	Drinking Fountains with Concrete Pad	CQ	EA	2		
22	Thrust Blocks	CQ	EA	77		
23	ARI D-040 1” Metal Air Release w/ Ball Valve with all Required Accessories, Connections, Valve Box & Lid	CQ	EA	3		
24	Pressure Booster Pump, Accessories, Pressure Tank, Meter, Air Release Valve, Pressure Switch, Gauges, Piping, Electric, Installation in Building, and New 36” Door & Frame.	CQ	EA	1		
25	Demolition & Disposal of Existing Waterlines, Valves, Meters, and Accessories. USFS to request some Hydrants, stockpile useable material for future use.	LS	EA	1		
26	Romort Water Tower Kit	CQ	EA	2		
27	Composite Waterline Markers	CQ	EA	110		
28					Total Cost	

Note: These Specifications have been Edited for the USFS Wedington Waterline Distribution Project 7/14/25

STANDARD SPECIFICATIONS FOR DESIGN AND CONSTRUCTION OF WATER IMPROVEMENTS

2023 EDITION

WASHINGTON WATER AUTHORITY

3685 E HERITAGE PKWY; FARMINGTON, AR 72730

TABLE OF CONTENTS

Contents

SECTION 1000

GENERAL REQUIREMENTS AND PROCEDURES

PART 1 – GENERAL

1.01 DIVISION 1 – SPECIFICATIONS – SUMMARY OF WORK

The Lake Wedington Water System Distribution project is located at the Lake Wedington Recreation Area, Boston Mountain Ranger District, which is approximately 16 miles west of Fayetteville , AR via highway 16. The contractor will be responsible for installing approximately 16,000 linear feet of distribution water line, valves, and associated appurtenances for Day Use and Campground Areas from the existing storage tank. All measurements will be field verified. This contract also includes flushing, testing, and disinfecting waterlines and tank. The US Contracting Officer Representative (COR) is responsible for all Contract Administration, the include scheduling, submittal approvals, construction, contract modifications, testing, and final approvals.

PROJECT COORDINATION

Notify the Forest Service Contracting Officer Representative (COR), two weeks prior to start of work. The contractor shall perform work in a manner that will protect the safety of public and employees. All work will be done in accordance with all OSHA Regulations.

All quantities, sizes, and lengths are approximate and shall be field verified before submitting quote. Pre-bid site visit will be conducted by Contracting Officer Representative (COR).

The contractor shall perform the work between the hours of 7:00 AM and 5:00 PM Monday through Friday excluding Federal Holidays unless approved by the COR. Access to the area will be coordinated with COR.

SUBMITTAL PROCEDURES

The contractor shall be required to furnish submittals on all items they intend to use. Submittals shall be made prior to commencing any work that relates to a respective submittal. Allow review time of 14 days for all submittals. Incomplete information on the proposed items will result in rejection.

The contractor shall submit the manufacturer's information. This information shall include:

Product name and model number, Manufacturer, Product description, Technical data, and Warranties for:

1. Pipe, valves, valve boxes and fittings
2. Compression couplings for similar and dissimilar pipes
3. Gradation for bedding material
4. Detectable trench tape “Water,” Blue in color
5. Method of disposing water used in testing and disinfection of waterlines
6. Trace wire

1.02 WORK INCLUDED

A. These Specifications for water improvements are provided to outline procedures and criteria for the design and construction of water facilities within Washington Water Authority's service area boundary.

B. In addition, all rules, regulations, and requirements of the Arkansas Department of Health are the minimum standard for all plan requirements, construction, and development practices to be approved by Washington Water Authority. In cases were the Specifications set forth exceeds the Arkansas Department of Health requirements, then this Specification shall govern.

C. Additionally, this Specification includes the requirement that the minimum standards of design and construction specified herein, that directly affects health and safety, must further comply with the recommendations of both (1) the 10 State Standards (GLUMRB) and (2) the American Water Works Association (AWWA). Any conflict that directly affects health and safety between these Specifications and the recommendations from 10 States Standards, or the recommendations from AWWA, must be resolved to provide the greatest protection of health and safety.

D. All rules, regulations and requirements of the Arkansas Board of Licensure for Professional Engineers and Professional Surveyors shall be the minimum standard for professional surveying and for professional engineering.

E. The Engineer of Record is responsible for design and compliance with the latest edition of local Ordinances and Specifications, Arkansas Department of Health minimum Standards and State of Arkansas Rules, Regulations and Laws, or any Federal Law, Rule, or Regulation.

F. These Specifications do not replace the Engineer of Record's specifications and contract documents, however these specifications set forth the minimum requirements for design and construction of all water and sanitary sewer with the jurisdiction of Washington Water Authority. These specifications may be included by reference only in the Engineer of Record's documents.

G. The Engineer of Record for any project designed and constructed under the requirements of these Specifications, shall provide to Washington Water Authority a sealed (PE) letter report stating that the design and construction of the water improvements met the minimum requirements of these Specifications. This sealed letter report will also contain documentation of any special conditions or additions to these Specifications for the specific project.

H. Comprehensive, full time construction observation services shall be provided by, or under the direction of, the Engineer of Record.

I. Washington Water Authority shall not be responsible, nor bears any liability for the Contractor's means, methods, techniques, sequences and procedures of construction, or safety precautions and programs incident thereto in performing or furnishing any of the work. Washington Water Authority shall not be responsible for any actions resulting from the direction of the Engineer of Record.

J. Special conditions may arise on projects that are not covered in these Specifications. Such special conditions, complete detail as to design, materials, method of construction, testing, and/or other procedures shall be submitted to Washington Water Authority for review and approval.

1.03 DEFINITIONS

A. Engineer of Record – A qualified Professional Engineer licensed in the State of Arkansas, for the Owner, City of Government agency.

B. Owner – Any Individual, partnership, firm, corporation, or other entity who, as property owner, initiating work.

C. Provide – Furnish and install, complete in place, operating, tested and approved.

D. Products – The materials, systems, and equipment provided by the Contractor.

E. Washington Water Authority – Division of the Rural Development of Authority Washington County Arkansas, having full and complete authority to manage, operate, improve, extend, and maintain water works and water distribution.

1.04 LAWS, REGULATIONS AND ORDINANCES

A. All Federal, State, County and Municipal Laws, Regulations, or Ordinances shall be complied with on all projects. Where the requirements of another jurisdictional authority having influence on work outside the purview of Washington Water Authority are greater than that provided by these Specifications, the work shall conform to the greater requirement of that respective jurisdictional authority.

1.05 PERMITS AND LICENSES

A. All permits required to accomplish the work shall be the responsibility of the Engineer of Record. Such permits may include but are not limited to permits for work within Arkansas

Department of Transportation R/W, railroad crossing permits, Arkansas Department of Environmental Quality "Notice of Intent", and or U.S. Army Corps of Engineers. Work shall not be started without appropriate permit(s) in place.

1.06 CONSTRUCTION LAYOUT

A. The layout and staking of the construction work shall be by trained and qualified survey personnel under the supervision of the Engineer of Record. Construction layout shall consist of staking necessary to determine alignment and elevations to properly construct the proposed facilities.

1.07 INSPECTION, OBSERVATION AND TESTING

A. Comprehensive, full-time construction observation services shall be provided by the US Forestry Service (USFS), with collaboration from Engineer-of-Record.

B. Only authorized personnel from Washington Water Authority are permitted to operate valves on the existing water system. Under no circumstances shall the Contractor operate any valve on the existing water system.

C. All field test required for a project shall be witnessed by Washington Water Authority in the presence of the Engineer-of-Record or his authorized representative and the Contractor.

D. Washington Water Authority representative will be determined at, or prior to, the pre-construction meeting.

E. Washington Water Authority requires 24-hour working day notice on all tests. Contractor shall pre-test the installation prior to notifying Washington Water Authority of the test.

F. Tests delayed by weather or other factors will be rescheduled on the same basis. If a test cannot by reasonably scheduled so that a representative of Washington Water Authority can be present, the Engineer-of-Record will witness the test and certify to Washington Water Authority the results.

G. In no case shall a test be made without the presence of the Engineer-of-Record's representative and the Contractor. It is the responsibility of the Engineer-of-Record and/or the Contractor to coordinate the scheduling of test with Washington Water Authority and with other parties involved.

H. All equipment, materials, and labor required for testing shall be furnished by the Contractor at his expense.

I. Generally, no Contractor or Engineer-of-Record involvement is required in the taking of samples for the Bac-T test except that the Contractor is responsible for the proper flushing of the line prior to samples being taken by Washington Water Authority. However, Washington Water Authority may require the presence of the Contractor or Engineer-of-Record when questions have been raised as to the methodology or techniques used the sampling process.

J. Bac-T samples are sent to the Arkansas Department of Health for testing. Results are available on-line at the ADH website. Results obtained by Washington Water Authority may be forwarded to the Engineer-of-Record either by email, fax, or mail upon request of the contractor.

K. Lines failing the Bac-T tests shall be re-sampled as soon as practicable. If a line fails two (2) consecutive Bac-T test, the line must be re-chlorinated before Bac-T samples can be taken again, Washington Water Authority shall not be responsible for rescheduling Bac-T tests.

L. The hydrant test shall consist of checking the operation of the hydrant valve and flowing the hydrant. This test will be performed jointly by the Contractor and Washington Water Authority representative. This test will be done after the pressure test has been completed. The hydrant valve shall be left in the open position during the test and after the test is complete.

1.08 FINAL ACCEPTANCE

A. USFS will be deemed to have accepted water lines and appurtenances for ownership upon completion of the following.

1. Acceptable Final Inspection by the following:

    a. A representative of Washington Water Authority

    b. The Engineer-of-Record

    c. The Contractor

2. Acceptable walk-through by Washington Water Authority for location, grade, and condition of water meter settings

3. Record Drawings (as-builts) are received by USFS & Washington Water Authority

4. Engineer-of-Record Certification is received by USFS & Washington Water Authority

5. Letter of Final Acceptance is provided by the USFS & Washington Water Authority to the Engineer-of-Record.

B. No water meter shall be set until all final acceptance requirements are met and the line is accepted by WWA.

1.09 CLOSEOUT SUBMITTALS

A. Furnish two copies of Operation and Maintenance Data for all valves, boxes, equipment, and miscellaneous appurtenances. Include recommended maintenance and testing procedures and frequencies.

B. List names, addresses, and telephone numbers of suppliers including local source of supplies and replacement parts.

1.10 NO-LEAD MATERIALS

A. All subsequent parts, materials, components found with the Specifications and/or Plans are to be “No-Lead” where applicable, even if not explicitly stated.

SECTION 2000

TRENCH SAFETY

PART 1 – GENERAL

1.01 WORK INCLUDED

A. The Contractor is responsible for ensuring that safe working conditions exist and safety procedures are being followed at the work site. The Contractor is responsible to notify OSHA of the commencement of all water construction.

B. If the Contractor is working for any public body (state agency, county, municipality, school district, or other local tax unit or improvement district), they shall be responsible for notifying the Arkansas Department of Labor Safety Division, (501) 682-9091 when water jobs are to begin. A copy of a letter or reporting form shall be sent to Washington Water Authority.

C. Regarding Trench Safety Systems, the Contractor shall design, install, and maintain a "Trench Safety System" in strict compliance with OSHA (Occupational Safety and Health Administration) Part 1926 of the Code of Federal Regulations and all other applicable federal, state, and local requirements.

SECTION 2100

EROSION AND SEDIMENT CONTROL

PART 1 – GENERAL

1.01 WORK INCLUDED

A. The Contractor shall accomplish temporary and permanent erosion protection related to grubbing, grading, excavation, paving, and other work as directed by the Engineer and as shown on the approved drawings. An erosion control plan shall be submitted to the Engineer prior to the commencement of construction.

B. The Contractor is responsible for implementing Best Management Practices (BMPs) during construction activities, (including, but not limited to, sediment and erosion control structures) and achievement of final stabilization upon completion of construction activities.

C. The Contractor shall be responsible for implementing all applicable requirements of the ADEQ General Stormwater Permit for Construction Activity, the Spill Prevention Control and Countermeasures Plan (SPCCP), as required by USEPA, local Municipal Separate Storm Sewer requirements, and all other environmental regulatory requirements that are associated with the construction activities that they are contracted to perform. The Contractor is responsible for managing all materials, equipment, and activities at the work site in a manner that is in compliance with local, State, and Federal environmental regulations.

PART 2 – PRODUCTS

A. NOT USED

PART 3 – EXECUTION

3.01 TEMPORARY EROSION CONTORL

A. The contractor shall construct temporary berms, dikes, dams, ditching or sediment basins, and maintain such control features until permanent erosion control features are placed all in accordance with Washington Water Authority or State “Erosion Control Standards”.

B. Water removed from open pits and/or trenches shall have silt removed prior to leaving the immediate site of construction. Silt shall be removed by natural vegetation, a straw bale trench dewatering inlet device, settling pond, filter bag, a rock/geotextile fabric sediment trap/basin, or other appropriate sediment control measure. Water filtered through a basin shall not violate any water quality standard and shall have efficient sediment/silt removal prior to discharge to a waterbody.

C. Contractor shall be responsible for providing adequate number of pumps for prompt and efficient dewatering. Ends of discharge hoses shall be provided with flow dispersion and filtration devices to prevent scouring of surface soils, discharge of turbid water, and/or washout of stream banks. Discharge from dewatering activities shall not be conveyed into or upon any roadside ditch, curb and gutter, street, or publicly used thoroughfare.

D. The direct discharge of silty/muddy water to a stream, offsite, or across areas of equipment access points and/or construction haul roads is strictly prohibited.

3.02 OTHER CONTROLS

A. A dedicated concrete truck was out area shall be maintained to include adequate containment to prevent runoff of concrete truck was water. Concrete truck drivers shall be notified to use wash out area.

B. Contractor shall follow the appropriate waste storage and disposal practices, as per applicable environmental regulatory requirements. Solid waste dumpsters/roll-offs, or other appropriate waste receptacles will be maintained and used at the site. Good housekeeping practices will preclude trash, construction wastes, and debris to be dumped or scattered on the construction site. There shall be no open burning of any waste material. No solid materials, including building materials, shall be discharge to water of the State.

C. No liquid waste chemicals, fuels, and/or oils are to be leaked or spilled on ground surfaces. Bulk storage of liquid chemical wastes will be provided with secondary containment with a capacity sufficient to contain the volume of the largest container within the secondary containment. All waste materials shall be stored in a manner to prevent releases and should be disposed of by a qualified waste disposal firm at an acceptable waste disposal facility. Records of the disposal of all solid, hazardous, non-hazardous, and liquid wastes are to be maintained by the Contractor. Contractor shall notify the Engineer of any spills or leaks that occur in spite of the preventive measures taken. Contractor shall notify the Engineer of any spills or leaks that occur in spite of the preventive measures taken. Contractor will prepare a report of any spills or leaks in accordance with the reporting and recordkeeping measures described in the BMPs. No contaminants from fuel storage areas, hazardous waste storage and truck wash areas shall be discharged to water of the State. These areas should not be located near a water body.

D. Contractor shall maintain compliance with applicable State and/or local sanitary sewer, septic system, and waste disposal regulations.

E. Off-site vehicle tracking of sediments and the generation of dust must be minimized. Measures such as stone at construction access points, parking areas, and unpaved roads, providing entrance wash racks or stations for trucks, and/or street sweeping shall be implemented where appropriate. Application of water to construction haul roads should be done, as appropriate, to control dust generation. Application of excessive levels of water that create mud should be avoided.

3.03 PERMANENT EROSION CONTROL

A. The Contractor shall incorporate permanent erosion control features into the project at the earliest practicable time as the construction progresses all in accordance with the approved plans and project specifications.

SECTION 2200

SITE PREPARATION

PART 1 – GENERAL

1.01 WORK INCLUDED

A. Excavation, grading, cutting and removal of trees, shrubs and underbrush, and the removal of any debris existing above natural ground surface and within the cleared area necessary to permit the construction of the improvements.

1.02 PROTECTION

A. In all cases the Contractor is responsible for protecting public and private property and protecting any person or persons who might be injured as a result of the Contractor's work.

B. All utilities shown on the drawings may not represent the exact location however, the Contractor is responsible for verifying these locations and contacting the Arkansas One Call System before excavating.

PART 2 – MATERIALS

A. NOT USED

PART 3 – EXECUTION

3.01 GENERAL

A. The Contractor will be required to submit a plan to build access roads/trails for approval by the Engineer.

B. It shall be the responsibility of each bidder to examine the site carefully and make his own calculations as to costs to be incurred by reason of the requirements of this section.

C. Trees, shrubs, underbrush, and debris removed from the improvement right of way shall be disposed of by the Contractor in a manner approved by Washington Water Utilities.

D. The Contractor shall not remove or disturb any vegetation except that required for the execution of the work.

3.02 SITE PREPARATION

A. Clear areas necessary for performance of the work and confine operations to that area provided through easements, licenses, agreements, and rights-of-way. Entrance upon any lands outside of that area provided by easements, licenses, agreements, or public rights-of-way, shall be at the Contractor's sole liability.

B. Do not occupy any portion of the project site prior to the date established in the Notice to Proceed without prior approval of the Owner.

C. Contractor shall be aware of certain easement considerations by individual property owners as stipulated in easement documents pertaining to the project.

D. Remove, relocate, reconstruct, or work around natural obstructions, existing facilities and improvements encountered during site preparation as herein specified. Take care while performing site preparation work adjacent to facilities intended to remain in place. Promptly repair damage to existing facilities. Dispose of waste materials in a satisfactory manner off the work site.

E. Protect, move, or brace public and private utilities as required by the affected utility.

F. Maintain mailboxes in the manner that the Postal Service requires to prevent interruption of mail delivery.

G. Site preparation includes the removal of trees, shrubs, brush, crops, and other vegetation within the limits of the easements (right-of-way), or as may be provided for in licenses, permits and agreements. All efforts shall be made to retain existing landscaping. In the event that trees, shrubbery, and hedges cannot be saved, then prior approval of the Owner and the Owner's Representative must be obtained before the existing landscaping is removed.

1. Trees

   a. All trees shall be saved unless removal is approved by USFS.

2. Shrubbery

   a. Shrubbery shall be saved unless removal is approved by USFS and the Owner’s Representative. Make reasonable efforts to save all shrubbery by trimming, in accordance with acceptable pruning practices, and treating wound surfaces with a commercial pruning compound.

3. Small Plants and Flowers

   a. At least two weeks prior to the start of construction, notify property owners of the proposed starting date so that the property owners can remove any small plants or flowers.

4. Protection of Existing Facilities

a. The Contractor shall notify all property owners in the immediate vicinity of the construction area that may be affected by the construction activities a minimum of one week before starting work in that area. The notification shall include a description of the work, work hours, and a 24-hour contact name and number for the contractor.

b. Fences interfering with construction, and located within public rights-of-way or as may be allowed for in permits or agreements, may be removed only if the opening is provided with a temporary gate which will be maintained in a closed position except to permit passage of equipment and vehicles, unless otherwise herein specified. Fences within temporary construction easements may be removed provided that temporary fencing is installed in such a manner as to serve the purpose of the fencing removed.

c. Fencing removed shall be restored to the condition existing prior to construction unless otherwise specified. The Contractor is solely liable for the straying of any animals protected or corralled or other damage caused by any fence so removed.

d. Driveways and driveway approaches removed or damaged during construction shall be restored to the original condition or better condition prior to construction.

e. Make every reasonable effort to protect private sewer facilities. Private sewer facilities may not be shown on the Plans.

f. Preserve property corners, pins, and markers. In the event any property corners, pins, or markers are removed by the Contractor, such property points shall be replaced at the Contractor's expense and shall be re-set by competent surveyors properly licensed to do such work. In the event such points are section corners or Federal land corners, they shall be referenced and filed with the appropriate authority.

g. Where existing utilities and service lines are encountered, notify the owner thereof at least 48 hours (not including weekends and/or holidays) in advance of performing any work in the vicinity.

h. Excavate, install pipeline, and backfill in the vicinity of such utilities in the manner required by the respective owner and, if requested, under his direct supervision. The Contractor shall be responsible for damages to a public or private utility that may occur as the result of the construction.

i. Protect, move, or brace public and private utilities as required by the affected utility.

j. Make a reasonable effort to ascertain the existence of obstructions and locate obstructions by digging in advance of machine excavation where definite information is not available as to their exact location. Where such facilities are unexpectedly encountered and damaged, notify responsible officials and other affected parties and arrange for the prompt repair and restoration of service.

SECTION 2300

EXCAVATION, BACKFILLING, AND COMPACTING

PART 1 – GENERAL

1.01 WORK INCLUDED

A. Excavation, backfilling, and compaction for water lines, appurtenances, and incidental construction.

1.02 QUALITY ASSURANCE

A. If requested by USFS, one moisture/density relationship test (AASHTO T-99 with Note 7, or AASHTO T-180 with Note 8) shall be taken at the beginning of the project, and one additional moisture/density relationship test shall be taken for every 5,000 cubic yards of imported material. ASTM testing methods for moisture/density Relationships may not be used for embankment and subgrade materials.

B. Determine the field density of backfill in accordance with AASHTO Designation T-147.

C. A minimum of one density test per 500 linear feet of pipe or portion thereof when the pipe is located in the street or under the curb and gutter.

1.03 PROTECTION

A. The Work included in this Project may require excavation and related activities in close proximity to existing buried and aerial utility lines and facilities, such as water lines, sewer lines, storm drains, natural gas lines, electrical power lines, telephone cables, and TV cables. Where their presence is known, the approximate location of such utilities is shown on the Drawings, but all such utilities and individual service lines are not shown. The Contractor shall be aware of the potential for such utility lines to conflict with intended construction efforts, and the Contractor shall use appropriate precautionary measures to locate and protect such utility lines and services so as to avoid damage and interruptions to service.

B. The Contractor shall contact the owners of the various existing utilities lines and services as may be affected by the construction and solicit their assistance in identifying, locating, marking, and protecting these facilities prior to the beginning of any excavation or other work which might endanger the existing utilities. If such utilities are damaged or impaired because of the Contractor's actions or omissions, the Contractor shall be responsible for the cost of repairs or replacements of the affected or damaged utility or service line.

C. The Contractor shall comply with the Arkansas One-Call System and shall alert potentially conflicting utility systems accordingly.

D. In all cases, the Contractor is responsible for protecting public and private property; and, protecting any person or persons who might be injured as a result of the Contractor's work.

1.04 DUST CONTROL

A. The Contractor shall be responsible for maintaining the site and adjoining paved surfaces in a dust free condition. Fugitive dust control is the sole responsibility of the Contractor.

1.05 SEDIMENT CONTROL

A. The Contractor shall be responsible for all sediment control in accordance with Section 2100 – Erosion and Sediment Control.

PART 2 – PRODUCTS

2.01 FOUNDATION MATERIAL

A. Foundation materials for trench over excavation shall be Class 7 Aggregate Base Course, or "B" stone with Class 7 aggregate cap as designed by the Engineer of Record.

2.02 EMBEDMENT MATERIAL

A. Embedment materials are restricted to materials as described below. The Engineer of Record shall provide as a submittal the gradation sieve analysis of the embedment materials proposed for each specific project.

B. Without regard to the pipe material, all embodiment materials include: bedding, pipe zone (including haunching), and initial backfill from six (6) inches below the bottom of the pipe to six (6) inches above the top of the pipe the full trench width with a minimum of six (6) inches of embedment materials all around the exterior of the pipe.

C. Special design considerations, including supplemental trench drains, geotechnical fabric, and/or specific aggregate, and/or filter gradations as recommended in AWWA C605 and ASTM D2321 may be required to prevent migration of embedment particles. The Engineer of Record is responsible to design the embedment system as needed for the specific project.

D. Embedment material shall be in accordance with ASTM D 2487, latest edition and shall conform to class 1A embedment materials in accordance with ASTM D 2321, latest edition. Material shall meet the grading requirements of a "MODIFIED" ASTM C 33, gradation 67, commonly referred to as ASTM #67 (3/4" concrete aggregate or 5/8" aggregate). Maximum aggregate size shall be 3/4 inch. This includes materials such as crushed stone or rock.

2.03 GRIT EMBEDMENT MATERIAL

A. Grit shall be the by-product of rock crushing, generally consisting of 1/4” and smaller limestone material.

2.04 AGGREGATE BACKFILL MATERIALS

A. Aggregate material for select backfill across streets, roads, driveways, and for placement of “gravel” or aggregates surfaced areas, shall be Class 7 Aggregate Base Course material conforming to the Standard Specifications of the Arkansas Department of Transportation, latest edition.

2.05 SELECT NATIVE BACKFILL MATERIAL

A. Select native material shall be suitable on-site material or imported good earth, sand, or gravel that is free from large rocks (3” maximum diameter) or hard lumpy materials. Never use materials of perishable, frozen, spongy, or otherwise unsuitable nature as select material.

PART 3 – EXECUTION

3.01 EXCAVATION – GENERAL

A. All excavation shall be carried accurately to the line and grade shown on the Drawings and as established by the Engineer of Record.

B. When excavation is necessary to be carried below or beyond that which is required, fill the over-excavated space with compacted with Foundation Material as approved by the Engineer of Record

C. The Contractor shall use a trench box or provide and install shoring where necessary to protect the labor, the work, or adjacent property. Shoring shall be maintained in place until the backfill has proceeded to a point where it can be safely removed.

D. Dewater all excavations before any construction is undertaken in accordance with Section 2100 of these specifications.

E. Install pipe only in dry trenches. Place concrete upon dry, firm foundation material only.

3.02 DISPOSAL OF EXCAVATED MATERIALS

A. The Contractor shall be responsible for disposal of excess material, or disposal of excavated material unsuitable for backfilling.

B. Disposal of excess material shall only be allowed on private property with written permission of the owner of the property. A copy of the written permission must be forwarded to the Engineer along with any permits as may be required by the governing authority, city, or county. Grading permits may be required.

3.03 EXPLOSIVES

A. The use of explosives/blasting is not permitted.

3.04 TRENCH DEWATERING

A. Dewater all trenches, in accordance with Section 2100, to the extent that water pipe can be placed on a dry and firm trench bottom. Never place pipe in a wet or unstable trench.

B. Pump, pipe, and drain all water resulting from dewatering operations into a properly designed dewatering structure or device in accordance with Section 2100. Prevent flooding of streets or private property.

C. Soil that cannot be properly dewatered: excavate and install 2” – 3” rock to provide a firm trench bottom.

D. Divert surface runoff water away from the excavation. Where the excavation crosses natural drainage channels, care should be taken to prevent unnecessary damage or delays. Route diverted surface water into existing drainage structures, such as storm sewers, ditches, or streams. Prevent flooding of streets or private property.

3.05 EXCAVATION – TRENCHES

A. During excavation, all pipe to be replaced shall be removed and disposed of offsite at a suitable landfill. Trench excavation for water lines shall be kept within the maximum width limits as shown on the Drawings. The specified maximum trench width from the bottom of the trench to 24-inches above the outside top of the pipe shall not be exceeded unless specifically authorized by the Engineer.

B. Prior to excavation in paved areas, the Contractor shall saw-cut (or other acceptable method approved by the Engineer) the existing pavement to minimize the destruction of the existing pavement outside the limits of the trench. The maximum trench width for the installation of water lines, up to 12-inches in diameter, in paved areas shall not exceed 36-inches without written approval from Engineer. Contractor is responsible for damage to paved areas by construction equipment outside the limits of trench excavation.

C. If necessary to prevent sliding and caving, cut the trench banks back on a slope above an elevation two (2) feet above the outside top of the pipe to reduce the earth load on the trench sides. Never exceed the specified maximum width until 2 ft. above the outside top of the pipe. See Section 2000 – Trench Safety.

D. Do not advance trench excavation more than three hundred (300) feet ahead of the completed pipe work and backfill.

E. Standard Trench Widths:

<u>Minimum</u>	<u>Maximum</u>
Outside pipe diameter + 12 inches	Outside pipe diameter + 24 inches

3.06 EMBEDMENT AND BACKFILLING – GENERAL

A. Install all water pipe using approved embedment materials only.

B. It is essential that the complete backfill be done in such a manner to minimize voids in the backfill.

C. Backfilling includes refilling and consolidating the fill in the excavation up to the surrounding ground surface or road grade.

D. Use select native materials for backfilling in unpaved areas.

E. Use mechanical compaction devices manufactured for that purpose to compact backfill materials in trenches.

F. Use mechanical compaction devices manufactured for that purpose to compact backfill materials in trenches.

3.07 BEDDING AND BACKFILLING PIPE

A. Bed pipe as described below and in accordance with the standard trench details shown in Standard Detail Drawings. The intent of the bedding is to create a uniform support which will protect the pipe from localized stress points and to provide for a well graded trench bottom.

B. Extend the trench excavation to a minimum depth of six (6) inches below the bottom of the pipe.

C. Install bedding materials in no greater than eight (8) inch compacted lifts. Install bedding from six (6) inches below the pipe to six (6) inches above the pipe. Shovel slice bedding beneath the pipe haunches.

D. Provide the following cover for water pipe:

1. Minimum depth to the top of pipe for all water mains lines less than 12-inches diameter shall be 3 feet.

2. Minimum depth to the top of pipe for water lines 12-inches diameter and greater shall be 4 feet.

3. Minimum depth of all water service lines from the main to the water meter shall be 30-inches.

E. The maximum depth of bury for PVC pipe is sixteen (16) feet. Any depths greater than sixteen (16) feet require ductile iron pipe, unless approved by Washington Water Authority.

SECTION 3100

CAST-IN-PLACE CONCRETE

PART 1 – GENERAL

1.01 WORK INCLUDED

A. This section covers cast-in-place concrete materials, reinforcing steel, forms, and finishing in conjunction with water and sewer construction.

B. Use Class B Concrete for all structures.

C. Use Class A Concrete for bedding and blocking only.

1.02 SUBMITTALS

A. Submit mix design, equipment details, and vendor name for field batched concrete.

PART 2 – PRODUCTS

2.01 CONCRETE

A. Concrete: composed of Portland Cement; fine and coarse aggregate; water; and an air entraining agent. Provide either Class A concrete or Class B concrete as described below.

B. For Class A and Class B concrete use ready-mixed concrete; conform to ASTM C 94, latest edition; deliver and place within one hour after all materials have been placed in the mixing drum.

C. The concrete mix shall be designed so that the proportions will produce results that will meet the requirements of Class A or Class B concrete. Proportion components, except water, by weight. Water may be measured by volume. One sack of Portland Cement consists of one cubic foot or 94 pounds. Proportion components to meet these requirements:

1. Class A Concrete:

    a. Maximum net water/cement ratio = 0.49

    b. Slump range: 1 - 4 inches

    c. Minimum 28-day compressive strength: 3,000 PSI

2. Class B Concrete:

a. Maximum net water/cement ratio = 0.49

b. Slump range: 1 - 4 inches

c. Minimum 28-day compressive strength: 4,000 PSI

d. Air Content: 4 – 7

D. Before beginning any concrete work, the Contractor shall have the concrete mix designed and the ingredients selected and proportioned by an approved independent testing laboratory meeting the requirements of ASTM E 329. Certified copies of all laboratory trial mix reports shall be sent to the Engineer from the testing laboratory for review. Do not place concrete prior to the Engineer's review and acceptance in writing of the concrete mix design.

E. Cement: Portland Cement conforming to AASHTO M 85, Type I. Use Type III cement (high early strength) only if approved by the Engineer.

F. Fly ash: Fly ash may be used as a partial cement replacement not exceeding 10% by weight of the cement when approved by Washington Water Authority. When fly ash is used, the total weight of both cement and fly ash will be used in design calculations.

G. Water: potable water free from injurious amounts of acids, alkalis, oils, sewage, vegetable matter and dirt.

H. Air entraining agent: use in all Class B concrete as required; conform to AASHTO M 154; add to the mixing water in solution; proportion to provide four (4) to seven (7) percent air in the concrete.

I. Fine aggregate: clean, hard, durable particles of natural sand free from injurious amounts of organic impurities; conform to the gradation requirements of AASHTO T 27.

J. Coarse aggregate: clean, hard, and durable crushed stone or washed gravel; reasonably well graded from course to fine; per AASHTO T 27.

2.02 REINFORCEING STEEL

A. Steel bars: deformed, conforming to ASTM A 615 or A 617.

B. Steel wire: conform to ASTM A 82, Cold-Drawn Steel Wire for Concrete Reinforcement.

C. Wire mesh: conform to ASTM A 185; gauge and mesh per plans.

D. Submit reinforcing steel bars shop drawings for approval.

E. All steel reinforcement: free from rust, scale, mortar, dirt, or other objectionable coatings.

PART 3 – EXECUTION

A. Perform excavation per Section 2300 - Excavation, Backfilling, and Compacting.

B. Build forms neat, square, and flat so concrete will have smooth finish when forms are pulled. Construct forms to provide finished concrete to dimensions shown on plans.

C. Place reinforcing steel accurately in accordance with details shown on the plans and properly secure in position.

D. Concrete shall not be placed when the temperature is below 40° F and dropping or below 35° F if the temperature is rising, unless approved by the Engineer.

E. Vibrate all structural concrete as it is placed using internal vibrators capable of transmitting vibration to the concrete at frequencies not less than 4,500 impulses per minute. Do not use form vibrators. Limit vibration to provide satisfactory consolidation without causing segregation. Do not insert vibrator more than six (6) inches into the lower courses previously vibrated. Use vibrators in a substantially vertical position; insert at uniformly spaced points no farther apart than the visible effectiveness of the vibrator.

F. Allow concrete to cure for at least 48 hours before stripping forms. If concrete is in a structural member, do not remove forms until the concrete can withstand safely all superimposed loads.

G. On all exposed surfaces, remove all fins and projections so the surface is smooth. Cut out and fill with grout any honeycombed areas. Extensive honeycombing is not allowable.

H. All concrete shall be properly protected from too rapid curing or from freezing while green. If the weather is hot or dry, all freshly placed concrete shall be covered with a matting or other suitable material and kept moist for at least ten (10) days after pouring, or an approved curing compound may be used with the approval of the Engineer of Record.

SECTION 4000

WATER PIPE, FITTINGS, AND MATERIALS

PART 1 – GENERAL

1.01 WORK INCLUDED

A. This section covers the manufacture, transportation, and storage of pipe, pipe joints, fire hydrants, fittings, and other materials for water lines and service lines.

B. Use only materials approved by Washington Water Authority.

1.02 SUBMITTALS

A. Use of materials other than those specifically listed below is prohibited.

B. Submit the manufacturer's certificate that the materials meet with these Specification requirements including material testing requirements.

1.03 LEAD-FREE BRASS

A. All brass shall be manufactured in accordance with the Safe Drinking Water Act as amended to be LEAD-FREE brass (< 0.25% Pb).

PART 2 – PRODUCTS

2.01 HIGH DENSITY POLYETHYLENE (HDPE) PRESSURE PIPE AND TUBING

A. HDPE tubing 3/4 inches through 3 inches in size may be used for service lines and shall be manufactured in accordance with AWWA C901, latest revision. Tubing shall be SDR 9 or SDR 7 depending on pressure conditions. Color shall be black or blue.

B. HDPE pipe 4 inches through 24 inches in size may be used for water mains and shall be manufactured in accordance with AWWA C906, latest revision. Pipe shall be SDR 9 or SDR 7 depending on pressure conditions. Color shall be black or blue. Use of HDPE pipe for water distribution will be considered on a case-by-case basis.

C. Marking on pipe and tubing shall include the following and shall be applied at intervals of not more than 5 ft.

1. Nominal size and diameter base (e.g., ID, OD, IPS, or CTS).

2. SIDR, SDR, or minimum wall thickness.

3. The manufacturer’s name or trademark.

4. The material designation code

5. The AWWA pressure class

6. The AWWA designation number for this standard (e.g., AWWA C901).

7. The manufacturer’s production (lot) code that includes information such as resin source, manufacturing location, extrusion outlet (line), and manufacturing date.

8. If not included in the manufacturer’s production code, the manufacturing date, including day, month, and year in a recognized standard format. The date format should be readily recognized as a date.

9. The seal or mark of the testing agency that certifies the material.

2.02 POLYETHYLENE ENCASEMENT (PIPE WRAP)

A. Polyethylene encasement shall be in conformance to ANSI/AWWA C105/A21.5, latest revision. The virgin linear low-density polyethylene film shall have a minimum normal thickness of .008 inches (8 mils) and shall be provided in either flat tube or sheet form.

B. The color shall be black with nominal 2% carbon black UV inhibitor and printed per the AWWA C105 standard.

C. Tape for field taping of polywrapped pipe, fittings, etc. or field repair of missing polyethylene encasement material shall be Polyken #900, Scotchrap #50 or equal, at least 2-inches wide, and installed as per the Polyethylene Encasement Installation Guide published by DIPRA. Duct Tape is not permitted.

D. All buried iron pipe, valves, and fittings shall be double wrapped.

2.03 PIPE RESTRAINTS

A. Bell restraints for AWWA C900 PVC sizes 4-inch through 12-inch shall be Series 1900 Restraint Harness, as manufactured by EBAA Iron, Inc. Devices shall have an approved coating system for corrosion resistance equivalent to MEGA-BOND® and manufacturing traceability.

B. Bell restraints for Ductile Iron Pipe sizes 4-inch through 48-inch shall be Series 1700 Restraint Harness, as manufactured by EBAA Iron, Inc. Devices shall have an approved coating system for corrosion resistance equivalent to MEGA-BOND® and manufacturing traceability.

C. When all-thread attachments are required, eye-bolt style attachments are not permitted. Romac “Ductile Lug” style attachments or approved equal shall be used. All-threads shall be made of 316 stainless steel.

2.04 RESTRAINED FLANGED COUPLING ADAPTERS

A. Flanged coupling adapters used to transition from plain end pipe to a flanged fitting, above ground, shall be EBAA Iron Series 2100 Megaflange.

B. Pressure rating shall be a minimum of 250 psi.

2.10 RESTRAINED COUPLINGS

A. Restrained couplings to connect two pieces of pipe, size on size, shall be EBAA Iron Series 3800 Restrained Coupling.

B. Pressure rating shall be a minimum of 250 psi.

2.11 BOLTS AND NUTS

A. All bolts and nuts for valves, fittings, and restraints shall be 316 stainless steel unless otherwise indicated. Anti-seize lubricant shall be used when assembling all stainless steel hardware to reduce galling.

2.12 GATE VALVES

A. Gate valves 2-inch through 12-inch nominal pipe size shall be resilient-seated type, non-rising stem gate valves, in conformance with the requirements of AWWA C509 or AWWA C515, latest revision.

B. Gate valves shall be Made in USA and shall be Mueller Series 2360, or Washington Water Authority approved equal.

C. All gate valves shall be designed for a minimum of 250 psi working pressure. All gate valves shall have 304 stainless steel bolts.

D. All gate valves shall have O-ring stem seals. The O-ring stem seal shall be so designed that the seal above the stem collar can be replaced with the valve under pressure in the full-open position.

E. Gate valves shall have standard mechanical joint ends unless otherwise indicated on the Drawings.

F. Buried gate valves shall be designed for operation with a nominal 2-inch square operating nut. The standard direction of opening shall be open left as viewed from the top.

G. Handwheels for gate valves shall be in conformance to AWWA C515, latest revision.

H. The interior and exterior of the valve body and bonnet shall have factory applied fusion bonded epoxy coating meeting AWWA C550, latest revision.

I. Valves shall be tested in accordance with AWWA C515, latest revision.

J. Markings shall be cast on the bonnet or body, or stamped on a permanently affixed corrosion resistant tag of each valve.

1. Manufacturer’s name or mark.

2. Year the valve casting was made.

3. Size of the valve.

4. Letters C509 or C515

5. Working water pressure (e.g., 250W)

2.13 BUTTERFLY VALVES

A. Butterfly valves 14-inch and larger shall conform with the requirements of AWWA C504, latest revision, for Rubber-Seated Butterfly Valves.

B. Butterfly valves shall be Pratt Groundhog Butterfly Valves, or Washington Water Authority approved equal.

C. Butterfly valves shall be designed for a minimum of 250 psi working pressure. Butterfly valves shall have 304 stainless steel bolts.

D. Butterfly valves shall be of the tight closing, synthetic rubber-seat type, as follows:

1. Valves 20 inches (nominal diameter) and smaller shall have bonded seats which are simultaneously molded in, vulcanized, and bonded to the body. Seat bond must withstand 75 pounds pull under test procedure ASTM D429, Method B.

E. Valve discs shall utilize an on-center shaft and symmetrical design and be cast from Ductile Iron ASTM A 126 Class B. The disc edge shall be stainless steel type 316.

F. Butterfly valves shall have standard mechanical joint ends unless otherwise indicated on the Drawings.

G. Buried butterfly valves shall be designed for operation with a nominal 2-inch square operating nut. The standard direction of opening shall be open left as viewed from the top. The valve shaft shall be constructed of stainless steel and the bearings shall be corrosion resistant and self-lubricating. The valves shall be equipped with a totally enclosed type operator, fully gasketed and grease packed, suitable for direct burial.

H. The interior and exterior of the valve body and bonnet shall have factory applied epoxy coating system meeting AWWA C550, latest revision.

I. Valves shall be tested in accordance with AWWA C504, latest revision.

J. Markings shall be cast on the bonnet or body or stamped on a permanently affixed corrosion resistant tag of each valve.

1. Manufacturer's name or mark.

2. Year the valve casting was made.

3. Size of the valve.

4. Class (e.g., 250B)

2.14 VALVE BOXES

A. Valve boxes shall be Made in USA and shall be East Jordan Iron Works 8550 Series or Tyler Union 6850 Series, screw type, and shall be of correct length to match the bury of the main.

B. The valve box and appurtenances shall include a base and a top section with a drop lid. The lid shall be marked with the word “WATER”. All lids shall have a concrete pad with a minimum of 24 inches round dimension and shall be grouted in place.

C. Lids on valves on fire lines shall be marked with the word “FIRE”.

D. Markings shall be cast on each part:

1. Manufacturer’s name or mark.

2. Model number

3. Year the casting was made.

4. Material of construction

5. USA

E. A valve box alignment device shall be provided and installed for each valve box installation. The device shall be of HDPE or Glass Filled Polypropylene construction. It shall be furnished in two pieces that will lock together under the operating nut of the valve without requiring the removal of the operating nut. The device shall not affect the operation of the valve. The device shall be AFC Alignment Ring as manufactured by American Flow Control.

2.15 OPERATING NUT EXTENSIONS

A. Operating nut extensions shall be used when the top of the operating nut is greater than 4 feet from the top of finished surface.

B. The stem shall be 1” SCH40 steel pipe with a 2-inch square bar steel operating nut attached to the upper end. The stem extension shall be of adequate length to reach from the valve operating nut to a point within 24-inches to 12-inches of the finished surface. A box wrench, 2 1/8" I.D. square, made from steel 3/16-inches thick shall be welded to the lower end of the stem extension which will fit over the valve operating nut. Two-inch valves with a tee-head operating nut will require a rectangular shaped box wrench on the end of the valve stem extension. The extension shall be secured to the valve operating nut by two 3/8” set screws. A round center guide made from 1/8-inch or 3/16-inch steel plate shall be placed on the valve stem extension approximately 6-inches from the upper end. The diameter of the guide shall be slightly less than the inside diameter of the valve box. The guide shall be affixed to the stem extension in such a way that it can rotate freely on the stem. Welds on stem extensions (top and bottom nut) shall be 1/8" - 3/16" fillet weld around full circumference.

C. Shop drawings shall be submitted to Washington Water Authority for approval prior to installing the stem extension pieces.

2.16 BLOW-OFFS

A. Blow-off hydrants shall be 2-inch post type with 2-1/2-inch hose nozzle having National Standard threads.

B. Hydrant shall have 2-inch shut-off valve and integral drain.

C. Hydrant shall have threaded inlet.

D. Hydrant shall be furnished for 3'-6" bury, with cap chain, and shall be painted red.

E. Blow-Off Hydrants shall be Maingaurd #77 or approved equal.

2.17 AIR RELIEF VAVLES

A. All water mains shall have 1”-2” single bodied combination air and vacuum valves where indicated on the drawings. Valves shall have epoxy coated and lined cast iron bodies with stainless steel or non-metallic internal parts. Valves shall have a minimum 250 psi working pressure. Valves shall be listed under NSF-61 and shall have ISO 9001 certificate. Valves shall be supplied with a male thread outlet or flanged outlet. All nipples and isolation valves for 1”-2” valves shall be brass or stainless steel. Ball style isolation valve shall be full port.

B. Air release valves shall be Val-Matic Model No. 201C for 1-inch or Val-Matic Model No. 202C for 2-inch or approved equal.

2.18 SERVICE SADDLES

A. Service saddles for 1”, and 2” NPT service taps shall be Romac 101NS or Mueller DR1S, and properly sized for use on the specific pipe being tapped.

2.19 TAPPING SLEEVES

A. Tapping sleeves shall be designed for a minimum 250 psi working pressure and the material being tapped. All bolts and nuts shall be stainless steel. Tapping sleeves for 4-inch through 18-inch shall be stainless steel. Tapping sleeves shall have a MJ outlet. Tapping sleeves shall be Mueller H-304SS or Ford FTSS.

2.20 SERVICE CONNECTION MATERIALS

A. Materials and standards for larger meters (2-inch and greater) are not listed in the standard specifications. Larger meter installations require a site-specific design. Please contact Washington Water Authority if you require a meter size 3-inch or greater.

Single Meter Set

Main diameter x 1” saddle	Romac 101 NS Mueller DR1S
1” corporation stop	Ford FB1000-4-NL Mueller P-25008N
1” HDPE pipe with inserts	
5/8” x 3/4” x 7” meter yolk	Ford VBH-72-7W-4F-44-NL Mueller 234P24981R2-N
1/2” x 16” SCH 40 PVC brace	
12”x1” Brass Nipple with PVC cap Tailpiece	
Meter Box	Carson Industries 2200
18” cast iron flat meter lid	East Jordan 109

Double Meter Set

Main diameter x 1” saddle	Romac 101 NS Mueller DR1S
1” corporation stop	Ford FB1000-4-Q-NL Mueller P-25008N
1” HDPE pipe with inserts	
1” x 7.5” x 3/4” U branch	Ford U48-43-6.5-NL Mueller P-15363N
5/8” x 3/4” x 7” meter yolk	Ford VBH-72-7W-11-34-NL Mueller 234P24981R2-N
1/2” x 16” SCH 40 PVC brace	
12”x1” Brass Nipple with PVC cap Tailpiece	
Meter Box	Carson Industries 2200
18” cast iron flat meter lid	East Jordan 109

1-inch Meter Set

Main diameter x 1” saddle	Romac 101 NS Mueller DR1S
1” corporation stop	Ford FB1000-4-Q-NL Mueller P-25008N
1” HDPE pipe with inserts	
1” x 12” meter yolk	Ford VB74-12W-44-44-Q-NL
3/4” x 16” SCH 40 PVC brace	
12”x1” Brass Nipple with PVC cap Tailpiece	
24” diameter x 24” deep SDR51 PVC meter	
24” cast iron flat meter lid	East Jordan 111

2.21 TRACER WIRE

A. Tracer wire shall be 12-gauge solid coated copper for underground burial.

B. Jacket color shall be BLUE and made of High Density Polyethylene (HDPE) or High Molecular Weight Polyethylene (HMWPE) designed for direct burial.

C. Connectors shall be used for all splices or repairs. Connectors shall be moisture displacement style as manufactured by 3M DBR, or equal.

1. A locate or conductivity test shall be performed prior to signing off on the project.

2. Wire shall be new and shall have the size, grade of insulation, voltage, and manufacturer's name permanently marked on outer covering at regular intervals. Insulation shall be type THHN, THWN, or XHHW and rated 600 volt, 75 degrees C or higher for wet locations.

2.22 MARKING TAPE

A. At the request of Washington Water Authority, marking tape may be a requirement of design and will be considered on a case-by-case basis.

A. Non-metallic water marking tape shall be warning tape as manufactured by Terra Tape "Extra Stretch", Rhino Marking and Protection Systems, Harris Industries, Inc., or approved equal.

B. Tape shall have a minimum thickness of 4 mils and manufactured with heavy metal-free polyethylene tape that is impervious to all known alkalis, acids, chemical reagents, and solvents found in soil. The minimum overall width of the tape shall not be less than 3-inches.

C. The tape shall be color coded Safety Blue and imprinted with the following message: Caution – Buried Water Line Below

2.23 WATER LINE MARKERS

A. Fiberglass post utility markers shall be fiberglass reinforced composite, 3.75 inches wide, and 72 inches long.

B. Fiberglass post utility markers shall be blue in color.

C. Fiberglass post utility markers shall have decals for visible identification of buried water line day or night.

D. Signs shall be by Carsonite International, or equal.

PART 3 – EXECUTION

3.01 INSTALLATION

A. Water Lines: Refer to Section 4100

B. Water Service Lines: Refer to Section 4100

SECTION 4100

INSTALLATION OF WATER PIPE, FITTINGS, AND MATERIALS

PART 1 – GENERAL

1.01 WORK INCLUDED

A. Installation of Water Lines

B. Installation of Water Services

1.02 SUBMITTALS

A. Submit to the Engineer and USFS for approval and procedures not described in these specifications

1.03 PROTECTION

A. In all cases, the Contractor is responsible for protecting public and private property and protecting any person or persons who might be injured as a result of the Contractor's Work.

B. All utilities shown on the drawings may not represent the exact location; however, the Contractor is responsible for verifying these locations and contacting "Arkansas One Call System" before excavating.

PART 2 – PRODUCTS

2.01 BEDDING AND BACKFILL

A. Refer to Section 2300 – Excavation, Backfilling, and Compacting

2.02 PIPE, FITTINGS, AND MATERIALS

A. Refer to Section 4000 – Water Pipe, Fittings, and Materials

2.03 CONCRETE

A. Refer to Section 3600 – Cast-In-Place Concrete

PART 3 – EXECUTION

3.01 EXCAVATION – GENERAL

A. Perform excavation and prepare in accordance with Section 2300 – Excavation, Backfilling, and Compacting.

B. Never lay pipe in a water filled trench, or when trench conditions or weather are unsuitable for such Work.

C. Divert surface water and de-water trenches during excavation.

D. Excavate for bells so that the entire barrel of the pipe will be uniformly supported on the pipe bedding before placing pipe in the trench.

3.02 LAYOUT

A. The Contractor shall install water lines, valves, fire hydrants, water meters, and other work as shown on the Drawings. Changes are not allowed without written notice from USFS.

3.03 PIERS

A. Install concrete piers as indicated on the plans per Section 3600 - Cast-In-Place Concrete.

3.04 COLD WEATHER INSTALLATION

A. Washington Water Authority reserves the right to order pipe installation discontinued whenever, in its opinion, there is danger of the quality of work being impaired because of cold weather. The Contractor shall be responsible for heating the pipe and jointing material so as to prevent freezing of joints. Do not lay any pipe on frozen ground. No flexible or semi-rigid pipe shall be laid when the air temperature is less than 32°F unless proper precautions per the manufacturer's recommendations are taken by the Contractor and the method is approved by the Engineer and Washington Water Authority.

B. When pipes with rubber gaskets or resilient-type joints are to be laid in cold weather, sufficiently warm the gasket or joint material so as to facilitate making a proper joint.

3.05 PIPE INSTALLATION – PUSH ON JOINTS

A. Inspect each joint of pipe carefully internally and externally before it is placed in the trench. Plainly mark and separate from the remaining pipe any joint found to be cracked, warped, or otherwise damaged. Remove these damaged joints from the project site as soon as possible.

B. Cut pipe in a neat and workmanlike manner without damage to pipe or pipe lining when trimming joint length. Cut ends shall be beveled according to the manufacturer’s recommendation to prevent damage to the bell gasket.

C. Use proper equipment for lowering sections of pipe into trenches. Lower pipe carefully into the trench so the spigot and bell will not become contaminated.

D. Lay each pipe joint to line and grade as shown on the drawings. Keep a minimum of six inches between the pipe and the trench wall.

E. Keep the pipe joints' interior clean from all dirt and other foreign matter as the Work progresses. Maintain the pipe's interior cleanliness until accepted or put in service.

F. Close the open ends of the pipeline temporarily with an appropriate manufactured watertight plug at the end of each day's Work or when discontinuing pipe installation for an appreciable period.

G. TRACER WIRE

1. Tracer wire shall be installed on all buried water lines and service lines.

2. Tracer wire shall be installed six inches above the pipe.

3. Run wire continuous from valve box to valve box, meter box, or other access points. If no access point is available and a splice is required, provide a tracer wire splice pedestal consisting of a tracer wire splice pole and cap at nearest property line or at a location as approved by the Engineer.

4. Pipe testing shall include trace wire. Contractor shall replace the trace wire along the pipe route for any areas in which the pipe cannot be located from the original trace wire installation.

5. Wire breaks shall be repaired with repair kit at no additional expense to the Owner. Repair kits are to be used only to repair wire breaks that occur on tracer wire which has already been backfilled. The splice kit is used because there is not enough slack available in the tracer wire to splice the wire in a tracer wire splice pedestal. If the tracer wire is broken during installation, the Contractor should splice the wire in a tracer wire splice pedestal.

I. WATER LINE MARKERS

1. Fiberglass post markers shall be installed at highway crossings, creek crossings, railroad crossings, and valves

3.06 PIPE INSTALLATION – MECHANICAL JOINT CONNECTIONS

A. All mechanical joint connections shall have Megalug joint restraints.

B. The spigot end of the pipe, the bell of the connecting pipe, and the rubber gasket shall be thoroughly cleaned using soapy water and cloth, removing all foreign materials from the bells, especially the gasket seats as specified for push-on joints. Clean the gland in a similar manner. An approved pipe lubricant shall be applied to the spigot end of the pipe and the gasket.

C. After the gland and gasket are placed on the spigot end of the pipe, a sufficient distance from the end to avoid fouling the bell, insert the spigot end in the fitting bell to the point of firm contact with the bell shoulder. Then advance the rubber gasket into the bell and seat in the gasket seat. Hammering the gasket into the seat is not permitted. Exercise care to center the spigot end within the bell. Bring the gland into contact with the gasket, enter all bolts, and make all nuts hand tight. Exercise continued care to keep the spigot centered in the bell.

D. Make the joints tight by turning the nuts with a torque wrench: First partially tighten a nut, then partially tighten the nut 180 degrees away from it. Work around the pipe with uniformly applied tension until the required torque is applied to all nuts.

E. The Contractor shall provide a torque wrench suitable for measuring tension on bolts for at least such a time as the workmen making the joints have gotten the "feel" of the required tension. At no time should handles longer than those supplied by the wrench manufacturer be permitted. The torque range shall be as follows:

Diameter Inches	Range of Torque Foot Pounds
5/8	45 – 60
3/4	75 – 90
1	85 – 100
1-1/4	105 – 120

F. Proper actuation of the gripping wedges for restraining glands shall be ensured with torque limiting twist off nuts. Tightening sequence shall be as follows. First partially tighten a nut, then partially tighten the nut 180 degrees away from it. Work around the pipe with uniformly applied tension until the required torque is applied to all nuts. Failure to follow proper the proper tightening sequence will result in the disassembly of the joint, removal of the current restraining gland, and installation of a new restraining gland. Any damaged PVC pipe shall be removed as well.

G. All buried iron pipes, valves, and fittings shall be double poly wrapped.

3.07 SOLID SLEEVE INSTALLATION

A. Solid sleeves shall be installed the same as mechanical joint connections for pipe installation.

B. The maximum gap between the two pipes being connected by a solid sleeve shall be one-half (1/2) inch.

3.08 VALVE INSTALLATION

A. All mechanical joint connections shall have Megalug joint restraints.

B. Valves shall be jointed in accordance with the methods of jointing pipe as specified elsewhere herein. Valve stems shall be plumb and there shall not be any obstructions that will prohibit the installation of valve boxes directly over the stem. Megalug joint restraints shall be installed on all valves with mechanical joint ends.

C. All valves shall be firmly supported from below with compacted crushed stone up to and including 12-inch valves, or concrete for all valves 14-inches and greater.

D. All valves shall be polywrapped.

E. Valve boxes shall be installed over the operating nut of each valve and be of adequate length to reach the finished ground or paved surface. Valve boxes shall be installed with a HDPE Valve Box Alignment Device of the proper size and manufacture to fit the valve and the valve box. Boxes shall be firmly supported, plumb, and centered over the valve operating nut. No part of the box shall rest on the valve. The box cover shall be flush with the finished surface.

F. Operating nut extensions shall be used when the top of the operating nut is greater than 4 feet from the top of finished surface. The stem extension shall be of adequate length to reach from the valve operating nut to a point within 24-inches to 12-inches of the finished surface.

G. Tracer wire shall be brought up on the OUTSIDE of the valve box. A “vertical” hole or notch shall be made through the valves box approximately 4-inches below the top. The tracer wire shall be pushed through the hole or notch. Approximately 12-inches of wire shall be coiled inside the valves box for traceability. The tracer wire shall not interfere with the insertion of the lid onto the valve box.

H. All lids shall have a pre-cast concrete collar with a minimum dimension of 24 inches round as appropriate. All collars shall have 3 inches of Class 67 Base or grit base under the collar and shall be grouted in place.

3.09 FIRE HYDRANT INSTALLATION

A. Pipe used to install fire hydrants from the main line to the base of the fire hydrant shall be 6- inch pipe, if less than 10’ in length. All pipe from the valve on the water main to the fire hydrant shall be fully restrained.

B. Pipe used to install fire hydrants from the main line to the base of the fire hydrant shall be 8- inch pipe, if equal to or greater than 10’ in length. A separate maintenance valve, in addition to the valve anchored to the main, shall be installed on all fire hydrant leads exceeding 10 feet.

C. Hydrants shall be thoroughly cleaned before setting, removing all dirt and foreign matter from the barrel and bottom section up to the main valve. The main valve shall be in the "closed" position and the waste outlet shall be free of any obstructions.

D. The Contractor shall take great care to protect the factory applied coating system. Means and methods for the protection of the fire hydrant are the responsibility of the Contractor. At no time shall chains or other abrasive materials come into contact with the factory applied coating system.

E. Minor touchup for “incidental” scratches is permitted using factory provided touchup kits.

F. When the factory applied coating system (from the bury line up), as identified by Washington Water Authority, has damage other than “minor scratches,” a new upper barrel section shall be ordered and delivered from the factory, inclusive of all internal working parts up to the operating stem breakaway. The upper barrel shall be replaced, and the damaged upper barrel returned to the factory at no cost to Washington Water Authority. This includes, but is not limited to, excessive scratches, appearance of rust, or other aesthetic flaws. Field repainting of new fire hydrant installations is not permitted.

G. Hydrants shall be located a safe distance from driveways, roadways, and narrow type sidewalks and in a manner to provide complete accessibility, and they shall stand plumb with nozzles at proper elevation. The hydrant’s “bury line” shall be set at or no more than four (4) inches above the finished grade elevation; therefore, the bottom of hydrant nozzles shall be 18 to 24 inches above the finished grade elevation.

H. Installation of fire hydrant extensions shall be made in the presence of the Engineer or the Engineer’s representative and shall be per the manufacturer’s instructions. The breakable flange and breakable stem coupling shall be removed and installed above ground level.

I. The Contractor shall, if necessary, rotate the hydrant barrel or nozzle section at the flanged joint to obtain the desired nozzle position as specified by the Engineer. The pumper nozzle shall be at a right angle to and face the street unless otherwise directed by the Engineer.

J. The bowl or bottom of the hydrant shall be supported firmly on the bottom and shall be braced against unexcavated earth at the end of the trench with concrete reaction backing. Solid concrete blocks may be used to support the bottom of the hydrant. If considered necessary by the Engineer, the hydrant shall be tied to the branch pipe with suitable Series 300 stainless steel rods or clamps. These rods or clamps shall be furnished by the Contractor without additional compensation.

K. A drainage bed shall be provided under and around the base of the hydrant of at least six (6) cubic feet in volume and extending at least six inches (6”) above the drain outlet and shall consist of ASTM #67 gravel. Under no circumstances shall the drain outlet on the hydrant or the drainage bed be connected to a sewer.

L. Backfilling and tamping around hydrant barrels shall be continuous in operation.

M. If a fire hydrant is not located inside a concrete paved area, then a 15.0" x 15.0" x 4.0" (deep) Class "B" concrete pad shall be poured just below the break-away joint of the fire hydrant barrel.

N. Fire hydrants, immediately after installation, shall be covered and wrapped with a heavy cloth, water-resistant sack, or black polyethylene sheeting, well taped in place around the hydrant, to identify the hydrant as being "not in service".

3.10 METER SETTINGS

A. Meter settings shall be installed where shown on the Drawings and/or as directed by USFS. Installation shall be as per the Standard Details. Meter setters shall be installed in a horizontal and plumb position within the meter box and at a depth to provide the required space between the top of the meter and the bottom of the meter box lid.

B. A drainage bed consisting of ASTM #67 stone shall be placed a minimum of 6-inches thick and 6- inches outside the edge of the bottom of the meter box.

C. Tracer wire shall be brought up on the INSIDE of the meter box with the water service line. Sufficient tracer wire shall be provided such that approximately 12-inches of wire will extend beyond the top of the box when pulled taunt. The tracer wire shall be loosely coiled and place inside the meter box. Do not wrap the tracer wire around the meter setter.

D. The bottom of the meter setter connections shall be visible at the bottom of the meter box at the time of testing and of final inspection. Any dirt or debris in the bottom of the meter box shall be removed before the project is released for final payment and/or acceptance.

E. The final grade at the meter box location shall be determined by the Engineer of Record and the meter box shall be placed at that grade. Final grade should take into account probable future installation of topsoil and/or sod. Any boxes falling in driveways or sidewalks shall be relocated at the expense of the developer or lot owner. No meters shall be set by Washington Water Authority until the meter box is adjusted to the proper grade.

3.11 BLOW-OFF INSTALLATION

A. Blow-off hydrants shall be thoroughly cleaned before setting, removing all dirt and foreign matter from the barrel and bottom section up to the main valve. The main valve shall be in the "closed" position and the waste outlet shall be free of any obstructions.

B. Blow-off hydrants shall be installed in such a manner to provide complete accessibility, and they shall stand plumb with nozzles at proper elevation. The discharge nozzle shall be a minimum of 24-inches above finished grade elevation.

C. The bowl or bottom of the blow-off hydrant shall be supported firmly on the bottom and shall be well braced against unexcavated earth on the backside of the blow-off hydrant. Solid concrete blocks, or other suitable material may be used to block the blow-off hydrant.

D. A drainage bed shall be provided under and around the base of the blow-off hydrant of at least six (6) cubic feet in volume and extending at least six inches (6") above the drain outlet and shall consist of ASTM #67 gravel. Under no circumstances shall the drain outlet on the hydrant or the drainage bed be connected to a sewer.

E. Backfilling and tamping around blow-off hydrant barrels shall be continuous in operation.

G. Blow-off hydrants, immediately after installation, shall be covered and wrapped with a heavy cloth, water-resistant sack, or black polyethylene sheeting, well taped in place around the hydrant, to identify the hydrant as being "not in service".

3.12 DRINKING WATER FOUNTAINS

A. Drinking water fountains settings shall be installed where shown on the Drawings and/or as directed by USFS.
B. Installation shall be as per the Standard Details and to be completed in accordance with manufacturer's specifications.

3.13 FROST FREE HYDRANTS

A. Frost Fee Hydrants shall be installed where shown on the Drawings and/or as directed by USFS.
B. Installation shall be as per the Standard Details and to be completed in accordance with manufacturer's specifications and with a new concrete slab.

3.14 PRESSURE SYSTEM FOR UPPER CABINS

A. Pressure system for upper cabins shall be installed as shown on the Drawings and/or as directed by USFS.
B. Installation shall be as per the Standard Details and in accordance with manufacturer's specifications.

3.15 TAPPING SADDLE INSTALLATION

A. Tapping saddles shall be used 1-inch and 2-inch service taps.

B. The pipe shall be free of dirt and other debris before attaching tapping saddle. That part of the pipe barrel, other than concrete pipe, which will be in contact with the gasket of tapping saddles, shall be smooth. All rough areas on the pipe barrel shall be smoothed. The Contractor shall field verify all pipe and fitting dimensions. Tapping saddles shall be installed at least twenty-four (24) inches from bell joints, fittings, end of pipe joint, or another tap.

C. Tapping saddles shall be bolted securely to the pipe. The face of the outlet shall be zero to ten (0-10) degrees from horizontal. The bolts for tapping saddles shall be alternately tightened "snug" and then alternately tightened to a torque as recommended by the manufacturer.

D. The tapping valve shall be attached securely to the tapping saddle to provide a watertight seal. Proper tools for installing brass hardware shall be used.

E. The pilot drill and shell cutter shall be in good condition. The pilot, shell cutter, and any other component of the tapping machine that will or may come into contact with the interior of the tap valve or potable water pipe, shall be thoroughly sterilized with straight bleach or super-chlorinated solution. The shell cutter shall be the size required to cut the full opening specified.

F. After the tap is complete and the tapping machine has been removed, the bolts for the tapping saddle must be re-torqued per the manufacturer's specifications to ensure a proper seal.

G. The tapping saddle shall be enclosed in polyethylene material

3.15. TAPPING SLEEVE INSTALLATION

A. The pipe shall be free of dirt and other debris before attaching tapping sleeve. That part of the pipe barrel that will be in contact with the gasket of tapping sleeve, shall be smooth. All rough areas on the pipe barrel shall be smoothed. The Contractor shall field verify all pipe and fittings.

B. Tapping sleeves shall be bolted securely to the pipe. The face of the outlet shall be plumb. The bolts for tapping sleeves shall be alternately tightened "snug" and then alternately tightened to a torque as recommended by the manufacturer.

C. The tapping valve shall be bolted securely to the tapping sleeve. The tapping valve shall be adequately supported from beneath. The weight of the tapping valve shall not be supported by the tapping sleeve. A concrete "mud slab" at least six (6) inches thick shall be poured under the location of all tapping valves 14 inches and larger and the weight of the valve shall be supported by the mud slab. The tapping machine shall be bolted securely to the valve.

D. After installation of the tapping sleeve and the tapping valve and before drilling through the pipe, the assembly shall be hydrostatically by introducing water through the sleeve test tap.

E. The pilot drill and shell cutter shall be in good condition. The pilot, shell cutter, and any other component of the tapping machine that will or may come into contact with the interior of the tap valve or potable water pipe, shall be thoroughly sterilized with straight bleach or super-chlorinated solution. The shell cutter shall be the size required to cut the full opening specified.

F. Openings in the pipe barrel for tapping saddles installed on dry pipe shall be cut with a pilot drill and shell cutter. Torch cutting is not permitted.

G. Tapping operations must not commence before inspection by the Engineer or his authorized representative. Tapping operations must not commence before the tapping assembly has passed a pressure test. The tapping assembly shall be tested with both the valve open and closed.

H. Only qualified operators shall operate the tapping machine. The "coupon" shall be withdrawn and be given to the Engineer for inspection. Care shall be exercised to avoid drilling or cutting the backside of the pipe by carefully assuring the engagement of the pilot drill and shell cutter shaft.

I. After the tap is complete and the tapping machine has been removed, the bolts for the tapping sleeve must be re-torqued per the manufacturer’s specifications to ensure a proper seal.

J. All taps 14-inch and larger shall require the installation of a butterfly valve immediately after the tapping valve. The tapping valve shall be fully opened and abandoned in place.

K. The tapping sleeve shall be enclosed in polyethylene material.

L. Tapping sleeves shall be adequately thrust blocked with concrete.

3.16 INSTALL OF POLYETHYLENE PROTECTION MATERIAL

A. Polyethylene material, either in tubing form or in the form of flat sheet or rolls, as specified herein, shall be placed around all buried mechanical joints of pipe and fittings, valves, fire hydrants, and all saddles, sleeves, and couplings, tapping saddles, and any other appurtenance with exposed bolts. Any and all iron or steel components installed below ground shall be wrapped with the polyethylene material.

B. Ductile iron and steel pipe and appurtenances shall be completely encased in polyethylene tubing material. It is not the intent that the material form an enclosure that is absolutely air or watertight, but to prevent pipe to soil contact.

C. Polyethylene tubing shall be applied to pipe by one of the following methods:

1. Method "A": Cut polyethylene tubes to a length approximately two feet (4’) longer than the length of the pipe section. Slip the tubes around the pipe, centering it to provide a one foot (2’) overlap on each adjacent pipe section, and bunching it accordion fashion lengthwise until it clears the pipe ends. Lower the pipe into the trench and make up the pipe joint with the preceding section of pipe. A shallow bell hole must be made at joints to facilitate installation of the polyethylene tube. After assembling the pipe joint, take bunched polyethylene from the preceding length of pipe, slip it over the end of the new length of pipe, and secure in place. Then slip the end of the polyethylene from the new pipe section of the end of the first wrap until it overlaps the joint at the end of the preceding length of pipe. Secure the overlap in place. Take up the slack width to make snug, but not tight, fit along the barrel of the pipe, securing the fold at quarter points with tape.

2. Method "B": Cut polyethylene tubes to a length approximately one foot (1’) shorter than the length of the pipe section. Slip the tubes around the pipe, centering it to provide six inches (6”) of bare pipe at each end. Make polyethylene snug, but not tight, secure ends. Before making up a joint, slip two sections of six foot (6’) length of polyethylene tube over the end of the preceding pipe section, bunching it accordion fashion lengthwise. After completing the joint, pull the two sections of six foot (6’) length of polyethylene over the joint, overlapping the polyethylene previously installed on each adjacent section of pipe by at least two feet (2’); make snug and secure each end.

D. Pipe-Shaped Appurtenances. Bends, reducers, offsets, and other pipe-shaped appurtenances shall be covered with double layers of polyethylene in the same manner as the pipe.

E. Odd-Shaped Appurtenances. Valves, tees, crosses, and other odd-shaped pieces which cannot practically be wrapped in a tube, shall be wrapped with two layers of flat sheets or split length of polyethylene tubes. The sheets shall be passed under the appurtenance and brought up around the body. Seams shall be made by bringing the edges together, folding over twice, and taping down. Slack width and overlaps at joints shall be handled as described in Paragraph 3.13.C.1 above. Tape polyethylene securely in place at valve stem and other penetrations.

F. Openings in Tubing Material. Openings for branches, service taps, blow-offs, air valves, and similar appurtenances shall be made by making an “X” shaped cut in the polyethylene and temporarily folding the film back. After the appurtenance is installed, tape the slack securely to the appurtenance and repair the cut, as well as any other damaged areas in the polyethylene with tape.

G. Junctions Between Wrapped and Unwrapped Pipe. Where polyethylene wrapped pipe joins a pipe that is not wrapped, extend the polyethylene tube to cover the unwrapped pipe a distance of at least two feet (2') and secure the end.

H. The polyethylene material shall be secured around the pipe and appurtenances by at least three (3) circumferential wraps of tape.

I. All tongs, cables, or chains that are used for lifting pipe and appurtenances that have been encased in polyethylene material shall be adequately padded to prevent damage to the material.

J. Repair any rips, punctures, or other damage to the polyethylene with tape or with a short length of polyethylene tube cut open, wrapped around the pipe, and secured in place.

K. Polyethylene material shall be stored on the job site in such a manner that it is not exposed to direct sunlight. Exposure during installation shall not exceed forty-eight (48) hours.

L. Backfill material shall be the same as specified for pipe without polyethylene wrapping. Special care shall be taken to prevent damage to the polyethylene wrapping when placing backfill.

M. Backfill material shall be free from cinders, refuse, boulders, rocks, stones, and/or other material that could damage polyethylene.

3.17 CONCRETE THRUST BLOCKS AND ANCHOR COLLARS

A. Concrete thrust blocks and anchor collars shall be provided along the water line in accordance with the construction details, plan sheets, or as directed by the Engineer. The concrete mix shall be Class "B" for anchor collars and Class "A" for thrust blocks. All bends, tees, caps, plugs, and fire hydrants shall be thrust blocked unless specifically detailed in the construction drawings that no thrust blocking is required. Anchor collars shall be constructed on ductile iron pipe only; if the pipeline being restrained is primarily PVC or other non-ductile iron pipe, at least one full joint of ductile iron pipe shall be inserted in the pipeline run to receive the concrete anchor collar.

B. Concrete for thrust blocks and anchor collars shall be placed against undisturbed soil. The excavation shall be hand shaped and free of loose material. Forms shall be used to confine the concrete in areas other than that part that is in contact with undisturbed soil in the direction of the thrust.

C. No concrete shall be placed around any part of a joint or placed so that it interferes with the removal of any joint accessories such as bolts, followers, threads, collars, couplings, etc. Fire hydrant drains shall not be restricted.

D. The top of the concrete thrust block or anchor collar shall be struck off with a wood straight edge or float.

E. Concrete shall not be placed when the temperature is below 40° F and dropping or below 35° F if the temperature is rising, unless approved by the Engineer.

F. Admixtures are not to be used without the approval of the Engineer.

G. All reinforcement shall be inspected by the Engineer prior to placement of concrete. All placement of concrete must be in the presence of the Engineer or his representative. The Contractor is cautioned that he may be required to remove, without compensation, any concrete placed in the absence of the Engineer or his representative.

H. Backfill over concrete thrust blocks or anchor collars shall not be placed before the concrete has attained initial set.

I. No thrust blocks shall be less than six inches (6") thick between the pipeline or appurtenances and undisturbed soil in the direction of thrust on pipes 12-inch diameter and smaller. On larger pipes, the thickness of thrust blocks shall be as directed by the Engineer. A thrust block with any component of its length to width to depth ratio exceeding two (2) shall be reinforced with steel reinforcement bars as directed by the Engineer. The Engineer will consider the size of the thrust block, the size of the water main, and the system pressure in the determination of the size and spacing of the steel reinforcement.

J. The excavation shall be free of water before concrete is placed. Steel reinforcement shall be placed as specified on the drawings.

K. The pipe or appurtenances to be in direct contact with concrete shall be cleaned before placing the concrete. The Engineer of Record is required to design the thrust blocks, anchors and other restraints and submit to Washington Water Authority for review and approval. If request by USFS, the engineer of record shall provide a geotechnical report to include soil valves for thrust blocking design.

L. Thrust blocks for vertical bends shall be adequate to resist the thrust by mass alone when the thrust is upward.

M. Thrust blocks and anchor collars shall be adequate to restrain the pipeline and appurtenances at design pressure equal to 150% of the static pressure at the lowest point with a minimum pressure of 200 psi. The Engineer of Record shall provide calculations for review for all thrust blocks, anchor collars and other thrust restraints.

N. Concrete thrust blocks and anchor collars on 12-inch and smaller pipelines shall have a minimum curing time of three days (72 hours) before any pressure is placed against the block or collar. Concrete thrust blocks and anchor collars on 14-inch and larger pipelines shall have a minimum curing time of seven days before any pressure is placed against the block or collar.

Q. Concrete thrust blocks or anchor collars that fail to restrain the pipe or appurtenances shall be replaced by the Contractor at his expense.

R. Reducers receiving an anchor collar shall be long bodied fittings.

S. All water lines with dead ends shall be installed with an upstream valve, one full joint of ductile iron pipe with a MJ restraining gland, concrete anchor collar, a MJ cap with restraint, and a blow off assembly. The seat of the MJ restraining gland on the ductile iron pipe shall face the valve.

T. Anchor collars subject to two-way thrust shall have two identical reinforcement steel mats. Anchor collars for 24” and smaller diameter pipe with two-way thrust shall have two (2) Mega-Lug retaining glands placed back-to-back, spaced just inside the two reinforcing steel mats. Anchor collars for 26” and larger diameter pipe shall have two weld-on thrust rings, spaced just inside the two reinforcing steel mats.

3.18 BACKFILLING AND INSPECTION

A. Before backfilling, install concrete thrust blocks and anchor collars in accordance with the details at the location and interval and shown on the Drawings. Use concrete as specified in Section 3600- Cast-In-Place Concrete.

B. After the pipeline is installed and visually inspected by the Engineer, backfill the trench per Section 2300-Excavation, Backfilling, and Compacting.

C. Test the pipeline per Section 5300-Inspection and Testing of Water Lines and Service Lines.

D. Repair all incidental damage to buildings, structures, utilities, pavements, landscaping, etc.

E. Repair all pavements per Section 6000-Pavement Repair.

F. Repair sodded and grass areas to original condition per Section 6100-Lawn and Grass Restoration.

3.19 SEWER LINE CROSSING

A. Sewer lines installed under a water line must have a clear distance between pipes of at least eighteen (18) inches.

B. The sewer line shall be installed such that a joint of pipe is centered along the water line and the joints are as far as possible from the water line.

C. If 18-inches of clearance cannot be provided or when the water main must pass under the sewer main, either the sanitary sewer main or the water main shall be encased in watertight steel encasement pipe a minimum of 10 feet either side of the line, centered over the point of crossing. The ends of the encasement pipe shall be sealed watertight. Refer to Section 3400 – Steel Encasement Pipe for end seals.

3.20 STORM SEWER CROSSING

A. All water lines crossing under all concrete storm drains, or any storm drain 30-inch diameter and larger, or all storm drains with multiple pipe runs, shall be steel encased a minimum of 5 feet either side of the storm drain.

3.21 CUT AND CAP

A. Water lines that are to be abandoned shall be cut and capped. Water mains abandoned under roadways shall be filled with flowable fill in addition to being capped.

B. Mechanical joint restraints and concrete shall be used to resist thrust loads.

3.22 ABANDON CORPORATION STOP

A. All corporation stops used for testing and/or chlorination need to be properly abandoned by fully closing the corporation stop, removing all service line materials, installing a solid copper disk, and reinstalling the corporation nut resulting in a watertight seal in the event that the corporation valve fails.

SECTION 5000

INSPECTION AND TESTING OF WATER LINES AND SERVICE LINES

PART 1 – GENERAL

1.01 WORK INCLUDED

A. This section covers the inspection and testing of water lines and services lines. Testing is required before final acceptance of water lines and service lines by USFS.

1.02 SCOPE OF WORK

A. All pipelines shall be inspected and tested before final acceptance. The methods to be used are as follows:

1. New Water Pipelines

    a. Visual inspection during installation and before backfill.

    b. Hydrostatic pressure test.

    c. Disinfection.

    d. Bacteriological sampling.

    e. Final Visual Inspection.

    f. Tracer wire continuity test.

2. Water Meter Setters

    a. Visual inspection during installation and before backfill.

    b. Final Visual Inspection.

3. Service Lines

    a. Visual inspection during installation and before backfill.

    b. Hydrostatic pressure test.

    c. Final Visual Inspection

d. Tracer wire continuity test.

e. Disinfection

1.03 DELIVERABLES

A. The Engineer shall provide a complete and comprehensive testing report summary complete with all inspection and testing dates and results.

PART 2 – PRODUCTS

A. NOT USED

PART 3 – EXECUTION

3.01 VISUAL INSPECTION DURING INSTALLATION AND BEFORE BACKFILL

A. The Engineer shall be responsible for inspecting water lines, water meter setters, and service lines during all phases of construction. The Engineer shall provide full time inspection services. All work not conforming to these specifications that is discovered during this inspection phase will be corrected by the Contractor.

3.02 PRESSURE TEST FOR TAPPING SLEEVES

A. The contractor shall provide all pumps or other equipment necessary to test the tapping sleeve before making a tap. The duration of the hydrostatic leakage test on tapping sleeves shall be thirty (30) minutes at 200 psi with zero leakage, or at static main line pressure, whichever is greater. The tapping sleeve assembly valve shall be tested in both the open and closed position.

3.03 PRESSURE TEST FOR WATER PIPELINES AND SERVICE LINES

A. After completion of construction of all water lines or sections thereof, the Contractor shall flush, test, and disinfect the new water lines and in accordance with the Engineer of Record’s plan as described below.

B. Flushing

1. All water for flushing, testing, and disinfecting water lines shall be supplied by the USFS.

2. The work shall be coordinated to ensure that it will not be carried on during periods of high water usage. Water valves or other appurtenances on the existing water system, new tapping valves, or valves or appurtenances that have been

accepted by Washington Water Authority, shall only be operated by, or under the direct supervision of, WWA.

3. The Contractor shall fill and flush the newly constructed lines and visually check all combination air release and vacuum valves, blow-off valve assemblies, line valves, fire hydrants, and meter setters to assure proper operation.

4. The Engineer of Record shall develop a flushing plan to ensure that all lines are properly flushed. The plan should specify the sequence in which valves and hydrants are to be opened and the duration thereof, ensuring complete flushing and protecting Washington Water Authority's water system from construction contaminated water. The volume to be flushed shall be no less than three (3) but no more than five (5) times the volume of the main to be flushed. The flow shall be such that a flushing velocity of not less than 2.5 feet per second and preferably 3.5 feet per second or greater is attained.

C. Hydrostatic Testing

1. All pipe shall be tested as set out in AWWA C600, latest revision, for ductile iron pipe and AWWA C605, latest revision, for PVC pipe. Tests will be conducted only after the line is completed, including all taps and meter settings as required and the backfill completed. These tests shall be performed by the Contractor in the presence of a USFS Representative and the Engineer of Record. The Contractor shall furnish all necessary pressure gauges, meters, and pumps and make all taps and connections.

2. Each valved section of pipe shall be slowly filled with water and the specified test pressure shall be applied by means of a pump connected to the pipe in a manner satisfactory to the Engineer. Before applying the test pressure, all air shall be expelled from the pipe by permanent taps or corporation cocks where necessary.

3. Test pressure shall be 200 psi or 150 percent of the normal static operating pressure, as determined by USFS, whichever is greater. The contractor shall provide all pumps or other equipment necessary to maintain the test pressure within +/-0 psi at the test point for a period of two (2) hours. All interior valves including guardian valves on fire hydrants and other appurtenances shall be open during all tests.

4. The test pressure shall not vary by more than ± zero (0) psi for the duration of the test. During the duration of the two-hour test, if the test pressure drops more than 0.0 psi from the start pressure, the test shall be terminated and considered failure due to assumed leaks in the tested pipe section. If the pressure rises beyond the allowed 0 psi variance, the test shall be terminated and remaining air shall be purged from the pipeline.

D. Leakage

1. The leakage test shall be conducted concurrently with the pressure test. Leakage shall be defined as the quantity of water that must be supplied into the newly laid pipe, or any valved section thereof, to maintain pressure within 5 psi of the above specified test pressure after the air in the pipeline has been expelled and the pipe has been filled with water.

2. The maximum allowable leakage volume shall be per AWWA standards for PVC and Ductile Iron pipes. For PVC pipe see AWWA C605 Table 4a, for Ductile Iron pipe see AWWA C600 Table 5A.

3. Upon completion of a two-hour test where the test pressure did not vary by more than ± five (5) psi, the contractor shall determine the leakage amount by measuring the amount of "make-up" water necessary to restore the original starting pressure.

4. Should any test of pipe laid disclose leakage greater than that specified, the leak(s) shall be located and repaired, and the line shall be re-tested at the Contractor's expense. All visible leaks shall be repaired regardless of the amount of leakage. It may be necessary to utilize leak detection equipment to locate not visible leaks at the Contractor's expense.

3.04 DISINFECTION

A. After successful pressure testing, the line(s) shall be flushed at a velocity equal to or greater than 2.5 feet per second. The line shall then be disinfected in accordance with AWWA C651, latest revision, for Disinfecting Water Mains, continuous feed method, except that the placing of hypochlorite granules into the main during construction will not be permitted.

B. The Contractor shall provide a test/chlorine tap no greater than 10' downstream from the beginning of the new water line. All excavation, installation, use of and proper abandonment of the test tap is the responsibility of the Contractor.

C. The final concentration of chlorine inside the main shall be 25 parts per million (ppm) at all locations and shall be maintained for a minimum of 24 hours. The chlorine residual at the end of the 24-hour period shall not be less than 10 ppm. The contractor shall supply all test kits necessary to verify chlorine concentrations.

D. The contractor shall operate all valves and hydrants in the treated section of water line during the initial 24 hours to ensure disinfection of the appurtenances.

E. The contractor shall take great care when flushing the line to assure proper drainage is available to prevent harm at any adjacent downstream location.

F. Disposal of the disinfecting water shall be in a manner that will protect the public and the receiving waters from harmful concentrations of chlorine. Dichlorination of the disinfecting water shall be in accordance with AWWA C655, Field Dichlorination, latest edition. A dichlorination plan shall be prepared by the Engineer of Record.

G. After disinfection is complete, USFS shall then flush the disinfecting solution from the lines to a point that the chlorine concentration is back down to the same level as the treated water from the distribution system. The treated water lines will then be tested before being placed into service.

3.05 BACTERIOLOGICAL SAMPLES

A. Bacteriological samples shall be taken by USFS personnel only. Samples shall be taken on two consecutive days and shall be taken only on Monday, Tuesday, Wednesday, or Thursday. Before a line is placed in service, two consecutive series of samples which are not collected on the same day and are taken no more than 4 days apart must show that the water is bacteriologically safe for drinking purposes.

3.06 FINAL VISUAL INSPECTION

A. Upon completion of the above tests, USFS and the Engineer will perform a final visual inspection of water pipelines and meters.

B. A punch list of defects (including obvious running leaks) will be prepared and sent to the Contractor for correction at the Contractors' expense.

3.07 TRACER WIRE CONTINUITY TESTING

A. Contractor shall perform a continuity test on all tracer wire in the presence of the Engineer or the Engineers’ representative. If the tracer wire is found to be not continuous during testing, Contractor shall repair or replace the failed segment of wire at their own expense.

B. A final continuity test shall be performed by a USFS representative before the project will be accepted by USFS. If the tracer wire is found to be not continuous during testing, Contractor shall repair or replace the failed segment of wire at his own expense.

SECTION 6002

GRAVEL SURFACE REPAIR

PART 1 – GENERAL

1.01 WORK INCLUDED

A. This section covers the materials and procedures used in the repair of gravel roads, streets, or other public rights-of-way where a water lines, sewer line, or structure is constructed.

1.02 RELATED WORK

A. Section 2300 - Excavation, Backfilling and Compacting.

1.03 REGULATIONS AND STANDARDS

A. All permanent repairs of streets, roads, sidewalks, other public rights-of-way, private drives, private parking lots, etc. shall comply with the requirements shown on the Standard Detail Drawings and Tables. The Contractor is responsible for following the requirements of all local Ordinances, Regulations, or Codes governing the repairs to roads, streets, or other public rights of way. In particular:

1. Repair of State Highways: per requirements of the Arkansas State Highway Commission.

2. Repair of county roads: per requirements of the County Roads Department.

3. Repair of local municipal streets, sidewalks, and driveways: per the requirements of local municipal code.

4. Permit for street cut and repairs shall be furnished by the Contractor.

5. Contractor shall secure permits and inspections, post necessary bonds, and pay necessary fees.

1.04 TESTING, OBSERVATION, AND INSPECTION REQUIREMENTS

A. Testing and inspection is required for pavement repairs in accordance with all local Ordinances, Regulations, or Codes governing the repairs. The inspection of street repairs shall be coordinated Washington Water Authority to ensure compliance with all applicable standards.

B. Gradation of stone materials shall be performed in accordance with ASTM C136.

C. The Contractor shall provide at least 24 hours of advance notice for any concrete or asphalt placement.

PART 2 – PRODUCTS

2.01 BASE COURSE MATERIALS

A. Base Course: Conform to AHTD Standard Specifications for Highway Construction, Section 303, Class 7.

B. Free from objectionable, deleterious, or other injurious matter.

PART 3 – EXECUTION

3.01 PLACING GRAVEL SURFACING

A. Trenches shall be backfilled with Class 7 base material compacted to 95% of maximum density at optimum moisture content as determined by ASTM D698 Standard Proctor Density.

B. Gravel surfacing shall be replaced to at least the compacted thickness of the original surface. All excavated material shall be removed from gravel surfaces affected by construction and sufficient new gravel material shall be placed to restore the original surfaced area. Minimum gravel thickness shall be six (6) inches.

C. For compacted depths exceeding 8 inches, place material in multiple courses of equal depth which do not exceed 8 inches.

D. Compact each course with mechanical compaction equipment. Compaction with wheel of backhoe or trackhoe is not acceptable. Compact to 98% Modified Proctor Density.

E. Finish grade to provide smooth transition with surrounding gravel. Avoid leaving any humps or ruts.

F. Repair settling as required.

3.02 TRAFFIC CONTROL

A. Whenever traffic flow restrictions of any kind are anticipated, the Contractor will be required to contact USFS to be given permission to obstruct traffic flow.

B. Street closing permits must be obtained from proper government agencies.

C. Suitable construction signs, barricades, lights, etc. shall be erected and the work outlined by adequate lighting at night in order to protect persons from injury and avoid property damage. Provide qualified flagmen to direct traffic while working upon a highway, street, or road over which traffic must pass.

D. If requested, Contractor will be required to submit a barricade plan to Washington Water Authority or traffic control agency having jurisdiction. Barricade plan shall be satisfactory to the traffic control agency having jurisdiction.

E. Maintain traffic and protect the public from all damage to persons and property in accordance with applicable State, City, and County regulations. Maintain and protect access, for vehicle and pedestrian traffic, to and from all properties adjoining or adjacent to those streets affected by operations, and to subject the public to a minimum of delay and inconvenience.

F. Traffic shall be detoured as required; however, no traffic shall be detoured without prior knowledge and approval of the traffic control agency having jurisdiction. Notify the traffic control agency having jurisdiction at least 24 hours in advance of the time proposed to detour traffic. No street shall be completely blocked, nor blocked more than one-half at any time without specific authorization.

G. Closing streets with one access route shall not be acceptable or permitted. One traffic lane shall remain open at any given time. Excavated areas within the traffic lanes of highways, streets, roads, and pedestrian walkways shall be backfilled as soon as possible and the area opened to traffic.

SECTION 6100

LAWN AND GRASS RESTORATION

PART 1 – GENERAL

1.01 WORK INCLUDED

A. This Section covers the replacement of sod in lawns disturbed by the construction.

B. Consists of furnishing and applying fertilizer, seed, mulch cover, and water at all other locations disturbed by the construction.

C. Maintenance service.

1.02 SCOPE OF WORK

A. This Section covers the furnishing and placing of sod to form solid mats on areas shown on the Drawings (generally lawns or commercial green spaces) or seed and mulch all other areas disturbed by the Contractor.

B. It covers the furnishing and applying of water.

C. It covers the furnishing and placing of four (4) inches of topsoil on all areas disturbed during construction.

D. It covers the furnishing and placing of fertilizer.

E. All work shall be in accordance with details shown on the Drawings and within these Specifications.

F. The Contractor is responsible for following the requirements of all local Ordinances, Regulations, or Codes governing re-vegetation and slope stabilization.

PART 2 – PRODUCTS

2.01 TOPSOIL

A. Topsoil shall be reasonably free from subsoil, clay, lumps, brush, objectionable weeds and/or other litter and shall be free from roots and toxic substances or other material or substances that might be harmful to plant growth or be a hindrance to grading, planting and maintenance operations.

2.02 FERTILIZER

A. Fertilizer shall be a standard commercial product complying with State and Federal laws and with the requirements issued by proper authorities.

B. Fertilizer shall be delivered to the site in the manufacturer's original container, on which shall be plainly marked the manufacturer's name and the guaranteed chemical analysis.

C. Except as noted in the following sentence, fertilizer shall contain not less than the percentages by weight of ingredients as follows:

Nitrogen - 13 percent
Phosphorus, P205 - 13 percent
Potash, K2 - 13 percent

Other 1:1:1 ratio fertilizers may be used, provided the available plant food remains the same as herein specified.

D. All fertilizer shall be solid and shall be in a condition which will permit proper distribution.

2.03 SEEDING

A. Contractor shall apply a mixture of various annuals and perennials to provide overlapping times of seasonal peak vegetative cover. Seeding shall only be done just prior to the vegetation's peak season for best results. Seeding shall be used when there is sufficient time in the season to ensure adequate vegetation establishment and erosion control.

B. To optimize soil stabilization, Contractor shall utilize a nurse crop of quick growing annuals within a mix of perennials appropriate for the season. The nurse crop germinates and grows rapidly, holding the soil until the slower-growing perennial seedlings become established. Permanent vegetation shall not be considered established until a ground cover of perennial vegetation is achieved that is uniform and mature enough to survive and be of sufficient density to preclude erosion.

C. Contractor shall conduct seeding activities to achieve stabilization that are generally congruent with the following schedule:

F.S. Seeding Specification

Seeding Season	From Date	To Date
Season 1	March 15	May 30
Season 2	September 1	October 15

Season 1

Fertilizer	Pounds/Acre	Seed	Pounds/Acre
12-24-12	500	Rye Grass	15
		Orchard Grass	20
		Brown Top Millet	6
		Ladino Clover (Inoc.)	8

Season 2

Fertilizer	Pounds/Acre	Seed	Pounds/Acre
12-24-12	500	Rye grass	15
		Orchard Grass	20
		Hairy Vetch	7
		Ladino Clover (Inoc.)	6

D. Contractor shall submit all labels/tags from bags and seed purchase invoices to Washington Water Authority.

2.04 MULCH

A. Mulching shall be used in conjunction with both temporary and permanent seeding practices to enhance their success by providing erosion protection prior to the onset of vegetative growth. Straw mulching shall be of oat, wheat, or rice straw mulch. Hay mulch shall be prairie grass, Bermuda grass, or other hay as approved. Mulch shall be dry and reasonably free from Johnson grass or other noxious weeds and shall not be excessively brittle or in an advanced state of decomposition. All material will be inspected and approved prior to use.

2.05 LIME

A. Lime shall be agricultural grade ground limestone or equivalent as approved by the Engineer.

2.06 WATER

A. Water shall be free from any substances, in solution or in suspension, which would inhibit the rapid growth of grass.

B. Contractor shall be responsible for any fees and water use charges as may be charged to the Contractor by Washington Water Authority.

PART 3 – EXECUTION

3.01 SEQUENCE

A. Unless otherwise requested in writing and acceptable to the Engineer, the sequence of work for seeding shall be:

1. Finish grading
2. Application of lime and fertilizer
3. Seeding
4. Firming and cultipacking
5. Mulching seeded areas
6. Vibratory rolling
7. Cleanup
8. Protection
9. Maintenance
10. Final acceptance

B. If trenching operations do not commence within 30 days after clearing and grubbing, the disturbed area shall be seeded.

3.02 FINISH GRADING

A. After trench backfilling and compacting activities have been completed, all areas which have been disturbed or damaged by construction operations shall be brought to original grade, or if directed by the Engineer, graded to secure effective drainage. Unless otherwise indicated, a slope of at least 1 percent shall be provided. All ruts, deep tracks, dead furrows, and ridges shall be eliminated.

B. In areas of maintained lawn grass, 6 inches of topsoil shall be placed. Imported topsoil may be substituted for stockpiled surface material. Topsoil shall be of a quality at least equal to the existing surface material in adjacent areas, free from trash, debris, and well suited to support plant growth.

C. Maintain finish grade until final acceptance. Repair damage caused by work operations or erosion. Add imported topsoil, if required.

3.03 CLEARING

A. Prior to seeding, vegetation that may interfere with operations shall be mowed, grubbed, and raked. The collected material shall be removed from the site. The surface shall be cleared of stumps, loose surface rocks, roots, cable, wire, and other materials that might hinder the work or subsequent maintenance.

3.04 FERTILIZER AND LIME APPLICATION

A. Fertilizers shall be applied at appropriate agronomic rate. If necessary to achieve final stabilization, fertilizer shall be applied at a minimum rate of 250 pounds per acre (0.0057 pounds per square foot) in advance of tilling/seedbed preparation operations. When soil samples are not practical, fertilizer shall consist of 13-13-13 (nitrogen, phosphorus, and potassium content). Lime shall be applied at a rate of one tone per acre or as recommended by soil test. The fertilizer distributor box shall be equipped with baffle plates to prevent downward movement of fertilizer when operating on a slope. The fertilizer shall be raked in and thoroughly mixed with the soil to a depth of approximately 2 inches prior to the application of seed or mulch.

B. Fertilizer shall be uniformly incorporated into the soil or combined with seed in the hydro-seeding operation.

3.05 APPLICATION OF SEED

A. Areas to be seeded shall be dressed to natural shape.

B. The Contractor shall obtain an approved topsoil from any available source and place uniformly on the designated areas and spread evenly to a minimum thickness of four (4) inches. Irregularities in the surface shall be corrected so as to prevent formation of depressions where water will stand. TOPSOIL SHALL NOT BE PLACED WHEN THE SUBGRADE IS FROZEN, EXCESSIVELY WET, OR IN A CONDITION DETRIMENTAL TO THE PROPOSED PLANTING AND PROPER GRADING.

C. Broadcast sowing of seed may be accomplished by hand seeders or by approved power equipment. Either method shall result in uniform distribution and no work shall be performed during high winds. The area seeded shall be lightly firmed with a cultipacker immediately after broadcast.

D. The contractor shall water and maintain seeded areas from time of completion until final acceptance of the project.

E. The contractor shall be responsible for establishing ground cover on all disturbed areas. Repeated seeding shall be required if necessary throughout the warranty period.

3.06 HYDROSEEDING

A. Hydroseeding shall be carried out by means of a proper hydroseeder where approved slurry of seeds, mulch, fertilizers, binders, and organic matter are sprayed on the prepared soil surface.

3.08 APPLICATION OF MULCH

A. Mulching shall be spread in a uniform continuous blanket, at a rate of 1 to 3 tons per acre (air dried weight) or to a uniform 2-inch depth. Mulch shall be spread by hand or by an approved blower type mulch spreader whereby the application of mulch cover and tackifier may be combined into one operation. If this method is used, no change in application rates will be allowed. Care shall be taken to remove all wire and/or twine from baled hay/straw when the control structures are removed from the site. Mulch shall be anchored in the soil to a depth of two to three inches to form a soil-binding mulch to prevent loss or bunching or held in place with a tackifier.

3.09 PROTECTION

A. Seeded areas shall be protected against traffic. If necessary, barricades or warning signs shall be erected. Such protective devices shall be maintained until acceptance of the work as specified herein.

3.10 SOD PLACEMENT

A. Solid sod or topsoil shall not be placed until all other items of work are complete.

B. Prior to placing the sod and topsoil in the areas designated, the ground surface shall be cleared of materials greater than ¾" that might hinder proper grading, tillage, or subsequent maintenance operations such as stumps, stones, roots, cable, wire, grade stakes, etc., and brought to four (4) inches below the finished grade. The areas shall then be thoroughly tilled to a depth of at least three (3) inches by plowing, disking, harrowing or other acceptable means. Lightly rake to provide a smooth, uniform, and fine surface texture. Remove ridges and fill depressions as required to drain.

C. The Contractor shall then obtain an approved topsoil from any available source and place uniformly on the designated areas and spread evenly to a minimum thickness of four (4) inches. Irregularities in the surface shall be corrected so as to prevent formation of depressions where water will stand. TOPSOIL SHALL NOT BE PLACED WHEN THE SUBGRADE IS FROZEN, EXCESSIVELY WET, OR IN A CONDITION DETRIMENTAL TO THE PROPOSED PLANTING AND PROPER GRADING.

D. After the topsoil has been spread and graded, the surface shall be cleared of stones, stumps or other objects that might hinder planting or maintenance preparations. Paved areas over which hauling operations are conducted shall be kept clean.

F. Where any portion of the surface becomes gullied or otherwise damaged, the affected areas shall be repaired to the aforementioned condition.

G. Areas to be sodded shall be shaped in such manner that they will, after placement of sod, conform to the typical sections. Lay sod uniformly, evenly, and parallel to the finished contour.

H. Minimize traffic on newly laid sod during installation. Provide plank or wood sheets as may be required to protect sod already laid during sodding operations.

I. Lay each strip with tightly-fitted joints against each other without voids. Do not overlap edges. Joints at the ends of sod strips shall be staggered with adjacent strips of sod.

J. Finish sod edges at walks, curbs, planting, mulch edges, and other vertical surface by cutting neatly and fitting tightly to edge and line.

K. Sod mat to be approximately one inch below finished height of walks, pavement, curbs, etc. Any sod not conforming to this requirement will be removed, the subgrade adjusted, and the sod re-laid.

L. Where new sod joins existing lawns, cut straight and neatly into existing lawn and level subgrade to allow height to match.

M. Soil shall be firmed or healed in along the edges of the sodded areas.

N. Stake sod on the slope ratio of 1 on 2 or steeper with minimum of two 8-inch stakes per square yard.

O. After all sod is laid and thoroughly watered, roll all sodded areas (except staked sod), either with a small mechanical or hand roller, sufficiently to set or press sod into soil.

3.11 WATERING

A. Prior to placement of seed or sod, areas shall be sprinkled with water sufficiently to make them moist, but not muddy. The initial application of water may be omitted if the area is sufficient moist from rainfall.

B. Immediately following the placing and tamping of sod, the covered area shall be wetted thoroughly. Subsequent applications of water shall be as required.

C. Immediately following the application of the mulch cover for seeding, water shall be applied in sufficient quantity to thoroughly moisten the soil to the depth of pulverization and then as necessary to germinate the seed and maintain growth.

3.12 MAINTENANCE

A. For the first two (2) weeks following the placing of the sod, all sodded areas shall be thoroughly watered daily, through the use on an on-site watering system.

B. Continue maintenance until sodded areas are established with sod knitted in place. Maintenance period shall be a minimum of 28 days, or to final acceptance.

C. Maintenance of sodded areas shall include watering, weeding, mowing to a two to two and one half (2-2 1/2") inches height after growth has exceeded three (3") inches and prior to a four (4") growth, and replacement and installation of sod as originally specified for sodded areas failing to survive. Clippings from mowing which mat on the grass are to be removed.

D. Water all sodded areas during maintenance period as necessary to maintain sod and soil moisture, supplement rainfall, to promote growth, proper rooting to ensure sod survival and to prevent dormancy.

E. All seeded areas shall be maintained and watered by the Contractor until acceptance of the work as specified herein. Maintenance shall include reseeding, watering, repair, or erosion damage, and maintenance of mulch.

F. Replanting: Prior to acceptance, seeded areas that show signs of substantial desiccation, as evidence by loss of color and distinct yellowing or by lack of germination, shall be considered failed and shall be reseeded and continue to be reseeded until acceptable cover is obtained. Replanting operations shall be as originally specified.

G. Maintenance of Grades: Original grades of seeded areas shall be maintained after commencement of planting operations and until acceptance. Any damage to the finished surface from construction operations shall be repaired within a reasonable time. In the event erosion occurs from rainfall, such damage shall be repaired within a reasonable time. Ruts, ridges, tracks, and other surface irregularities shall be corrected, and re-seeded where required.

H. Maintenance of Mulch: Mulch shall be maintained until covered with growing grass seedlings. Material that has been removed from the site by wind or other causes shall be replaced and secured.

3.13 COMPLETENESS

A. The lawn and grass operations shall not be considered complete until a uniform (e.g., evenly distributed, without large bare areas) perennial vegetative cover with a density of 80% of the native background vegetative cover for the area has been established on all unpaved areas and areas not covered by permanent structures.

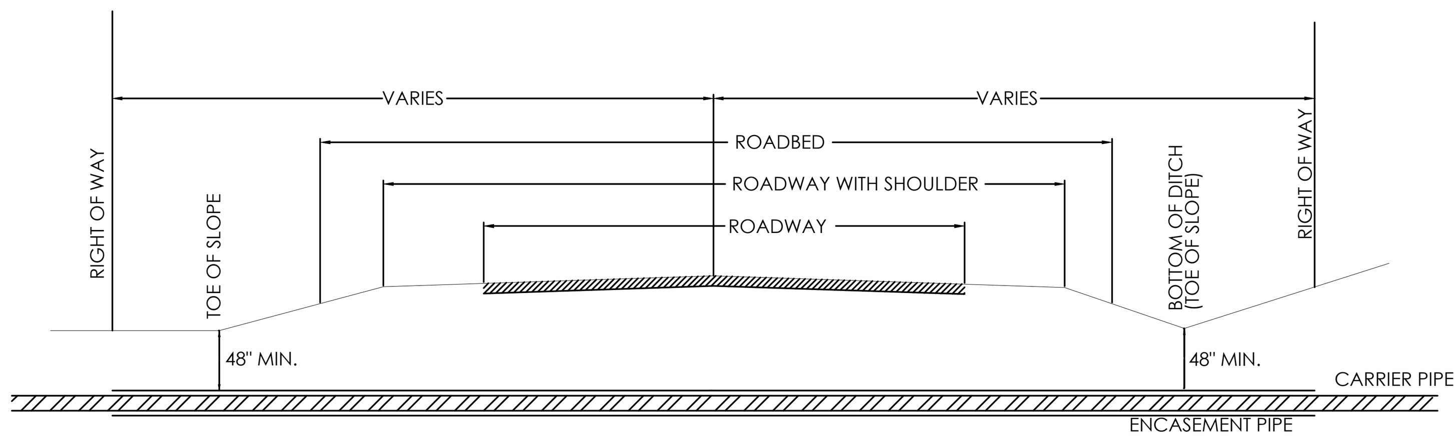
B. If a satisfactory stand of grass has not been produced, the Contractor shall renovate and reseed the grass or unsatisfactory portions thereof.

C. A satisfactory stand is defined as grass or section of grass that has:

1. No bare spots larger than 1 square foot.
2. Not more than 10 percent of total area with bare spots larger than 1 square foot.
3. Not more than 15 percent of total area with bare spots larger than 6 inches square.

3.14 INTERMITTENT CLEANUP

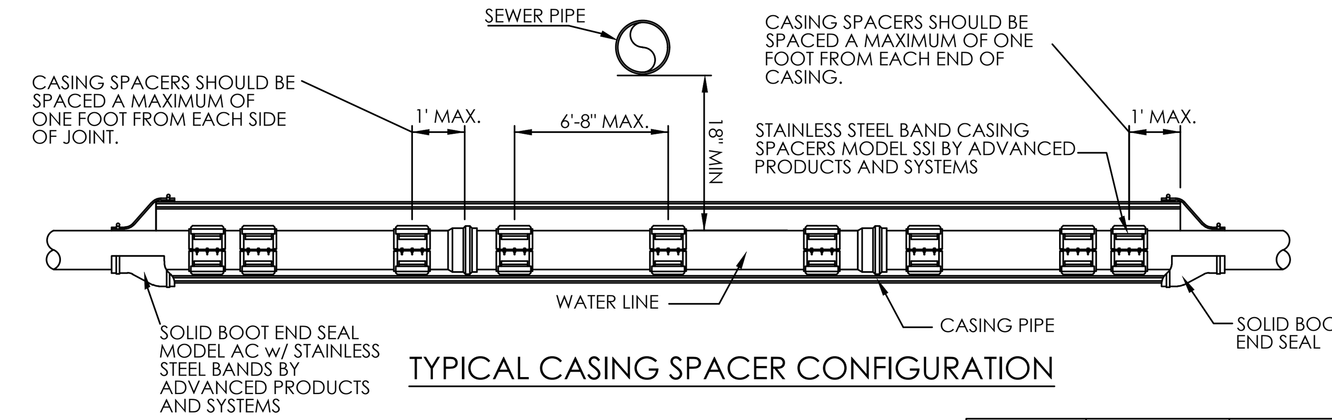
A. Immediately following the lawn and grass restoration operations, all gutters, sidewalks, driveways, street pavement, yard or other areas shall be cleaned of all debris, excess sod, topsoil, or other objectionable matter. All such cleanup operations shall be completed before sodded areas are measured for payment.



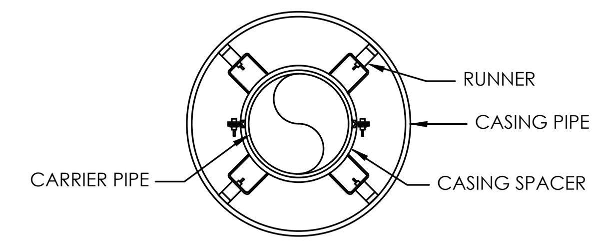
NOTES:
1. CASING SHALL EXTENDED A MINIMUM OF SIX (6) FEET BEYOND THE FLOW LINE OF PARALLEL DITCHES, TOE OF SLOPE, OR FACE OF CURB AS APPLICABLE FOR THE HIGHWAY SECTION.
2. ENCASEMENT MAY BE OF METALLIC OR NONMETALLIC MATERIAL.
3. THE STRENGTH OF THE ENCASEMENT MATERIAL SHALL EQUAL OR EXCEED STRUCTURAL REQUIREMENTS FOR HIGHWAY DRAINAGE CULVERTS.
4. A MARKER BEARING THE UTILITY OWNERS NAME SHALL BE PLACED AT EACH RIGHT OF WAY LINE WHERE CROSSED. (NOT REQUIRED FOR SERVICE LINES)
5. OWNERSHIP OF THE LINES SHALL BE SHOWN ON VENTS.
6. CASING PIPE SHALL BE SEALED AT THE ENDS WITH A FLEXIBLE MATERIAL.
7. CASING SHALL EXTEND RIGHT OF WAY TO RIGHT OF WAY.

ROAD CROSSING

NTS



TYPICAL CASING SPACER CONFIGURATION



TYPICAL END VIEW

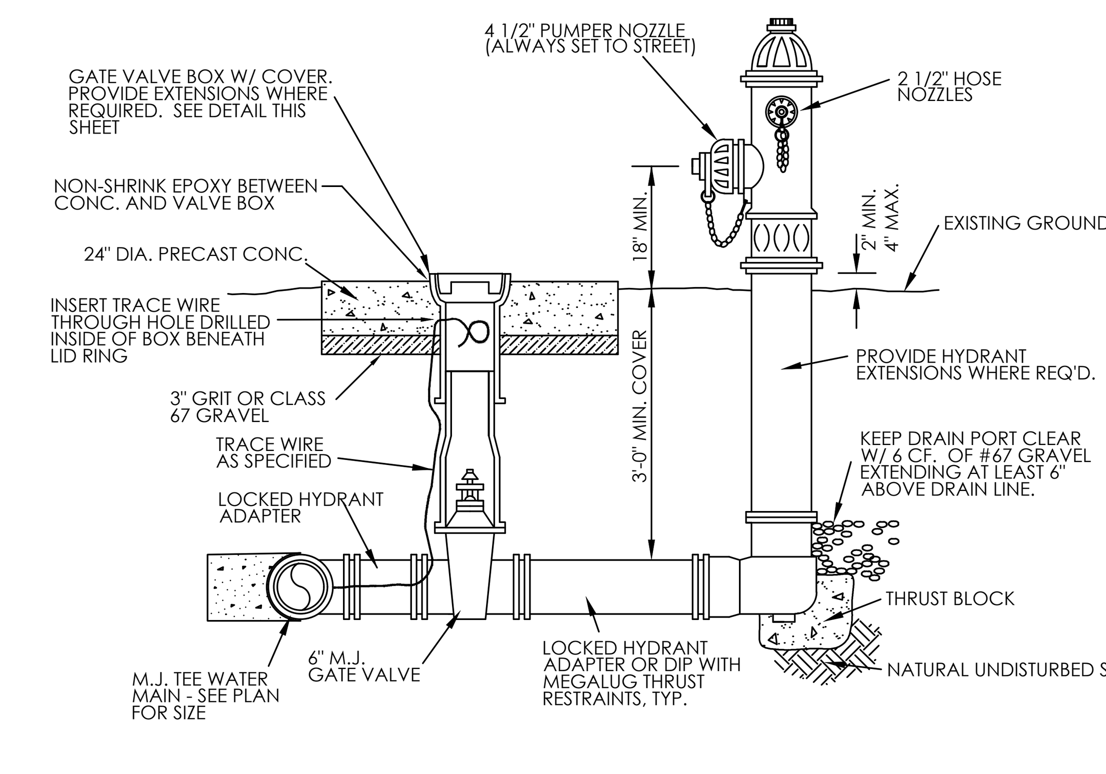
NOTES:
1. SPACERS SHALL BE PREFABRICATED STEEL SPACERS, SMITH-BLAIR MODEL 883.
2. QUANTITY OF RUNNERS SHALL BE IN ACCORDANCE WITH CARRIER PIPE SIZE AS FOLLOWS:
   TO 14" DIAMETER - 4 RUNNERS
   16" TO 36" DIAMETER - 6 RUNNERS
   38" TO 48" DIAMETER - 8 RUNNERS
2. DIMENSIONS BETWEEN SPACERS SHALL BE AS FOLLOW:
   STAINLESS STEEL - 8 TO 12 FEET.
   CARBON STEEL - 8 TO 12 FEET.
3. WIDTH OF SPACERS SHALL BE IN ACCORDANCE WITH CARRIER PIPE SIZE AS FOLLOWS:
   TO 14" DIAMETER - 8" WIDTH
   16" AND LARGER DIAMETER - 12" WIDTH
4. A MINIMUM OF THREE SPACERS PER FULL JOINT OF PIPE. MAXIMUM SPACING BETWEEN SPACERS SHALL BE 6'-8".
5. ALL D.I. CARRIER PIPE MUST HAVE FIELD LOCK GASKETS INSTALLED WITHIN THE CASING PIPE. ALL PVC CARRIER PIPE MUST HAVE BELL RESTRAINTS ON EVERY PIPE BELL WITHIN THE CASING PIPE.
6. WATER LINES SHALL BE "CENTERED" IN CASING.
7. MAINTAIN 18" MINIMUM BETWEEN WATER AND SEWER MAINS. SEE "WATER CROSSING AT SEWER LINE" DETAIL.
8. ALL ENCASEMENT PIPE SHALL BE NEW STEEL & HAVE A MINIMUM WALL THICKNESS AS SHOWN IN THE ENCASEMENT/CARRIER PIPE SIZING CHART.

NOMINAL WATER LINE SIZE	NOMINAL ENCASEMENT SIZE (MIN.)	MIN. WALL THICKNESS (IN.)
3"	6"	0.250" (1/4")
4"	8"	0.250" (1/4")
6"	12"	0.250" (1/4")
8"	16"	0.250" (1/4")
10"	20"	0.250" (1/4")
12" to 14"	24"	0.3125" (5/16")
16" to 18"	30"	0.375" (3/8")
20" to 24"	36"	0.500" (1/2")

ENCASEMENT / CARRIER PIPE SIZING CHART

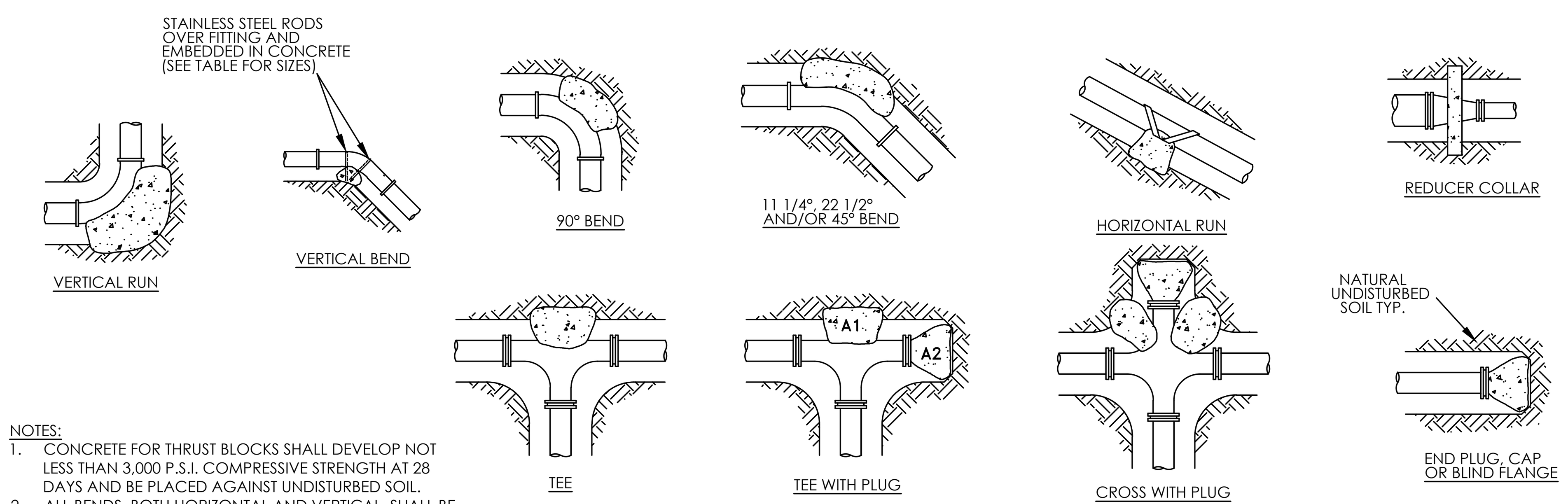
WATER MAIN ENCASEMENT

NTS



FIRE HYDRANT ASSEMBLY

NTS



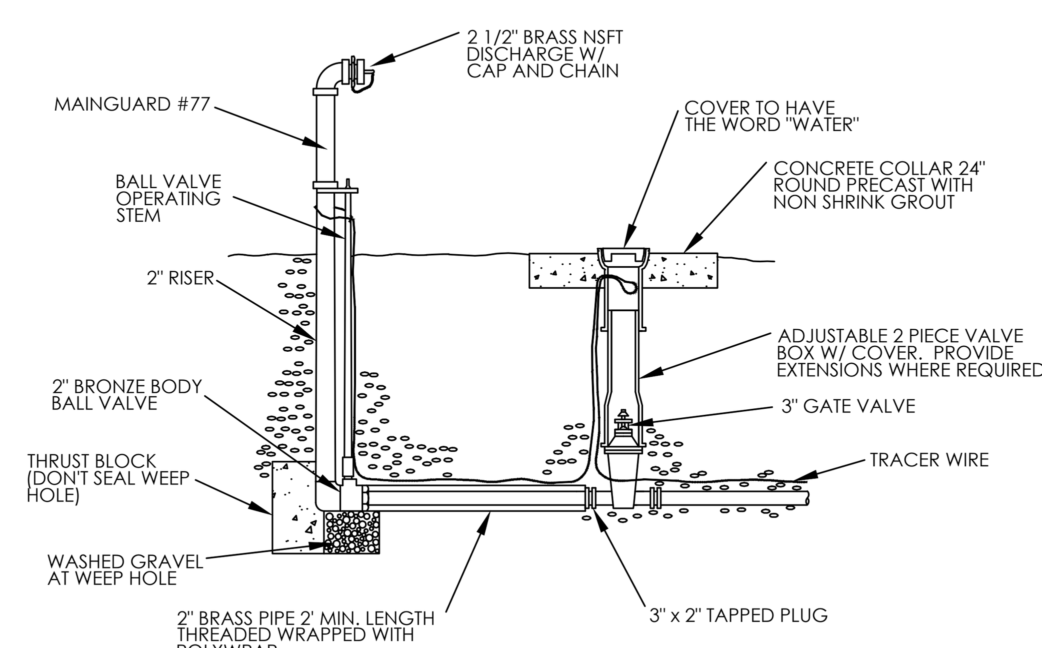
NOTES:
1. CONCRETE FOR THRUST BLOCKS SHALL DEVELOP NOT LESS THAN 3,000 P.S.I. COMPRESSIVE STRENGTH AT 28 DAYS AND BE PLACED AGAINST UNDISTURBED SOIL.
2. ALL BENDS, BOTH HORIZONTAL AND VERTICAL, SHALL BE BACKED WITH CONCRETE. VERTICAL BENDS SHALL BE PLACED ON CONCRETE PADS WHERE BENDS TURN UP, OR LOADED WHERE BENDS TURN DOWN.
3. WRAP PIPE JOINTS IN 8 MIL POLYETHYLENE BEFORE PLACING CONCRETE. USE LONG-RADIUS FITTINGS WHEREVER POSSIBLE.
4. BEARING AREA SHOWN IN TABLE, IS BASED UPON A 2,000 LB/SF. SOIL BEARING, AND UPON A PIPE LINE PRESSURE OF 250 psi PLUS WATER HAMMER. AREAS SHOWN SHALL BE ADJUSTED, SHOULD FIELD CONDITIONS VARY.
5. UTILIZE MEGALUG THRUST RESTRAINTS ON MECHANICAL JOINT FITTINGS AND VALVES, IN ADDITION TO THESE THRUST BLOCKS.

THRUST BLOCK SCHEDULE
(2000 PSF SOIL, 250 PSI WATER PRESSURE)

BEARING AREA OF THRUST BLOCKS IN SQ. FT. (HORIZONTAL BENDS)									VOLUME OF THRUST BLOCK IN CU. YDS. (VERTICAL BENDS)							
FITTING SIZE	TEE, WYE PLUG OR CAP	90° BEND, PLUGGED CROSS	TEE PLUGGED ON RUN (A1)	TEE PLUGGED ON RUN (A2)	BEND ANGLE 45°	22 1/2°	11 1/4°	5 5/8"	FITTING SIZE	BEND ANGLE 45°	22 1/2°	11 1/4°	5 5/8"	ROD SIZE	EMBED-MENT	CUBIC YARDS
2, 3, & 4	1.3	1.8	1.3	1.8	1.0	1.0	-	-	2, 3, & 4	1.5	0.5	0.3	-	#6	30"	-
6	2.8	4.0	2.8	4.0	2.2	1.1	1.0	-	6	3.6	1.3	0.5	-	#6	30"	-
8	5.0	7.1	5.0	7.1	3.8	2.0	1.0	-	8	5.3	2.0	0.8	-	#6	30"	0.6
10	7.9	11.1	7.9	11.1	6.0	3.1	1.6	-	10	8.0	3.1	1.2	-	#6	30"	-
12	11.3	16.0	11.3	16.0	8.7	4.4	2.3	-	12	11.3	4.3	1.7	-	#6	30"	1.3

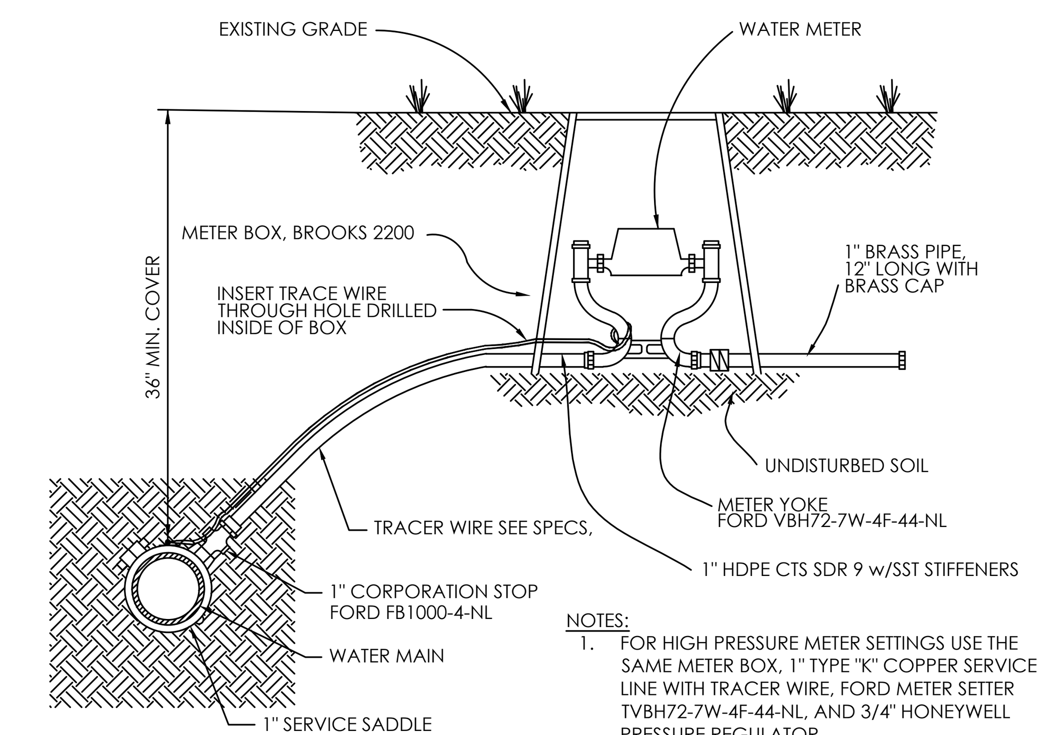
THRUST BLOCKING

NTS



BLOWOFF ASSEMBLY

NTS

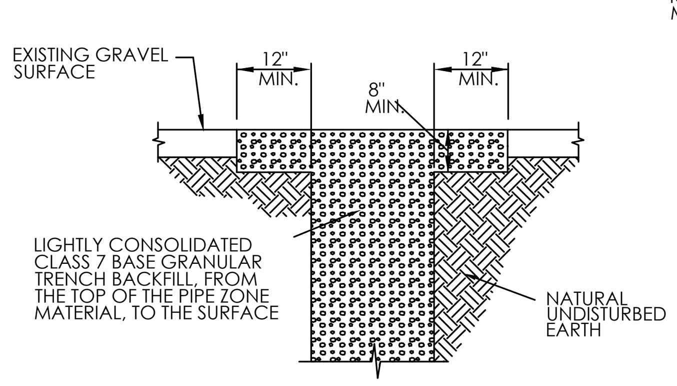


NOTES:
1. FOR HIGH PRESSURE METER SETTINGS USE THE SAME METER BOX, 1" TYPE "K" COPPER SERVICE LINE WITH TRACER WIRE, FORD METER SETTER TVBH72-7W-4F-44-NL, AND 3/4" HONEYWELL PRESSURE REGULATOR.
2. TRACER WIRE ON METER SERVICE LINES MUST BE A CONTINUOUS LOOP FROM THE WATERMAIN TO THE METER SETTER AND BACK. TRACER SHALL BE SPLICED WITH GREASE FILLED DIRECT BURY WIRE NUTS.

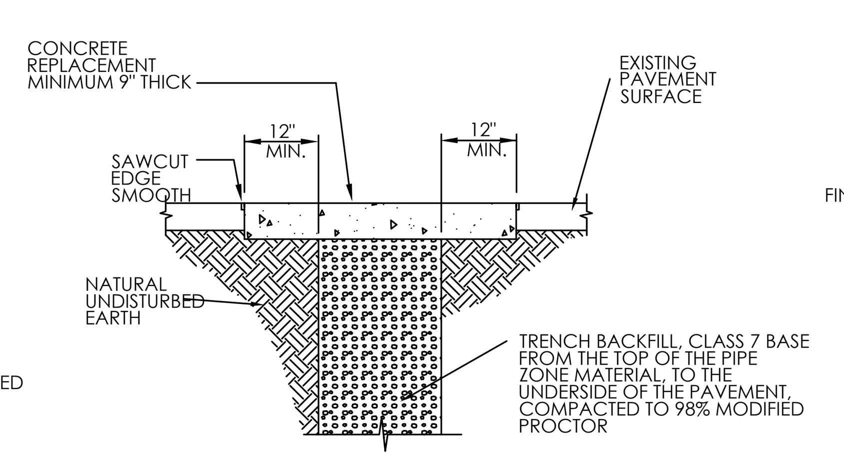
***ALL COMPONENT/FITTINGS SHALL MEET THE FEDERAL DEFINITION OF "LEAD-FREE"**

SINGLE METER SETTING

NTS



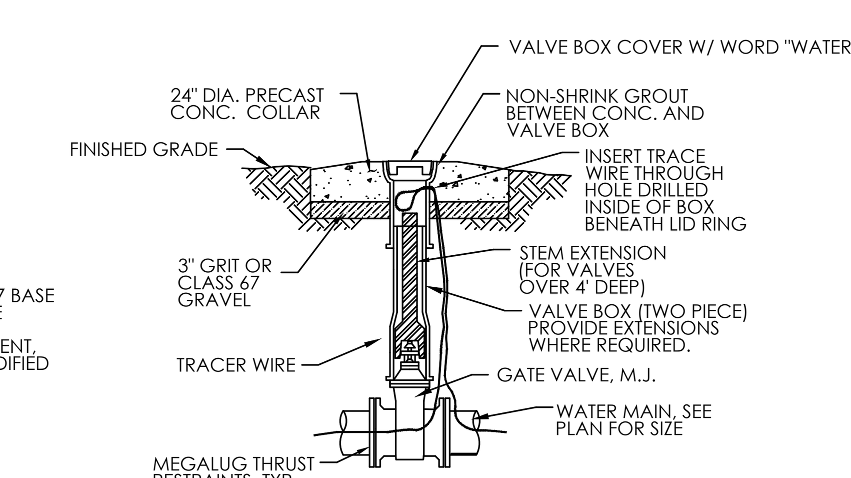
GRAVEL DRIVEWAY REPAIR



CONCRETE OR ASPHALT UPAVEMENT REPAIR

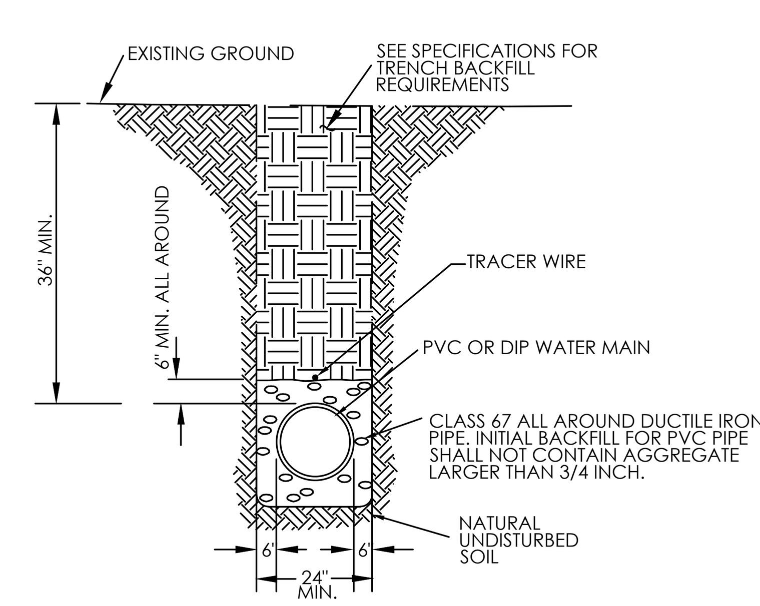
REPAIR DETAILS

NTS



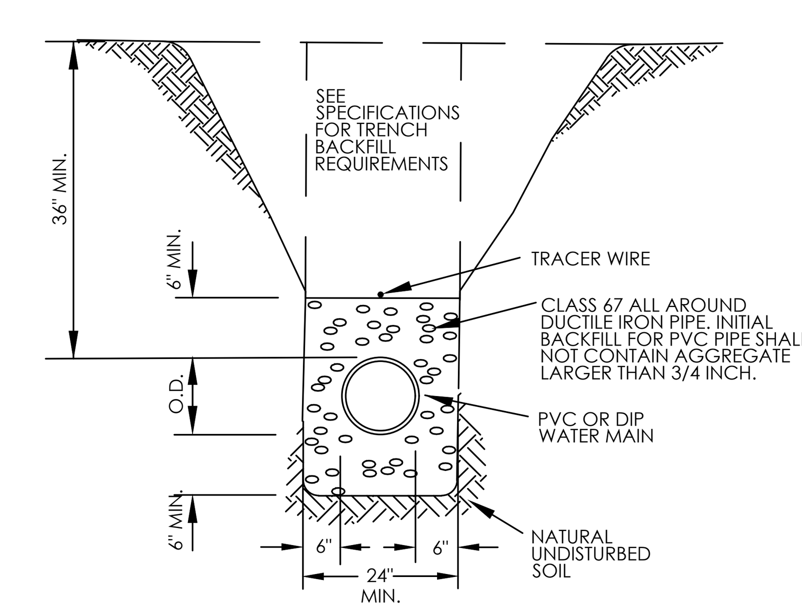
GATE VALVE

NTS



TYPICAL SOIL TRENCH

NTS

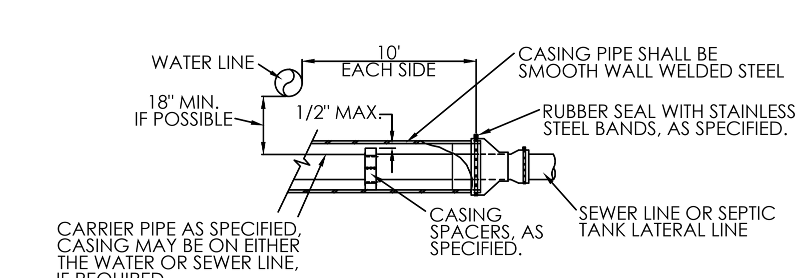


ROCK EXCAVATION TRENCH

NTS

NOTES:
1. INSTALL STEEL CASING ON THE WATER PIPE OR SEWER PIPE WHERE 18-INCHES OF VERTICAL SEPARATION CANNOT BE MAINTAINED OR WHENEVER SEWER PIPE IS ABOVE WATER LINE.
2. SEE "WATER MAIN ENCASEMENT" DETAIL.
3. SEWER LINES SHALL BE HELD TO SPECIFIED SLOPE.

* SEE "WATER MAIN ENCASEMENT" DETAIL FOR REQUIRED CASING SIZES AND THICKNESSES.



WATER CROSSING AT SEWER LINE

NTS

Washington Water Authority
12567 West Highway 62
Farmington, AR 72730
479-267-2111

WWA STANDARD DETAILS
2023
WASHINGTON COUNTY, ARKANSAS

Drawn By: RZJ
Checked By: LFG
Date: 08/07/2023
Scale: NTS
Revisions: 08/07/2023
Sheet
1 of 1

SECTION 22 1123.13

HYFAB eMVP-D10F

DOMESTIC-WATER PACKAGED BOOSTER PUMPS

PART 1 – GENERAL

1.1 SUMMARY

A. This section includes the following types of systems:

1. Variable speed booster pump packages.

1.2 REFERENCED STANDARDS

A. National Electrical Code Compliance: Components shall comply with NFPA 70, "National Electrical Code."

B. UL Compliance: Package shall be listed and labeled by UL, and comply with UL QCZJ. "Packaged Pumping Systems."

C. UL Compliance: Controller shall be listed and labeled by UL, and comply with UL 508A, “Industrial Control Panels.”

D. NEMA Compliance: Electric motors and components shall be listed and labeled NEMA.

E. ANSI/HI 1.1-1.4, 1.6 – 2000, Nomenclature, Definitions, Applications, Operation and Testing.

F. ANSI/HI 9.6.1-1998, NPSH Margin

G. ANSI/HI 9.6.3-1997 Operating Region

H. ASTM 48, Standard Specification for Gray Iron Castings

1.3 SUBMITTALS

A. General: In addition to procedures specified in Division 1, comply with the following.

1. Tag submitted data with the corresponding designation of that item as scheduled in the construction documents. All information submitted shall be individually "tagged" to easily identify it with the corresponding material or equipment. Failure to comply with this requirement will result

in that item or, at the discretion of the Engineer, all items submitted, being rejected without being reviewed.

B. Product data: Furnish catalog information, manufacturers package curve, rated capacities, final impeller dimensions, and accessories provided for the product indicated.

1. A general arrangement drawing showing overall dimensions and all piping layouts.
2. Complete submittal data for all major equipment (pumps, motors, valves, electrical controls, etc.).
3. An electrical schematic, that provides power and control wiring.
4. Sequence of operation.
5. Indicate operating point of each pump on curves.
6. Furnish composite curve for booster package.

C. Operation and Maintenance Data: Provide installation, operation and maintenance (IOM) manuals on all equipment, including assembly drawings as required for a complete explanation and description of all equipment.

1. Include installation instructions, maintenance instructions, assembly views, and replacement parts lists.

1.4 QUALITY ASSURANCE

A. Single Source Responsibility: Obtain pumps from a single manufacturer.

B. Each booster package shall be hydrostatically and flow tested prior to shipment to verify system integrity.

C. Pumps shall operate:

1. At specified system fluid temperatures without vapor binding and cavitation.
2. With variable speed controller to maintain specified discharge pressure and prevent motor overloads.

D. The pump NPSH shall conform to the ANSI/HI 9.6.1-1998 standards for Centrifugal and Vertical Pumps for NPSH Margin.

E. Pump pressure ratings shall be at least equal to the system's maximum operating pressure at point where installed, but not less than specified.

F. ASME Compliance: Comply with ASME B31.9 for piping.

G. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

H. Provide certified equipment start-up and, when noted, an on site training session. Pump start-up shall be for the purpose of determining pump rotation, lubrication, voltage, and amperage readings. Start-up shall also include verification of proper electrical connections, pump's balance, and recording of discharge and suction gauge readings. A copy of the start-up report shall be submitted to the Engineer.

I. UL Compliance for Packaged Pumping Systems:

1. UL 508, "Industrial Control Equipment."
2. UL 508A, "Industrial Control Panels."
3. UL 778, "Motor-Operated Water Pumps."
4. UL 1995, "Heating and Cooling Equipment."

1.5 DELIVERY, STORAGE, AND HANDLING

A. Manufacturer's Shipping Preparation: Provide flange covers for protection during shipping.

B. Store booster in a dry indoor location.

C. Retain shipping flange protective covers and protective coatings during storage.

D. Protect bearings and couplings against damage from moisture, sand, grit, and other foreign matter.

PART 2 – PRODUCTS

2.1 VARIABLE SPEED BOOSTER PUMPING PACKAGE

A. Subject to compliance with requirements below, provide pumps from one of the following manufacturers:

1. Basis of design: HYFAB
2. Bell & Gossett
3. Other pre-approved manufacturer.

B. General Description: Factory-assembled and tested variable speed domestic water pressure booster system with pumps, piping, isolation and check valves, controls, specialties and accessories mounted on a structural steel base. Pump and electrical protection shall be integrated into the pump controller package. The package shall be arranged in a compact design to allow it to be moved through a standard 36” doorway.

C. Pumps:

1. Pumps shall be multistage centrifugal pump unit/s, Model e-SV as manufactured by Goulds Pumps or equal. All pump units shall be from one manufacturer and provided complete including electric motor drive.

   a. Casing Construction: The pump casing shall be of deep drawn, laser welded AISI 304L stainless steel on pump and/or cast iron pedestal (triplex) with discharge up to two inch. Pump with larger discharge shall be furnished with cast iron pedestals unless specified to have all stainless construction and shall be capable of withstanding maximum working pressures of 360 psi or 580 psi, based on pump staging and flange selection. Piping connections shall be in-line and shall be compatible with ANSI raised face flanges optional NPT.

   b. Wear Ring: Wear rings shall be provided within each stage. Wear rings must be self centering and easily replaceable.

   c. Impeller: Impellers shall be of enclosed design and constructed of AISI 316L stainless steel. Impellers shall provide internal thrust balance in each stage.

   d. Diffuser Bowl: Each stage shall have a bowl with attached diffuser and be constructed of AISI 304L stainless steel.

   e. Seal Housing: The seal housing shall be of concave design and shall hold the seal faces below the topmost part of the pump casing.

   f. Mechanical Seal: Mechanical seals consisting of high temperature carbon rotating ring, stainless steel spring, silicon carbide graphite filled seat, and viton elastomer.

   g. Shaft Sleeve and Bearing: The pump shall have shaft sleeves made of Tungsten Carbide and ceramic bearings. Shaft height shall be set with a standard spacer.

   h. Electric Motor: The pump drive motor shall be NEMA standard design TC frame suitable for vertical mounting and close coupled to the pump unit. Motors shall be of standard manufacturers catalog design and must not use special bearings as a thrust handling device. The motor rating shall be as specified on the pressure booster schedule and shall have a 1.15 service factor. Motor enclosure and efficiency shall be as specified on the booster pump schedule.

i. Testing: Each pump shall be hydrostatically tested by the manufacturer in accordance with Hydraulic Institute Standards at a minimum of 350 PSI. Production performance testing shall also be conducted by the manufacturer on each pump unit. Flow and head shall be measured at three operating to verify performance.

j. NPSH: Unit shall be provided with low NPSH option (NPSHR = 3.5 FT or less) for open system applications.

D. Check valves: Angle style, pilot operated check valves shall be installed on the discharge of each pump. The valve shall have a fusion bonded epoxy coated ductile iron body with a bronze disc. The check valves shall be able to be serviced or rebuilt without removing them from the pressure booster. Direct acting check valves are not acceptable.

E. Isolation Valves: Two-piece, bronze body, full port ball valves shall be provided to isolate each pump and check valve.

F. Thermal protection: Self contained thermal-relief valves factory set to discharge at 125°F shall be mounted on each pump.

G. Piping: All piping shall be constructed from Type 304 stainless steel. Suction and Discharge headers shall be designed and constructed for minimal friction loss.

H. Pressure gauges: Pressure gauges shall be adjustable LED displays. Displays shall be supplied for both the suction and discharge headers.

I. Package Base: The pressure booster base shall be designed and fabricated to provide proper structural support for all attached equipment, and provide for anchorage to the structure. The base shall include a rigid structural member for pump and motor support. Main members shall be constructed from heavy weight A-36 structural steel members with reinforcing channels for larger boosters. Steel base shall be painted with machinery enamel.

J. Control Panel: Manufacturer shall provide a UL-508A listed complete electrical system including main disconnect, variable speed pump controller, pressure transducers, instrumentation and controls to automatically start, stop and modulate pump speed to smoothly, efficiently and reliably provide pump variable flow rates at a constant discharge pressure. The booster package shall include full pump, motor and drive safety features required to protect the equipment and piping system.

1. Main Control Panel: A door-operated non fused main disconnect shall be provided which shall remove power from the entire pressure booster when switched off. Each Variable Speed Drive (VFD)/Motor shall be protected by a molded case circuit breaker or MSP enclosed in the main control panel.

2. Motor Controller: NEMA ICS 7.1 variable-frequency, variable torque, solid-state pulse-width modulating type. VFD protective features shall include:

    a. Ground fault

    b. Short Circuit

    c. Motor overload

3. Suction and discharge pressure transducers.

    a. Pressure transducers shall be utilized for providing all pressure signals for the pump control logic. Pressure switches are not acceptable.

    b. Pressure transducer shall be a solid-state bonded strain gage type with an accuracy of plus/minus 1 %. Transducer shall be constructed of non-ferrous metal suitable for used with domestic water. Pressure transducers constructed of plastic are not acceptable.

    c. Transducer shall be rated for package discharge and suction pressures as shown on submittal.

    d. Pressure transducer shall utilize an M12 type connector to prevent moisture intrusion.

4. Temperature Sensor

    a. A NTC thermistor or RTD shall be used for sensing the booster pumps' discharge temperatures for temperature limit control logic. Thermostats are not acceptable.

    b. The temperature sensor shall be of the immersion type and shall be immersed in water from the pumps' discharge. It shall be sealed off by a compression type fitting to prevent accidental dislodgement.

5. Controller: Touchscreen variable speed pump logic controller in a NEMA 1 enclosure with door operated disconnect to include, power distribution and overload protection for each AFD and the following control features:

    a. Unit shall utilize user-friendly front panel programming that displays parameters in clear text.

    b. Display shall show all system variables in plain English.

    c. Program settings shall be changeable and stored in non-volatile memory. Program settings shall be retained in memory in the event of loss of power to the controller, without the use of a backup battery.

    d. System operating pressure shall be clearly displayed in PSI or feet of head for ease of use and to provide an operator friendly interface.

    e. Additional parameters, where applicable, shall be displayed in units consistent with pumping systems.

f. The settings and program in whole or part may be locked out with the use of an operator selectable password. Standard system settings shall include at a minimum the following functions:

1) Low suction pressure shutdown with auto restart
2) High discharge pressure shutdown with lockout
3) High discharge temperature shutdown
4) Pump failure alarm
5) Constant pressure setting with variable flow capability
6) Multiple pump operation with alternation
7) Pump starting point with allowable, adjustable pressure drop
8) Minimum speed
9) No flow standby

6. Hydro-pneumatic Tank:

a. Furnish hydro-pneumatic tank separately for field installation.

b. Furnish tank of size pressure rating and construction specified on pressure booster schedule.

c. Furnish hydro-pneumatic tank connection on discharge header with union, isolation valve and drain valve on discharge header of packaged pressure booster.

7. Sequence of Operation:

a. The pump controller shall receive a 4-20mA signal from each pressure transducer.

b. The pressure transducer shall monitor system pressure and provide an analog signal to the pump control software, and allow the VFD motor controller to provide a variable Volts/Hz output to the motor.

c. Whenever the pressure drops below the set system pressure, the lead pump shall start and run at speed required to maintain system pressure setpoint (as set by the operator.) If the pressure setpoint cannot be maintained by one pump, the lag pump shall start to provide the extra flow and pressure automatically.

d. When demand decreases to a level which can be met by one pump and an adjustable minimum run-timer has elapsed, the lag pump will be stopped.

e. The lead pump shall alternate based on elapsed run time.

f. When the system experiences low demand the controller shall test for a no flow condition without the use of external switches or controls. The controller will stop the lead pump after verifying a zero demand condition exists and a minimum run-timer has elapsed. The hydro-pneumatic tank shall supply water to the system until the pressure falls below the restart pressure, at which point the lead pump shall restart

g. All program settings shall be based on centrifugal pump language.

h. Program settings shall be password protected. With proper password entry, settings shall be field adjustable to allow changes by the user.

i. DemandSet Control (Optional, requires optional flowmeter): To meet ASHRAE 90.1-2010 standard and building code requirement the variable speed pressure booster shall employ logic that recalculates the setpoint based on the actual system friction loss between booster discharge and the remote critical fixture. The controller will monitor system demand and continuously change the discharge pressure setpoint based on flow. The controlling set point and flow shall be displayed on the digital screen.

PART 3 – EXECUTION

3.1 EXAMINATION

A. Examine areas, equipment foundations, and conditions, for compliance with requirements for installation tolerances and other conditions affecting performance of pumps.

B. Examine rough-in for piping systems to verify actual locations of piping connections prior to installation.

C. Examine equipment foundations and/or inertia bases for suitable conditions where pumps are to be installed.

D. Correct unsatisfactory conditions prior to installation of pumps.

3.2 INSTALLATION

A. General: Comply with the ANSI/HI 1.4-2000 and manufacturer's written installation and alignment instructions.

B. Refer to the drawings for details of booster installation.

C. Install booster package in specified location. Provide access for periodic maintenance, including removal of motors, impellers, and accessories.

D. Verify proper pump rotation at start-up.

E. Perform start-up procedures per manufacturer’s instructions.

3.3 ADJUSTING

A. Adjust pressure and temperature setpoints.

END OF SECTION 22 1123.13

Series 34
Air Release Valve

- Ductile Iron Body
- Stainless Steel Trim and Float
- Easily serviced without removal from pipeline
- Working pressures to 800 psi
- Engineered for drip tight seal at low pressures

Cla-Val Series 34 Air Release Valves are designed to vent entrained air that collects at high points in a pipeline. This valve continuously eliminates air from a system by releasing small quantities of air before large air pockets can occur. In many installations, continuing accumulations of air in the pipeline (lacking air release valves); cause flow capacity to slowly decrease; power consumption slowly increases; unnoticeable at first, until flow drops dramatically, even stopping due to air blocks in the piping. Another problem resulting from excessive air accumulation is unexplained pipeline rupture. These ruptures are passed off as the result of ground settling or defective pipe, Where as in reality its large air pockets that greatly increase pressure surges (normally occurring) when flow stops and starts causing the rupture. During normal pipeline operation, air accumulation at the high point will displace the liquid within the air valve and lower the water level in relation to the float. As level of the liquid lowers, where the float is no longer buoyant, the float drops and opens the valve orifice seat and permitting accumulated air to be exhausted to atmosphere. After air is released, the liquid level in the air valve rises and closes the valve orifice seat. This cycle automatically repeats as air accumulates inside the air release valve, thereby preventing the formation of air pockets.

Installation

Series 34 Air Release Valves are typically installed at high-points in pipelines and at regular intervals, of approximate 1/2 mile, along uniform grade line pipe.

Mount the unit in the vertical position on top of the pipeline with an isolation valve installed below each valve in the event servicing is required. A vault with adequate air venting and drainage is recommended.

Note:

Vacuum check valves can be supplied on the discharge of all size air release valves to prevent air re-entering the system; during negative pressure conditions

Purchase Specifications

The air release valve shall be of the float operated, simple lever or compound lever design, and capable of automatically releasing accumulated air from a fluid system while the system is pressurized and operating.

An adjustable designed orifice button shall be used to seal the valve discharge port with drip-tight shut-off. The orifice diameter must be sized for use within a given operating pressure range to insure maximum air venting capacity.

The float shall be of all stainless steel construction and guaranteed to withstand the designed system surge pressure without failure. The body and the cover shall be ductile iron and valve internal parts shall be stainless steel and Viton™ or Buna-N® (standard)for water tight shut-off.

The air release valve shall be manufactured per ANSI/AWWA C512-04 Series 34 from Cla-Val in Newport Beach, CA, USA.

Product Specifications

Sizes
1/2", 3/4", 1", 2", 3" NPT

Pressure Ratings (see note)
150 psi
175 psi
300 psi
600 psi
800 psi

Temperature Range
Water to 180°F

Materials
Body and Cover:
Ductile Iron ASTM 536 65-45-12

Float:
Stainless Steel

Internal Parts:
Stainless Steel

Seal:
Viton™ or Buna-N® (Standard)

Note: Specify when operating pressure below 10 PSI

visit www.cla-val.com to see our complete line of air and check valves.

Series 34
Air Release Valve

CLA-VAL

Air Release Valve Sizing

Air release valve sizing requires determining the volume of air that must be released from pipeline high points during normal operation and the diameter of the pipeline. Series 34 Air Release Valves are primarily used to continuously release pockets of air (as they develop) from high point, hence it is not critical to determine exact volume of air to be released.

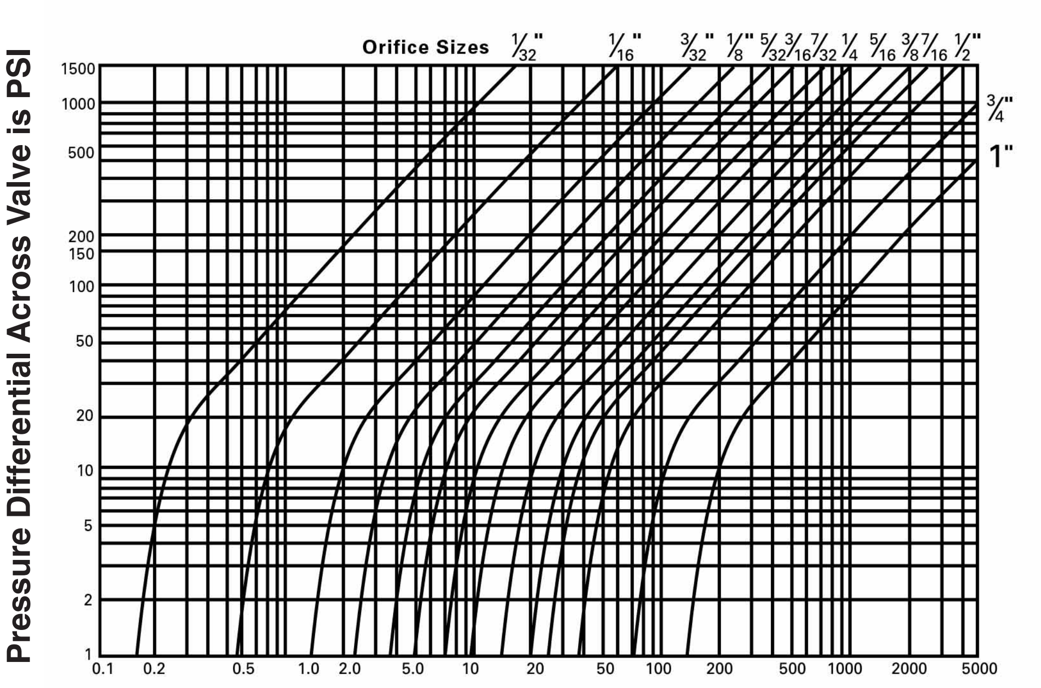
See chart on page 3 for sizing based on venting capacity.

Air Release Valve Sizing Chart For Water Pipelines

Figure A	Model No.	Inlet Size	Outlet Size	Orifice Size	GPM	MWP	Height	Width	Wt. (lbs.)
	**UL Listed • FM Approved** 3450-AR332 3475-AR332 3410-AR332	1/2", 3/4", 1"	1/2"	3/32"	200 - 2200	175	5-7/8"	3-3/4"	6
	FM Approved UL Listed 3450-AR116.3 3475-AR116.3 3410-AR116.3	1/2", 3/4", 1"	1/2"	1/16"	200 - 2200	300	5-7/8"	3-3/4"	6
Figure B	Model No.	Inlet Size	Outlet Size	Orifice Size	GPM	MWP	Height	Width	Weight
	3410-AR316C 3420-AR316C	1", 2"	1/2"	3/16"	2200 - 15000	150	10"	7"	20
	3410-AR532.3C 3420-AR532.3C	1", 2"	1/2"	5/32"	2200 - 15000	300	10"	7"	20
	3410-AR332.6C 3420-AR332.6C	1", 2"	1/2"	3/32"	2200 - 15000	600	10"	7"	20
Figure C	Model No.	Inlet Size	Outlet Size	Orifice Size	GPM	MWP	Height	Width	Weight
	3420-AR038C 3430-AR038C	2", 3"	1"	3/8"	15000 - 50000	150	12-1/2"	9-1/2"	45
	3420-AR732C 3430-AR732C	2", 3"	1"	7/32"	15000 - 50000	300	12-1/2"	9-1/2"	45
Figure D	Model No.	Inlet Size	Outlet Size	Orifice Size	GPM	MWP	Height	Width	Weight
	3420-AR-HP500	2"	1"	7/32"	2200 - 50000	500	13"	12"	75
	3420-AR-HP800	2"	1"	1/8"	2200 - 50000	800	13"	12"	75

Series 34
Air Release Valve

Venting Capacity Graph for Air Release Valves



Venting Capacity in Cubic Feet of Free Air Per Minute

Valve Selection Based on Venting Capacity

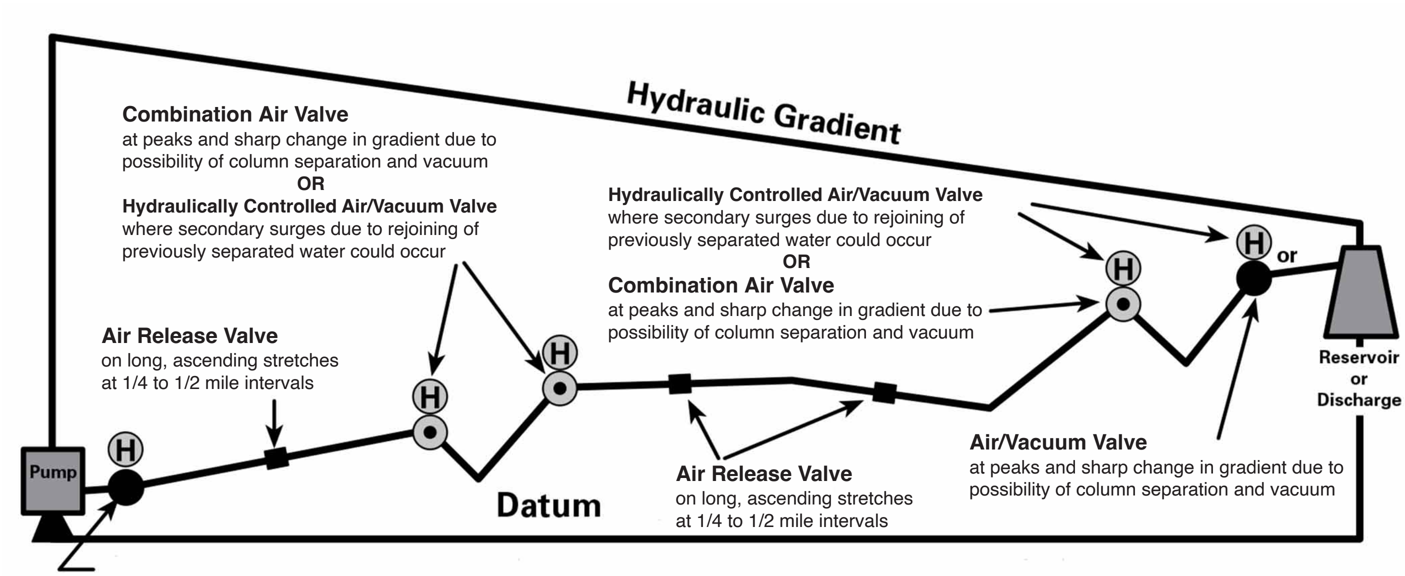
Follow these steps to select and size an air release valves when a specific venting capacity is required:

A) Enter graph with required system pressure and venting capacity

B) Read off nearest orifice diameter to intersection of pressure and capacity lines on graph

C) Enter table above with orifice diameter and select valve that can use this orifice diameter with the corresponding pressure

Series 34 Technical Data



Air/Vacuum Valve

Air Release Valve

Combination Air Valve

Hydraulically Controlled Air/Vacuum Valve

Installation Tips

1. The effectiveness of Series 34 Air Release Valve is dependent upon it being located at appropriate high points in a pipeline and at uniform intervals of approximately 2500 feet on horizontal pipelines.

2. There are four variables that can cause an air pocket to form slightly downstream of the true high point in a piping system:
   1. Severity of the slope adjacent to the high point or change of gradient
   2. Velocity of the liquid
   3. Texture of the inside surface of the pipe being used
   4. Viscosity of the fluid

   It is recommended where an air pocket can form slightly downstream of the high point, to install additional Series 34 Air Release Valve at this point.

3. Cla-Val has available, upon request, a Slide Rule Air Valve Calculator. It will greatly reduce the amount of time to size valves for pipeline service.

Other typical applications include:
1. Centrifugal pumps
2. Hydropneumatic tanks
3. Enclosed systems
4. Sewage lines

When Ordering, Please Specify:
1. Model Number
2. Inlet Size (NPT)
3. Inlet Pressure Rating
4. Orifice Size

1701 Placentia Ave • Costa Mesa CA 92627 • Phone: 949-722-4800 • Fax: 949-548-5441 • E-mail: info@cla-val.com • www.cla-val.com

SINCE 1908
wessels company

SUBMITTAL

FX-SERIES

HYDRO-PNEUMATIC TANKS

Models: FX-8 thru FX-2000V
Submittal Sheet No. C-1101A Date: 4/20

Job Name	______________	Submitted By	______________	Date	________
Location	______________	Approved By	______________	Date	________
	______________	Order No.	______________	Date	________
Engineer	______________	Notes	______________		
Contractor	______________		______________		
Sales Rep.	______________		______________		

Description

Wessels type FX-Series Tanks are non-ASME replaceable bladder type pre-charged hydro-pneumatic tanks for residential and commercial well and water systems, booster systems, irrigation systems, etc. They are designed to deliver water under pressure between pump cycles to provide sufficient flow to meet demands. The water is contained in a butyl bladder. The exterior has a baked epoxy finish. Products comply with NSF/ANSI Standard 61.

Construction

Shell: Carbon Steel
Bladder: Heavy Duty Butyl
FDA Approved
NSF 61 Listed
System Connection: Stainless Steel

Performance Limitations

Maximum Design Temperature: 240°F
Maximum Design Pressure: 150 PSIG

Products Comply
NSF / ANSI
Standard 61

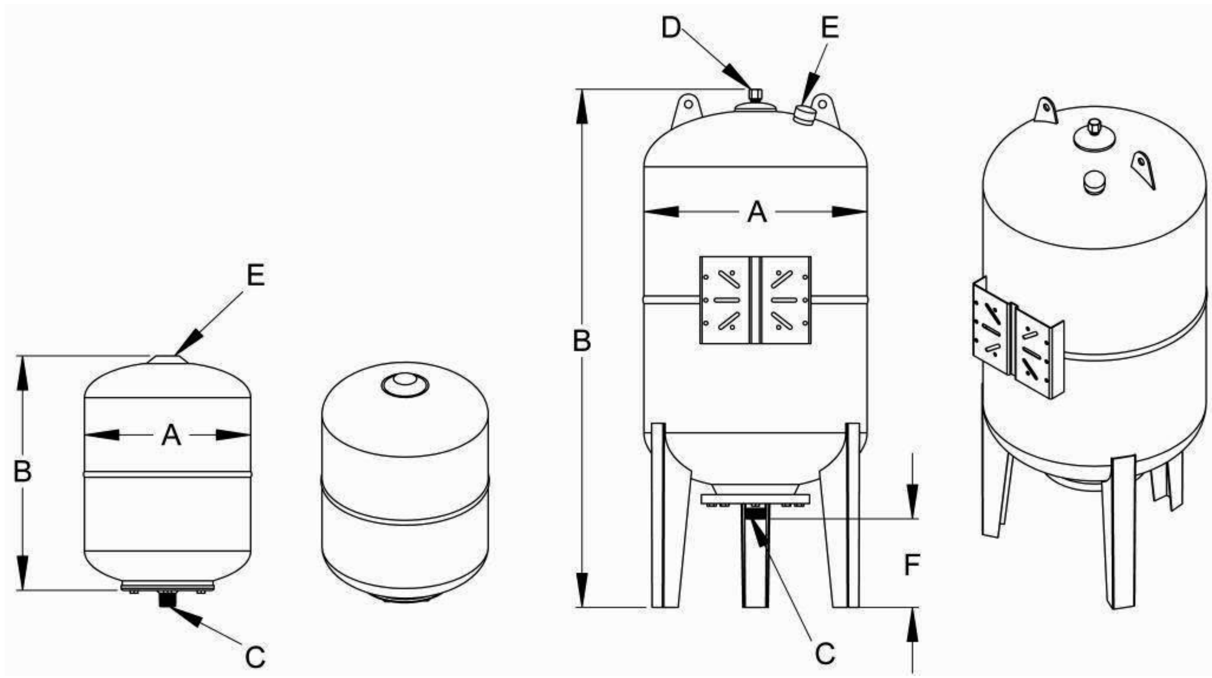
Model Number	Part Number	Tank Volume (Gallons)	Tagging Information	Quantity
FX 8	30011008	2.1		
FX 19	30011019	5		
FX 24	30011024	6.3		
FX 60V	30011060	16		
FX 80V	30011080	21		
FX 100V	30011100	26		
FX 200V	30011200	52		
FX 300V	30011300	80		
FX 500V	30011500	132		
FX 750V	30011700	198		
FX 1000V	30011000	264		
FX 2000V	30012000	528		

Typical Specification

Furnish and install, as shown on plans, a 16 gallon 15" diameter X 34" (high) pre-charged hydro-pneumatic tank with heavy-duty butyl bladder. The tank shall have stainless steel NPT system connections and a 0.302"-32 charging valve connection (standard tire valve) to facilitate the on-site charging of the tank to meet system requirements. Products comply with NSF/ANSI Standard 61.

Each tank shall be Wessels model number FX- 60V or approved equal.

101 TANK ST • GREENWOOD, IN 46143 • (317) 888-9800 • (317) 888-9988 FAX • www.westank.com



FX-8 thru FX-24

FX-60V thru FX-2000V

Dimensions & Weights

<table>
<tr><th rowspan="3">Model Number</th><th colspan="6">Dimensions in Inches</th><th rowspan="3">Approx. Ship Wt. (lbs)</th></tr>
<tr><th rowspan="2">A</th><th rowspan="2">B</th><th colspan="2">System Connection</th><th>Charging Valve</th><th rowspan="2">F</th></tr>
<tr><th>C</th><th>D</th><th>E</th></tr>
<tr><td>FX 8</td><td>8</td><td>13</td><td rowspan="3">3/4</td><td rowspan="5">-</td><td rowspan="12">0.302"-32NC</td><td rowspan="3">-</td><td>7</td></tr>
<tr><td>FX 19</td><td>11</td><td>16</td><td>13</td></tr>
<tr><td>FX 24</td><td>14</td><td>13</td><td>15</td></tr>
<tr><td>FX 60V</td><td>15</td><td>34</td><td rowspan="3">1</td><td>7</td><td>39</td></tr>
<tr><td>FX 80V</td><td>18</td><td>34</td><td>7.7</td><td>49</td></tr>
<tr><td>FX 100V</td><td>18</td><td>38</td><td rowspan="7">1/2</td><td>7.2</td><td>61</td></tr>
<tr><td>FX 200V</td><td>22</td><td>49</td><td rowspan="4">1 1/2</td><td rowspan="2">8.5</td><td>112</td></tr>
<tr><td>FX 300V</td><td>25</td><td>55</td><td>141</td></tr>
<tr><td>FX 500V</td><td>31</td><td>61</td><td rowspan="4">9</td><td>265</td></tr>
<tr><td>FX 750V</td><td>31</td><td>79</td><td>330</td></tr>
<tr><td>FX 1000V</td><td>37</td><td>77</td><td rowspan="2">2</td><td>398</td></tr>
<tr><td>FX 2000V</td><td>50</td><td>84</td><td>835</td></tr>
</table>

Notes

- Tanks are factory pre-charged at 30 psi and field adjustable.
- All vertical models furnished with three mounting legs.
- Lifting rings are standard on models FX-100V thru FX-2000V.

A PRODUCT SHEET OF NEPTUNE TECHNOLOGY GROUP

T-10® METER

SIZES: 1 ½" and 2"

Every Neptune® T-10® water meter meets or exceeds the latest AWWA C700 Standard. Its nutating disc, positive displacement principle has been time-proven for accuracy and dependability since 1892, ensuring maximum utility revenue.

The T-10 water meter consists of three major assemblies: a register, a lead free, high-copper alloy maincase, and a nutating disc measuring chamber.

The T-10 meter is available with a variety of register types. For reading convenience, the register can be mounted in one of four positions on the meter.

The corrosion-resistant maincase will withstand harsh service conditions: internal water pressure, rough handling, and in-line piping stress.

The innovative floating chamber design of the nutating disc measuring element protects the chamber from frost damage while the unique chamber seal extends the low-flow accuracy by sealing the chamber outlet port to the maincase outlet port. The nutating disc measuring element utilizes corrosion-resistant materials throughout and a thrust roller to minimize wear.

KEY FEATURES

Register

- Magnetic-driven, low-torque registration ensures accuracy
- Impact-resistant register
- High-resolution, low-flow leak detection
- Bayonet-style register mount allows in-line serviceability
- Tamperproof seal pin deters theft
- Date of manufacture, size, and model stamped on dial face

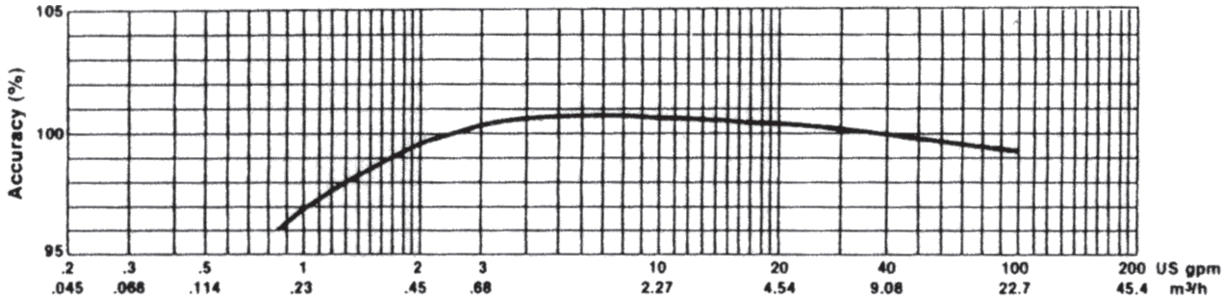
Lead Free Maincase

- Made from lead free, high-copper alloy
- NSF/ANSI 61 Certified
- NSF/ANSI 372 Certified
- Lifetime guarantee
- Resists internal pressure stresses and external damage
- Handles in-line piping variations and stresses
- Provides residual value vs. plastic
- Electrical grounding continuity

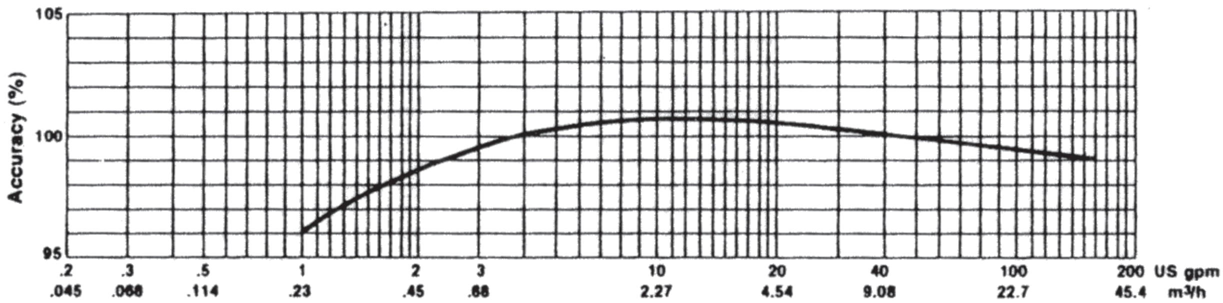
Nutating Disc Measuring Chamber

- Positive displacement
- Widest effective flow range for maximum revenue
- Proprietary polymer materials maximize long-term accuracy
- Floating chamber design is unaffected by in-line piping stresses

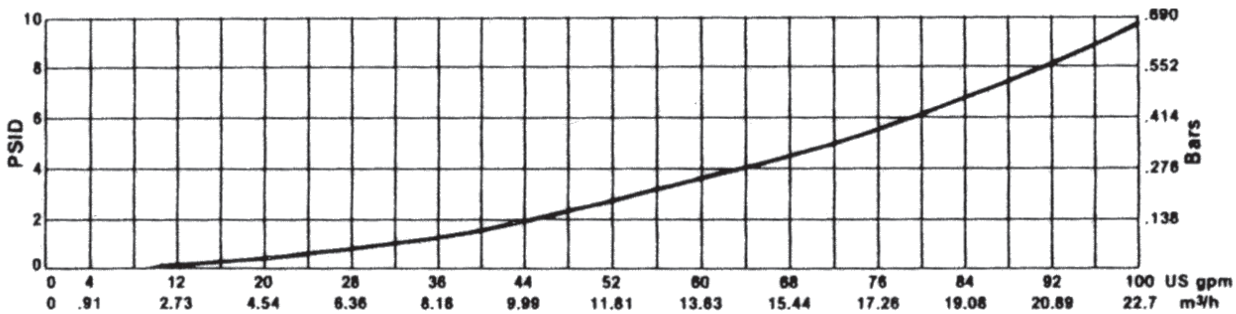
1 ½" Accuracy



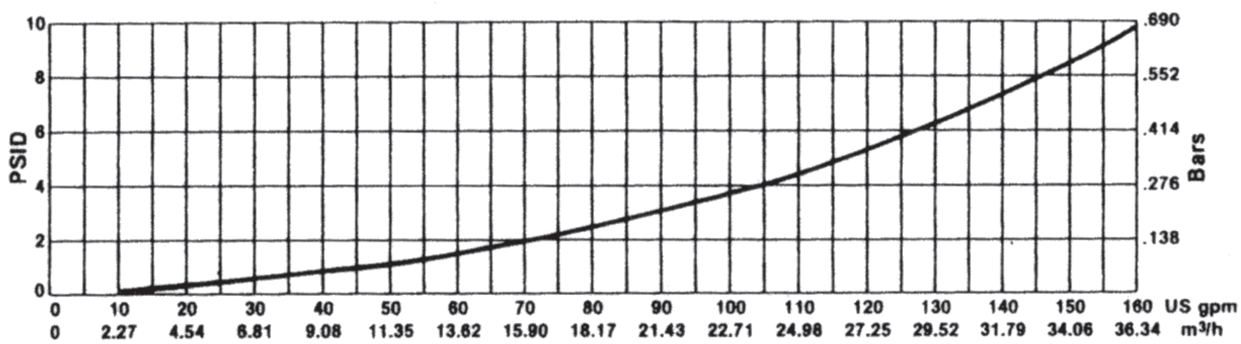
2" Accuracy



1 ½" Pressure Loss



2" Pressure Loss



These charts show typical meter performance. Individual results may vary.

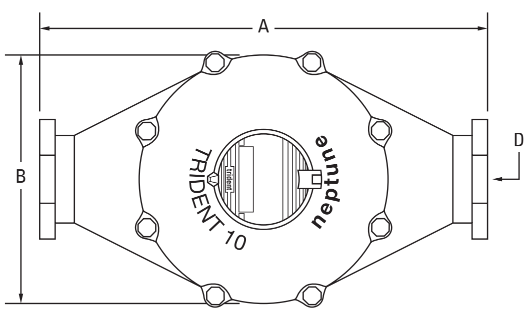
Operating Characteristics

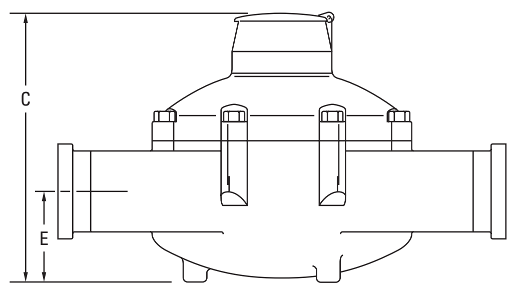
Meter Size	Normal Operating Range @100% Accuracy (±1.5%)	AWWA Standard	Low Flow @ 95% Accuracy
1 ½”	2 to 100 US gpm 0.46 to 22.73 m^3/h	5 to 100 US gpm 1.1 to 22.7 m^3/h	$^{3}/_{4}$ US gpm 0.17 m^3/h
2”	2 $^{1}/_{2}$ to 160 US gpm 0.57 to 36.36 m^3/h	8 to 160 US gpm 1.8 to 36.3 m^3/h	1 US gpm 0.23 m^3/h

Dimensions

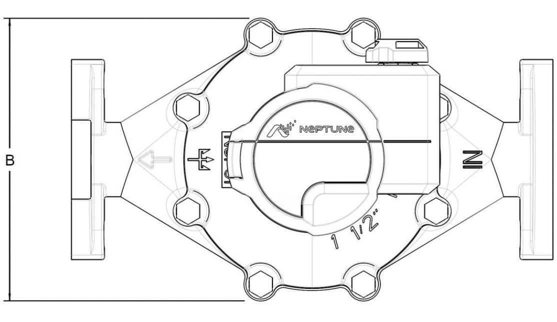
Meter Size	A in/mm	B in/mm	C-Std. in/mm	C-ARB in/mm	C-E-CODER®) R900*i*™ or ProCoder™) R900*i*™	D-Threads per inch	D-Thread Type	E in/mm	Weight lbs/kg
1 ½” Screw End	12 $^{5}/_{8}$ 321	8 $^{1}/_{16}$ 205	8 $^{1}/_{8}$ 206	8 $^{13}/_{16}$ 220.3	8 $^{3}/_{8}$ 213	11 $^{1}/_{2}$	1 $^{1}/_{2}$ NPT	2 $^{9}/_{16}$ 65	31 14.1
1 ½” Flanged End	13 330	8 $^{1}/_{16}$ 205	8 $^{1}/_{8}$ 206	8 $^{13}/_{16}$ 220.3	8 $^{3}/_{8}$ 213	—	—	2 $^{9}/_{16}$ 65	35 15.9
2” Screw End	15 $^{1}/_{4}$ 387	9 $^{7}/_{16}$ 240	9 $^{5}/_{16}$ 237	9 $^{15}/_{16}$ 248.4	9 $^{1}/_{2}$ 241	11 $^{1}/_{2}$	2” NPT	3 $^{1}/_{8}$ 79	40 18.1
2” Flanged End	17 432	9 $^{7}/_{16}$ 240	9 $^{5}/_{16}$ 237	9 $^{15}/_{16}$ 248.4	9 $^{1}/_{2}$ 241	—	—	3 $^{1}/_{8}$ 79	44 20.0

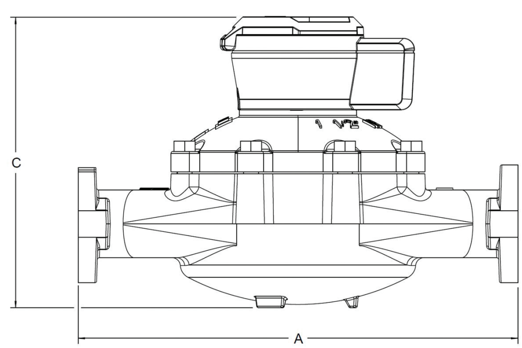
T-10 With Standard Register





T-10 With E-CODER®)R900*i*™ or ProCoder™)R900*i*™ Pit Register





Guaranteed Compatibility

All T-10 meters are guaranteed adaptable to our ProRead™, ProCoder™, E-CODER®, E-CODER®)R900*i*™, E-CODER®)R450*i*™, ProCoder™)R900*i*™, TRICON®/S, TRICON/E®3, and Neptune Utility Systems™ without removing the meter from service.

Specifications

Certification

- NSF/ANSI 61, NSF/ANSI 372

Application

- Cold water measurement of flow in one direction

Maximum Operating Water Pressure

- 150 psi (1,034 kPa)

Maximum Operating Water Temperature

- 80°F

Measuring Chamber

- Nutating disc technology design made from proprietary synthetic polymer

Registration

ProRead Registration (per sweep hand revolution)		1 ½"	2"
100	US Gallons	✓	✓
100	Imperial Gallons	✓	✓
10	Cubic Feet	✓	✓
1	Cubic Metre		✓
.01	Cubic Metre	✓	
Register Capacity ProRead, ProCoder, and E-CODER		**1 ½"**	**2"**
100,000,000	US Gallons	✓	✓
100,000,000	Imperial Gallons	✓	✓
10,000,000	Cubic Feet	✓	✓
100,000	Cubic Metres	✓*	
1,000,000	Cubic Metres	✓**	✓
E-CODER High Resolution (8-digit reading)		**1 ½"**	**2"**
1	US Gallons	✓	✓
1	Imperial Gallons	✓	✓
0.1	Cubic Feet	✓	✓
0.01	Cubic Metres		✓
0.001	Cubic Metres	✓	
ProCoder High Resolution (8-digit reading)		**1 ½"**	**2"**
1	US Gallons	✓	✓
1	Imperial Gallons	✓	✓
0.1	Cubic Feet	✓	✓
0.01	Cubic Metres	✓	✓

**ProRead and E-CODER only* ***ProCoder only*

Options

Sizes

- 1 ½" flanged or threaded end
- 2" flanged or threaded end

Units of Measure

- U.S. gallons, imperial gallons, cubic feet, cubic metres

Register Types

- ProCoder, E-CODER, E-CODER)R900*i*, ProCoder)R900*i*

Measuring Chamber

- Synthetic polymer

Companion Flanges

- Lead free, high-copper alloy

Environmental Conditions

- Operating temperature: +33ºF to +149ºF (0ºC to +65ºC)
- Storage temperature:
- +33ºF to +158ºF (0ºC to +70ºC)

Test Ports

- 1" (optional)

Warranty

Neptune provides a limited warranty for performance, materials, and workmanship. See warranty statement for details.

NEPTUNE TECHNOLOGY GROUP

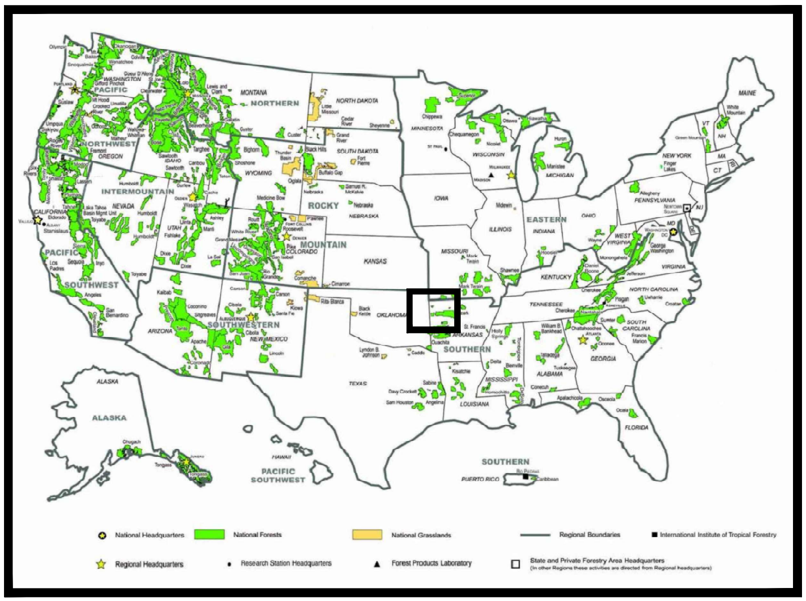
#winyourday
neptunetg.com

Neptune Technology Group
1600 Alabama Highway 229
Tallassee, AL 36078
800-633-8754 f 334-283-7293

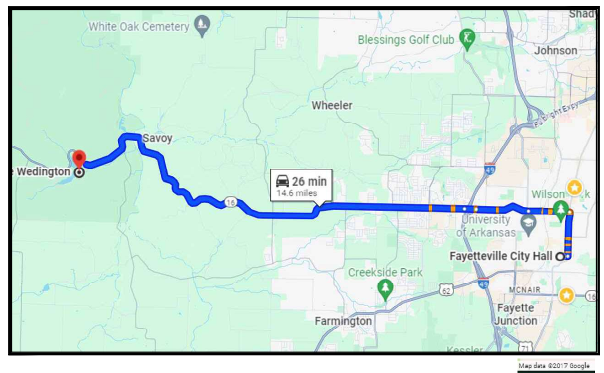
United States Department of Agriculture
Forest Service

ARKANSAS
WASHINGTON COUNTY
FOREST SERVICE REGION 8
SOUTHERN REGION

WEDINGTON RECREATION AREA
WATER DISTRIBUTION CONSTRUCTION PLAN



FOREST LOCATION MAP



PROJECT LOCATION MAP

TRAVEL
DIRECTIONS:

Head east on W Mountain St toward S Block Ave 164 ft, Continue straight to stay on W Mountain St 0.1 mi Turn left onto US-71 BUS N/S College Ave Continue to follow US-71 BUS N 1.1 mi Turn left onto E North St 1.0 mi Continue onto AR-112 Spur W/W Wedington Dr 1.3 mi Continue straight onto AR-112 Spur W 0.3 mi Continue onto AR-16 W

36.090364, -94.376313

INDEX OF SHEETS

RECOMMENDED BY:

SAMMIE MCDOWELL Digitally signed by SAMMIE MCDOWELL Date: 2025.08.04 12:27:40 -05'00'
FOREST ENGINEER DATE

WILLIAM SMITH Digitally signed by WILLIAM SMITH Date: 2025.08.04 16:07:13 -05'00'
DISTRICT RANGER DATE

ROBERT MCBROOME Digitally signed by ROBERT MCBROOME Date: 2025.08.04 17:13:05 -05'00'
FOREST SUPERVISOR DATE

KIMBERLY BARNETT Digitally signed by KIMBERLY BARNETT Date: 2025.08.05 13:13:51 -04'00'
RO WATER AND WASTEWATER PROGRAM MANAGER DATE

APPROVED:

BRADLEY ASHER Digitally signed by BRADLEY ASHER Date: 2025.08.05 13:22:00 -04'00'
DIRECTOR OF ENGINEERING DATE

General Notes

1. In general, the work for this project is shown on the plans as specified.
2. The Contractor shall have full access to grounds for construction operations during the construction period. FS may occupy some of the buildings during construction.
3. Contractor shall minimize the disturbance of the work area.
4. This design was compiled from available information from the USFS, locations are approximate and should be field verified.
5. The new water lines are to be installed in proximity to the existing water lines where applicable or existing roads due to environmental and historical clearances.
6. All existing waterlines, fire hydrants, valves, and accessories are to be demolished & properly disposed.
7. Applicable sections of the Washington Water Authority, Standard Specifications for Design and Construction of Water Improvements, 2023 Edition will be followed for the construction of this project. All references to the Washington Water Authority for inspections or decisions will be made by the USFS Contracting Officer Representative and/or the Engineer of Record.
8. All rules, regulations, and requirements of the Arkansas Department of Health are the minimal standard for this plan for all construction.
9. Work Shall be performed by a licensed contractor with 5 years experience with installation of HDPE potable water supply line systems. Work shall be done using applicable building codes, and industry accepted standard practices.
10. Contractor shall comply with requirements of all applicable regulatory agencies.
11. The contractor shall verify all dimensions, elevations, and conditions on these drawing prior to start of construction. Contractor will notify Engineer of any discrepancies prior to start of construction.
12. Contractor shall contact Arkansas One Call prior to any Excavation at 811, 1-800-482-8998, or file an eTicket at arkansas 811.com. Locate all utilities prior to beginning construction.
13. Construction methods are not indicated on these drawing. The contractor shall be solely responsible for all methods, sequences, safety, and procedures of construction. The contractor shall provide adequate shoring, bracing, formwork, etc. required for the protection of life and property during construction.
14. Excavation procedures, including shoring and protection of adjacent property, structures, street and utilities, shall be performed in compliance with local building codes, regulations and safety requirement, and shall be the contractor's responsibility.
15. Trenches for this project are relatively shallow, exact bury depth of existing waterlines varies, contractor shall comply with applicable OSHA requirements.
16. Contractor is responsible for implementing Best Management Practices (BMP's) for stormwater and erosion controls including construction entrance.
17. Contractor is responsible for excavation, grading, cutting and removal of trees, shrubs and underbrush as required for the construction of this project. Contractor shall dispose of excess soils, excess vegetation, and construction materials off of the Government's property.
18. The installation of the waterline will be constructed using approved embedment materials only.
19. HDPE Pipe shall be manufactured in accordance with AWWA C901, latest revision.
20. HDPE Pipe and valves shall use mechanical joint connections and fuse butt weld joints.
21. Valves shall be plumb without obstructions that will prohibit the installation of the valve boxes directly over the stem.
22. Air release valves shall be ARI D-040 for 1" or approved equal.
23. Concrete thrust blocks shall be poured in place and constructed along the waterline as indicated on plans.
24. Waterline shall be installed & tested prior to backfilling.
25. Contractor is responsible for flushing, testing, and disinfecting the waterline. Disinfection by chlorination shall be performed after flushing and pressure testing are completed and approved by the COR. COR will be present during all testing/disinfecting operations.
26. Contractor shall fill and flush the newly constructed line and visually check air release to assure proper operation.
27. Contractor shall perform a continuity test on all tracer wire. Any tracer wire found to be not continuous shall be repaired at contractors expense.
28. Trench backfilling should bring disturbed areas back to the original grade plus a mound to allow for settling. After trench backfilling and compacting activities have been completed, all areas shall be reseeded per USFS requirements including fertilizer application. Straw or mulch used for soil stabilization will be native to the area and weed free. Broadcast sowing of seed may be accomplished by hand seeders or by approved power equipment.
29. Contractor is responsible for establishing ground cover on all disturbed areas. Repeated seeding shall be required if necessary throughout the warranty period.
30. All cleanup operations shall be completed prior to final payment. To the extent that any road, vegetation, utilitiy, or other items are damaged or displaced by the Contractors operations, these shall be restored to their original or better condition prior to final inspection.
31. Periodic observations and monitoring shall be be conducted by the A & E Engineer.
32. Period of Performance is 270 days.
33. The Contractor warrants the products delivered and installed to be free from defects or damage prior to and after installation for a period of two years.
34. Contractor to supply USFS with 4 of the existing demo'd functional fire hydrants and 4 frost free hydrants to store onsite. Hydrants to be saved will be marked by the USFS. COR to provide on-site storage location.

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Region 8
Southern Region

STAMPS, LOGOS, AND SEALS

Hodges Engineering

CERTIFICATE OF AUTHORIZATION HODGES ENGINEERING No. 3352 ARKANSAS

STATE OF ARKANSAS LICENSED PROFESSIONAL ENGINEER No. 8289 KEVIN E. HODGES 6/20/25

NO.	REVISION / ISSUE	DATE
4		
3		
2		
1	USFS Comments	7/14/25

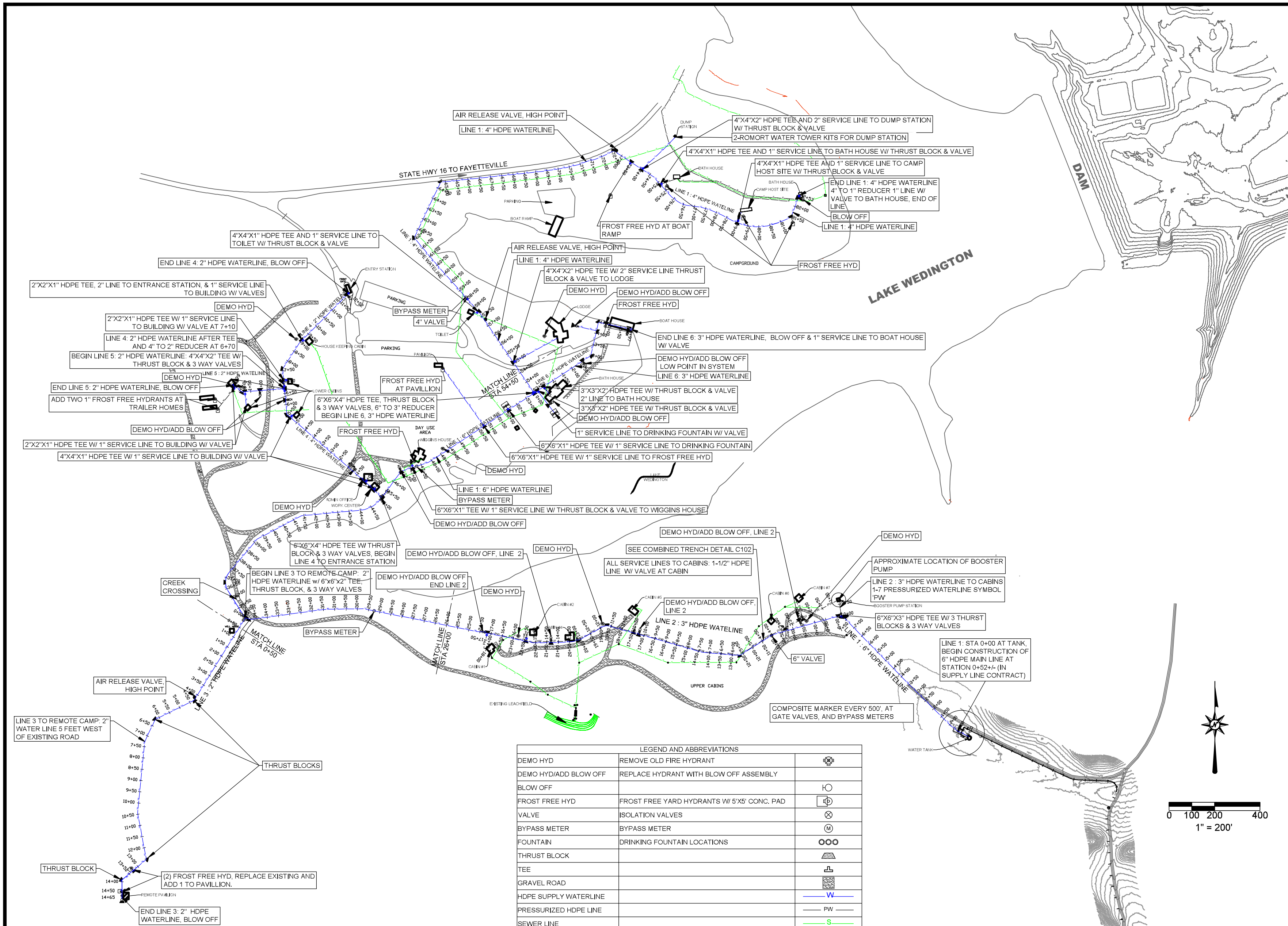
PROJECT NAME

LAKE WEDINGTON WATER DISTRIBUTION

DRAWING TITLE

GENERAL NOTES AND SPECIFICATIONS

DATE 06.20.2025	ARCHIVE NO.
DESIGNER K.HOD	DRAWING SHEET NO. C101
DRAWN H.HOD	
CHECKED K.HOD	
PROJECT NO. 12445123C0075	SHEET 2 OF 15



LEGEND AND ABBREVIATIONS		
DEMO HYD	REMOVE OLD FIRE HYDRANT	
DEMO HYD/ADD BLOW OFF	REPLACE HYDRANT WITH BLOW OFF ASSEMBLY	
BLOW OFF		
FROST FREE HYD	FROST FREE YARD HYDRANTS W/ 5'X5' CONC. PAD	
VALVE	ISOLATION VALVES	
BYPASS METER	BYPASS METER	
FOUNTAIN	DRINKING FOUNTAIN LOCATIONS	
THRUST BLOCK		
TEE		
GRAVEL ROAD		
HDPE SUPPLY WATERLINE		W
PRESSURIZED HDPE LINE		PW
SEWER LINE		S

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Forest Service

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Southern Region

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NO.	REVISION / ISSUE	DATE
4		
3		
2		
1	USFS Comments	7/14/25

PROJECT NAME

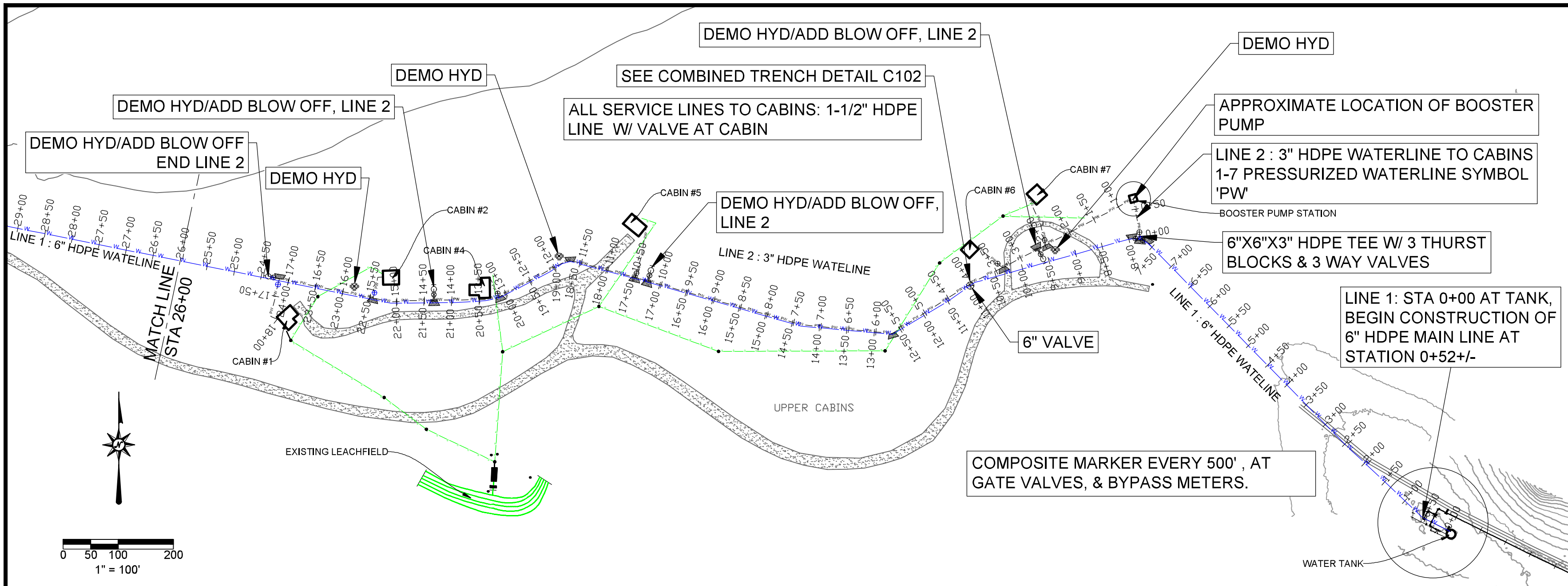
LAKE WEDINGTON WATER DISTRIBUTION

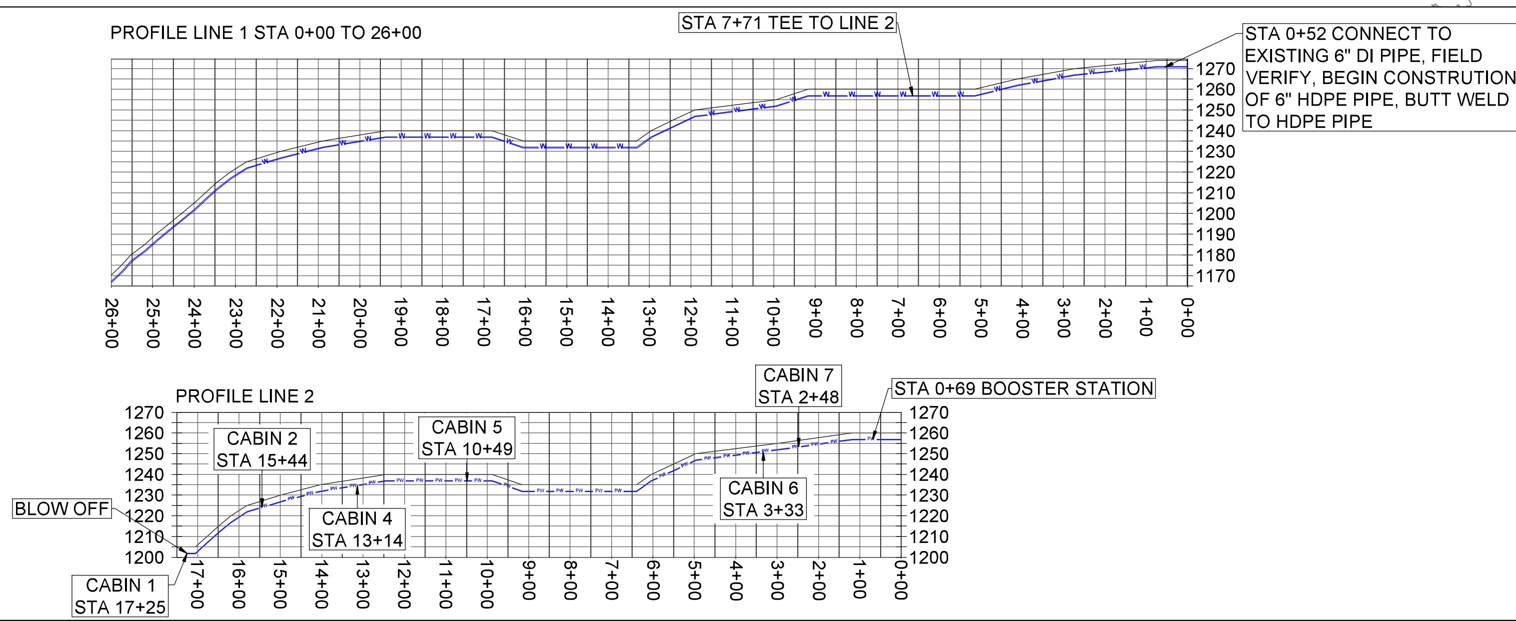
DRAWING TITLE

ALIGNMENT PLAN OVERVIEW

DATE 06,20,2025
ARCHIVE NO.
DESIGNER K,HOD
DRAWN H,HOD
CHECKED K,HOD
DRAWING SHEET NO. C102
PROJECT NO. 12445123C0075
SHEET 3 OF 15

7/16/25 09:29 KHODG C:\USERS\KHODG\ONEDRIVE\DOCUMENTS\HODGES ENGINEERING\USFS\WEDINGTON\DWG\C\DISTRIBUTION\LAKE-WEDINGTON-WATERLINE-DISTRIBUTION-CD-7-14-25 FINAL.DWG;





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NO.	REVISION / ISSUE	DATE
4		
3		
2		
1	USFS Comments	7/14/25

PROJECT NAME

LAKE WEDINGTON WATER DISTRIBUTION

DRAWING TITLE

ALIGNMENT PLAN LINE 1: STA 0+00 TO 26+00 AND LINE 2: STA 0+00 TO 18+15

DATE 06,20,2025	ARCHIVE NO.
DESIGNER K,HOD	DRAWING SHEET NO. C103
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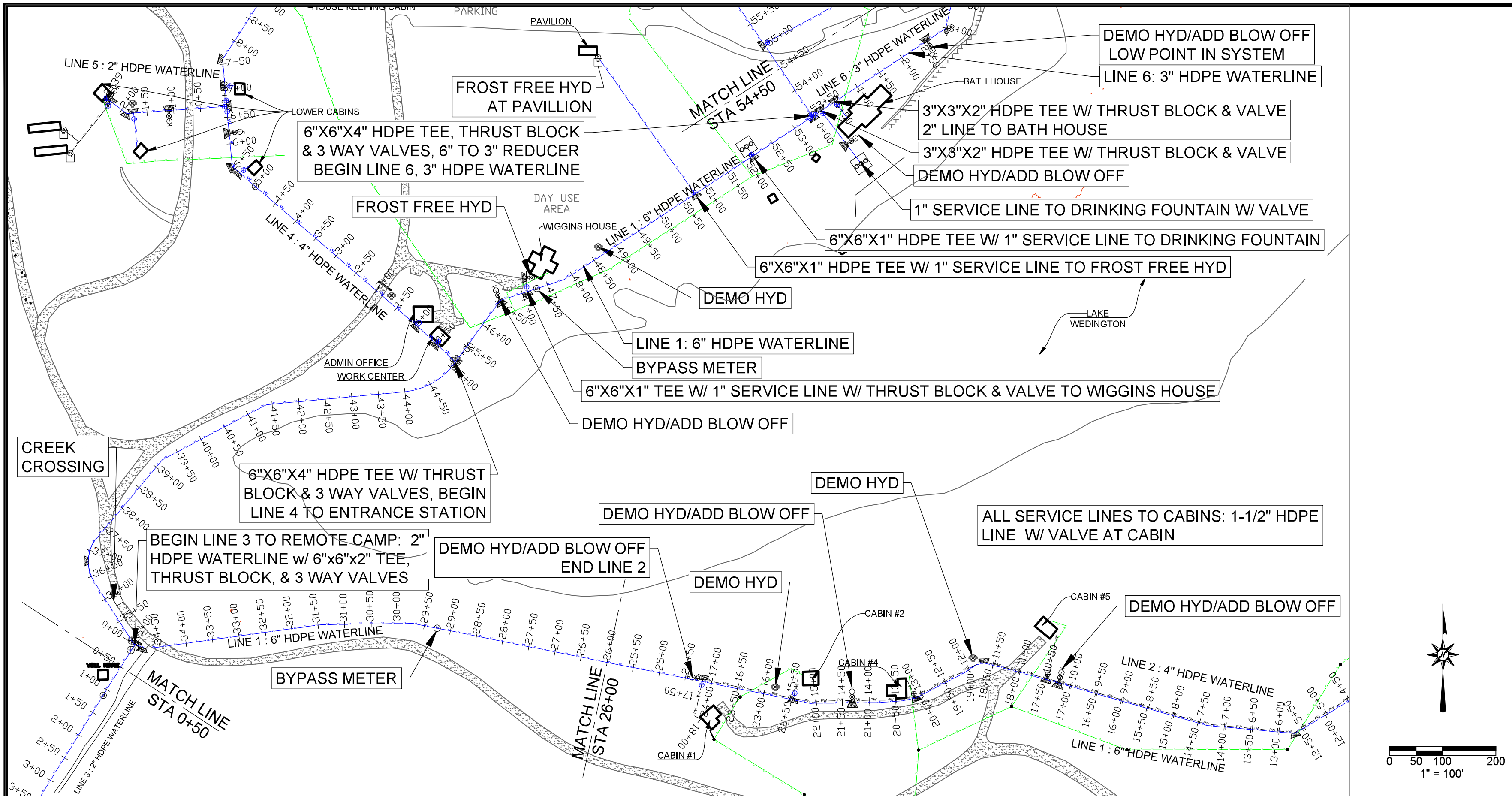
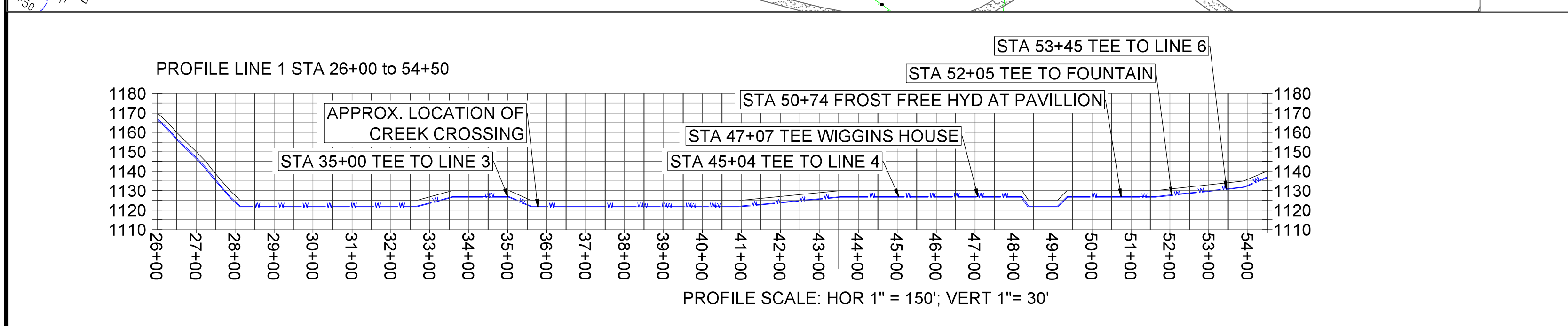
CHECKED K,HOD	
PROJECT NO. 12445123C0075	SHEET 4 OF 15





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HODGES ENGINEERING No. 3352 ARKANSAS CERTIFICATE OF AUTHORIZATION

STATE OF ARKANSAS LICENSED PROFESSIONAL ENGINEER No. 8209 KEVIN E. HODGES

NO.	REVISION / ISSUE	DATE
4		
3		
2		
1	USFS Comments	7/14/25

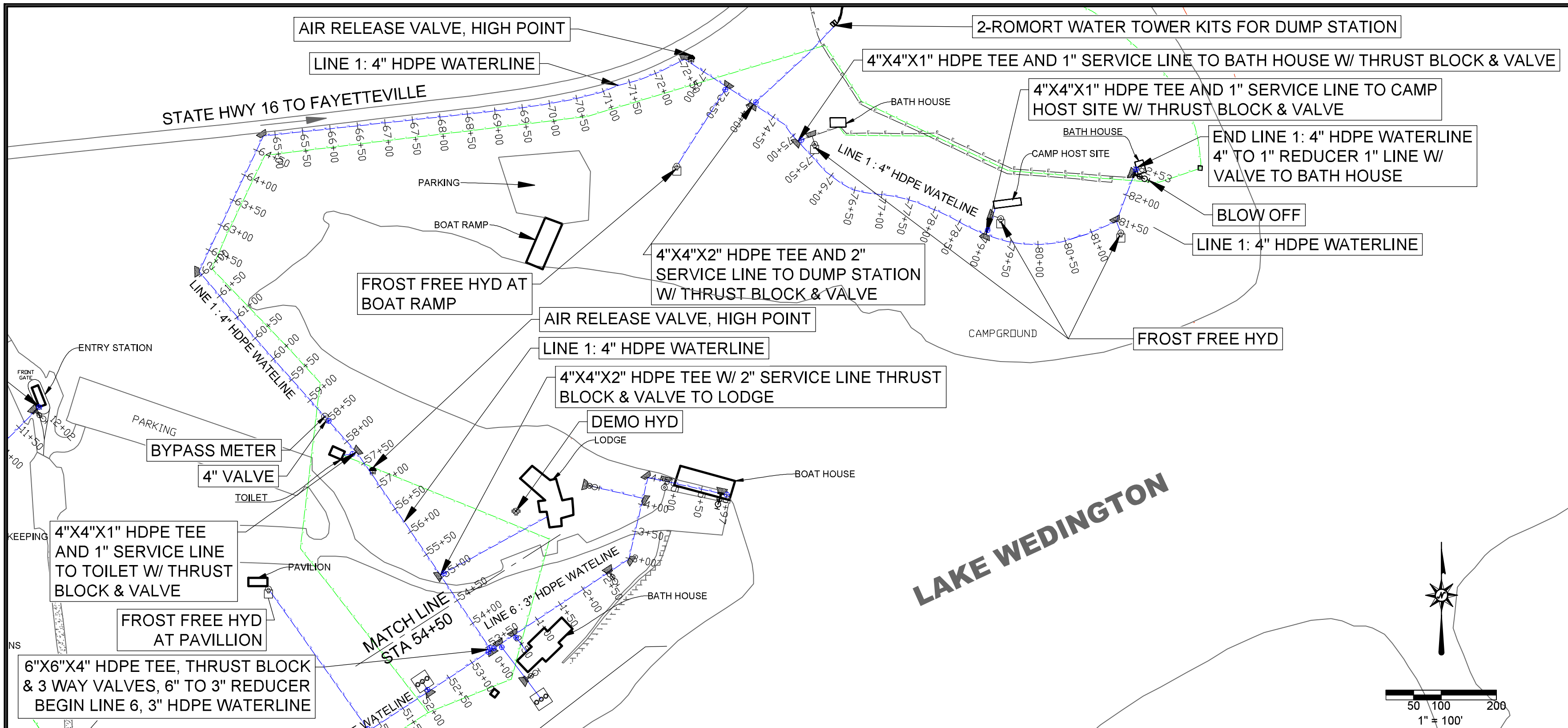
PROJECT NAME

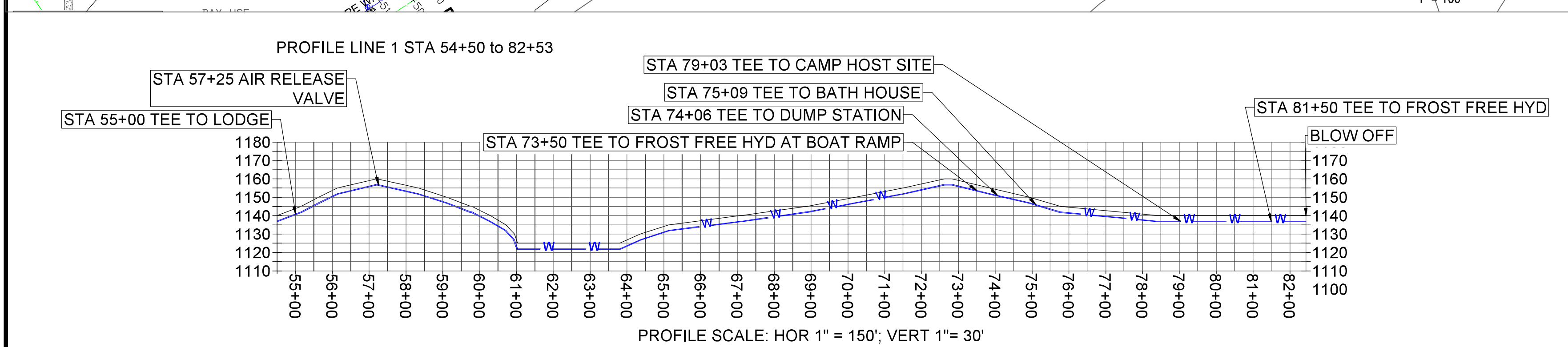
LAKE WEDINGTON WATER DISTRIBUTION

DRAWING TITLE

ALIGNMENT PLAN LINE 1: STA 26+00 TO 54+50

DATE 06,20,2025	ARCHIVE NO.
DESIGNER K.HOD	DRAWING SHEET NO. C104
DRAWN H.HOD	
CHECKED K.HOD	
PROJECT NO. 12445123C0075	SHEET 5 OF 15





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NO.	REVISION / ISSUE	DATE
4		
3		
2		
1	USFS Comments	7/14/25

PROJECT NAME

LAKE WEDINGTON WATER DISTRIBUTION

DRAWING TITLE

ALIGNMENT PLAN LINE 1: STA 54+50 TO 82+53

DATE 06,20,2025	ARCHIVE NO.
DESIGNER K.HOD	DRAWING SHEET NO. C105
DRAWN H.HOD	
CHECKED K.HOD	
PROJECT NO. 12445123C0075	SHEET 6 OF 15

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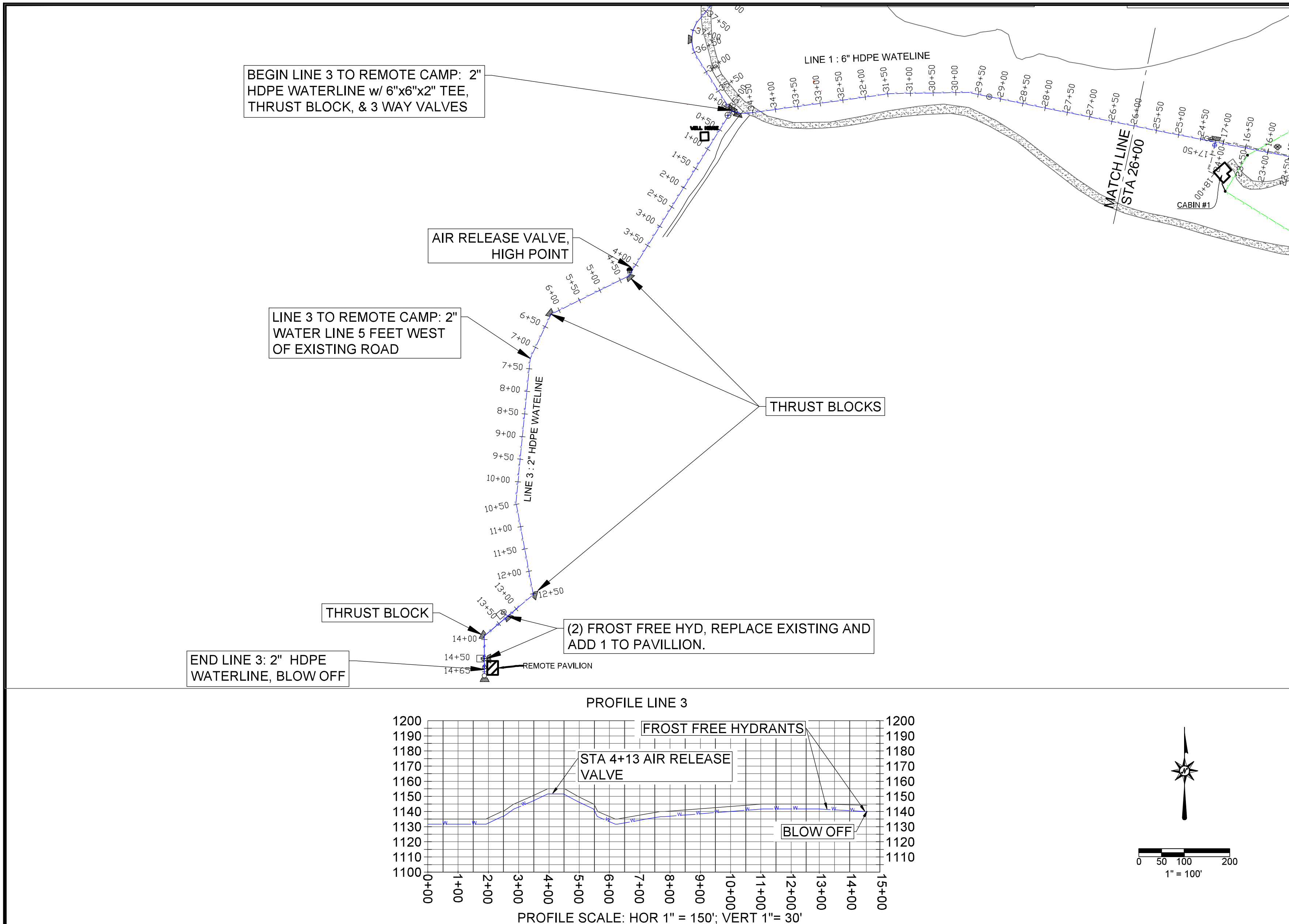

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4		
3		
2		
1	USFS Comments	7/14/25

PROJECT NAME

LAKE WEDINGTON WATER DISTRIBUTION

DRAWING TITLE

ALIGNMENT PLAN LINE 3: STA 0+00 TO 14+65

DATE 06,20,2025	ARCHIVE NO.
DESIGNER K.HOD	DRAWING SHEET NO. C106
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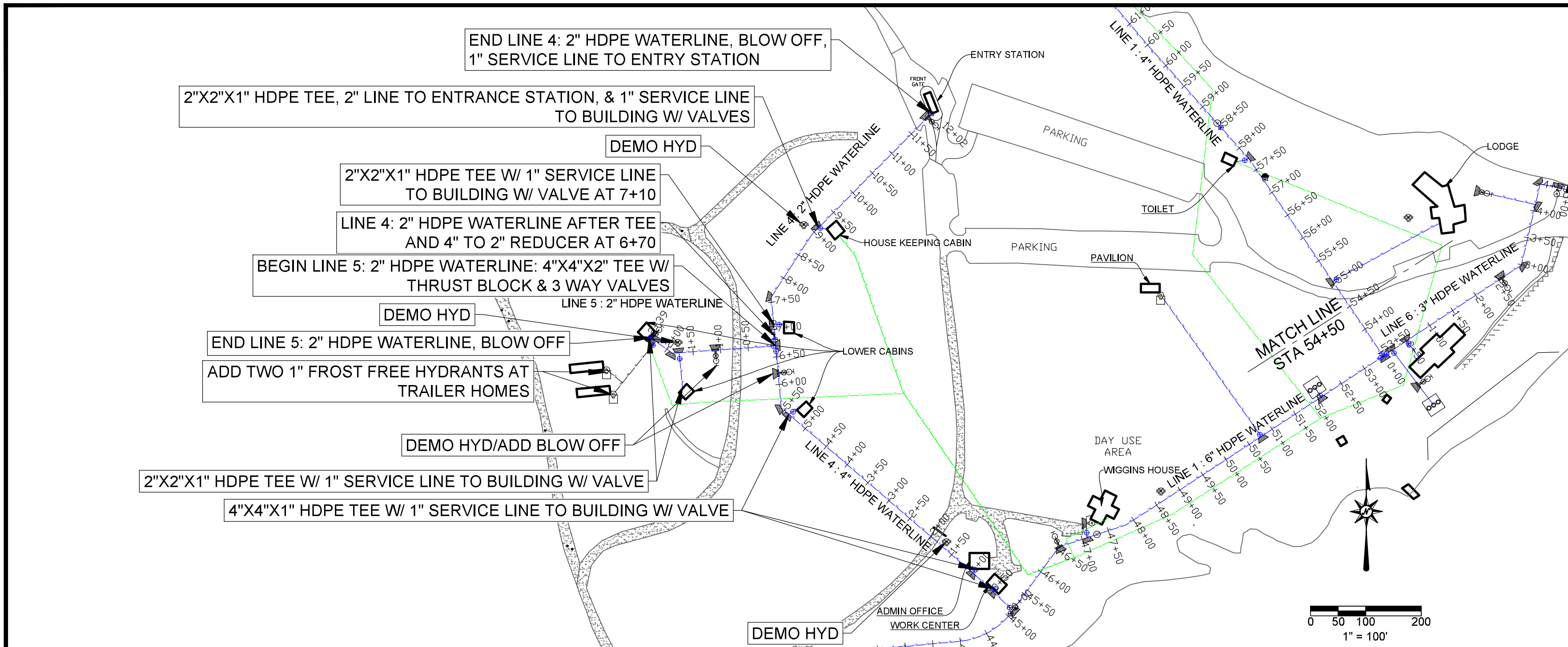
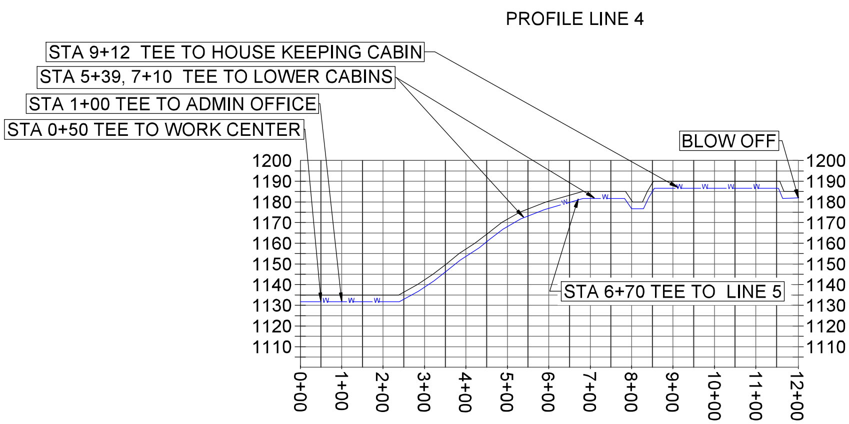
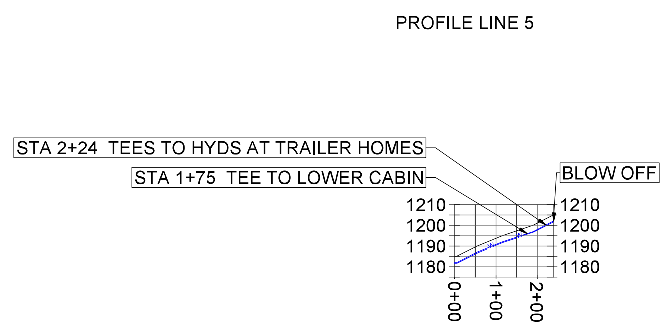
CHECKED K.HOD	
PROJECT NO. 12445123C0075	SHEET 7 OF 15







PROFILE SCALE: HOR 1" = 150'; VERT 1"= 30'

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NO.	REVISION / ISSUE	DATE
4		
3		
2		
1	USFS Comments	7/14/25

PROJECT NAME

LAKE WEDINGTON WATER DISTRIBUTION

DRAWING TITLE

ALIGNMENT PLAN LINE 4: STA 0+00 TO 12+02 AND LINE 5: STA 0+00 TO 2+39

DATE 06,20,2025
ARCHIVE NO.
DESIGNER K,HOD
DRAWN H,HOD
CHECKED K,HOD
DRAWING SHEET NO. C107
PROJECT NO. 12445123C0075
SHEET 8 OF 15

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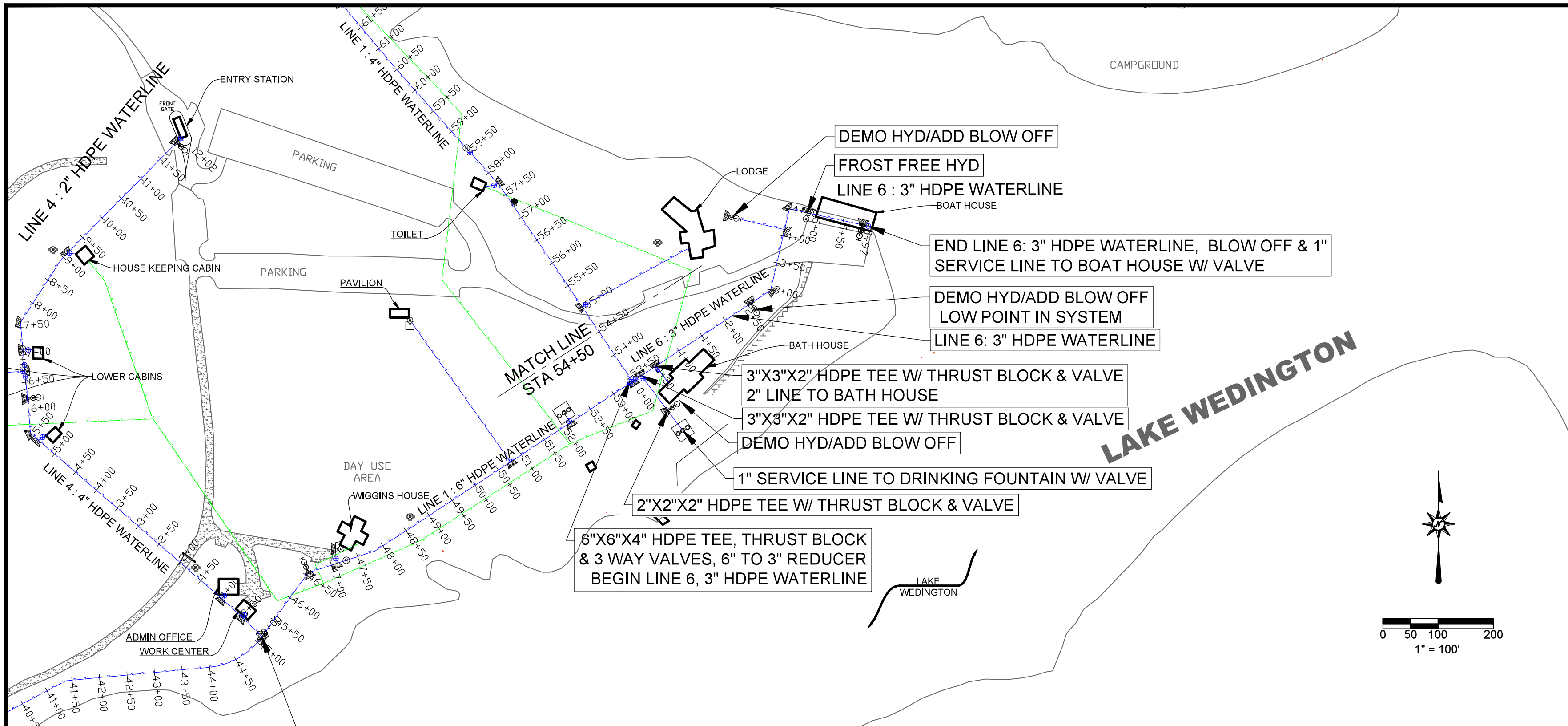
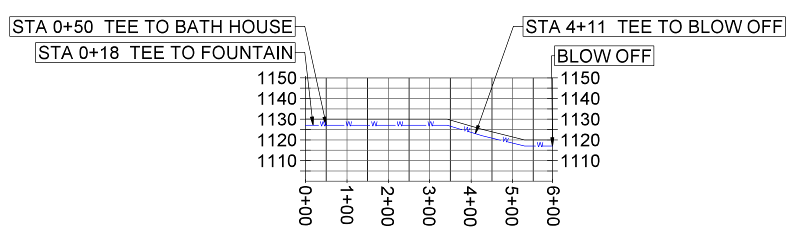


PROFILE LINE 6



PROFILE SCALE: HOR 1" = 150'; VERT 1"= 30'

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Region 8
Southern Region

STAMPS, LOGOS, AND SEALS

Hodges Engineering

NO.	REVISION / ISSUE	DATE
4		
3		
2		
1	USFS Comments	7/14/25

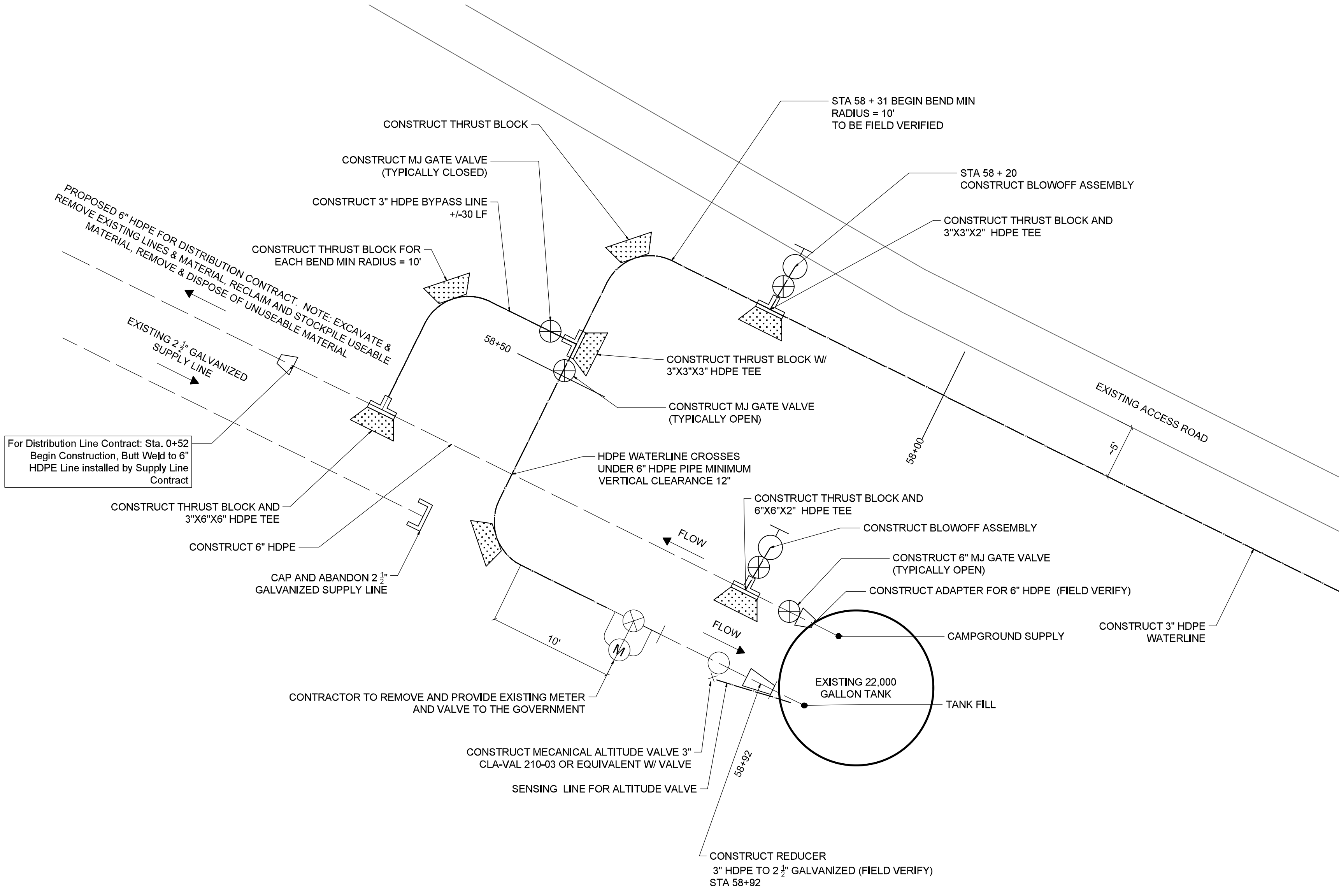
PROJECT NAME

LAKE WEDINGTON WATER DISTRIBUTION

DRAWING TITLE

ALIGNMENT PLAN LINE 6: STA 0+00 TO 5+97

DATE 06,20,2025	ARCHIVE NO.
DESIGNER K.HOD	DRAWING SHEET NO. C108
DRAWN H.HOD	
CHECKED K.HOD	
PROJECT NO. 12445123C0075	SHEET 9 OF 15



PIPING PLAN AT TANK
NTS

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Region 8
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STAMPS, LOGOS, AND SEALS

Hodges Engineering

CERTIFICATE OF AUTHORIZATION HODGES ENGINEERING No. 3352 ARKANSAS

STATE OF ARKANSAS LICENSED PROFESSIONAL ENGINEER No. 8209 KEVIN E. HODGES

NO.	REVISION / ISSUE	DATE
4		
3		
2		
1	USFS Comments	7/14/25

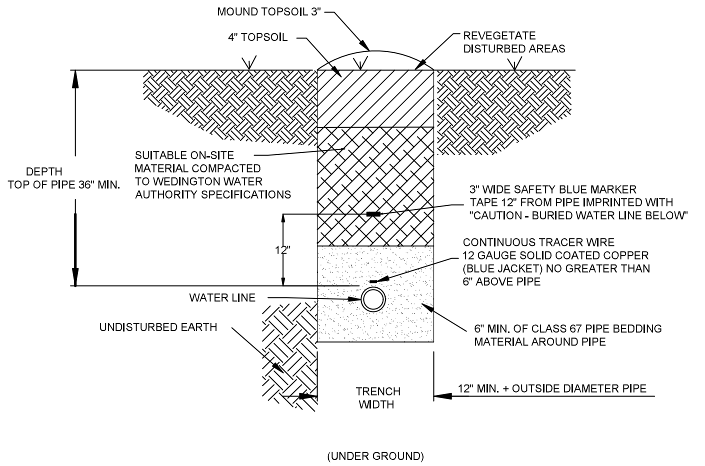
PROJECT NAME

LAKE WEDINGTON WATER DISTRIBUTION

DRAWING TITLE

PIPING PLAN AT TANK

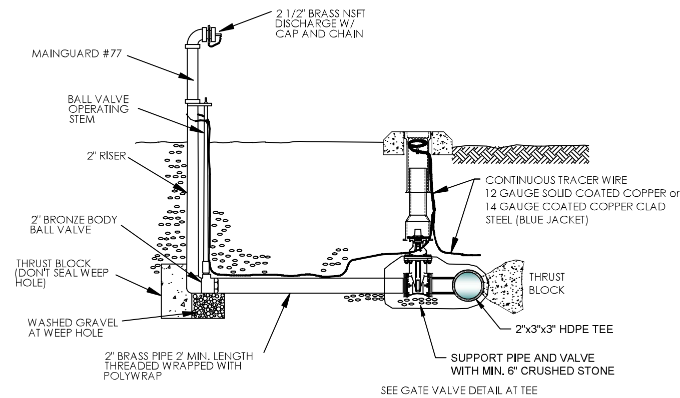
DATE 06,20,2025	ARCHIVE NO.
DESIGNER K.HOD	DRAWING SHEET NO. C109
DRAWN H.HOD	
CHECKED K.HOD	
PROJECT NO. 12445123C0075	SHEET 10 OF 15



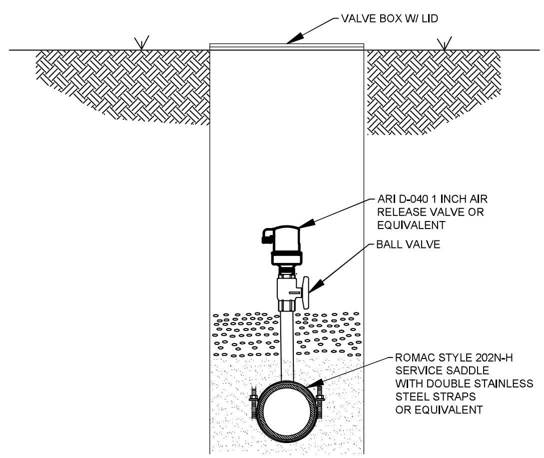
NOTES:

1. ALL VALVES, BENDS, ETC. SHALL BE RESTRAINED.
2. THE CONTRACTOR SHALL PROVIDE ALL ITEMS NECESSARY TO CONNECT WITH ANY PART OF THE EXISTING WATER SYSTEM THAT WILL REMAIN IN ORDER TO ESTABLISH A SATISFACTORY AND ACCEPTABLE WATER SYSTEM.
3. CONTRACTOR TO CONSTRUCT ALL TRENCH EXCAVATION IN ACCORDANCE WITH ALL OSHA REGULATIONS (29 CFR CH.XVII, SUBPART B)
4. TRENCH SHALL BE EXCAVATED BELOW GRADE REQUIRED TO PROVIDE A MINIMUM 21" OF PIPE COVER.

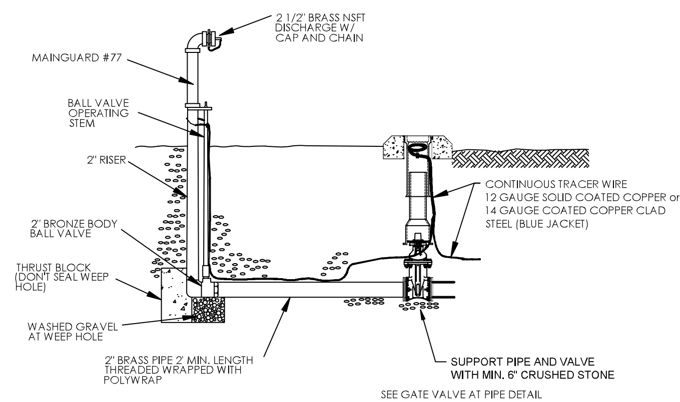
TYPICAL WATER LINES TRENCH DETAILS
NTS



BLOWOFF ASSEMBLY
NTS



AIR RELASE VALVE DETAIL
NTS



BLOWOFF ASSEMBLY IN LINE WITH SEVICE LINE
NTS

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Region 8
Southern Region

STAMPS, LOGOS, AND SEALS

Hodges Engineering

CERTIFICATE OF AUTHORIZATION HODGES ENGINEERING No. 3352 ARKANSAS

STATE OF ARKANSAS LICENSED PROFESSIONAL ENGINEER No. 8289 KEVIN E. HODGES

NO.	REVISION / ISSUE	DATE
4		
3		
2		
1	USFS Comments	7/14/25

PROJECT NAME

LAKE WEDINGTON WATER DISTRIBUTION

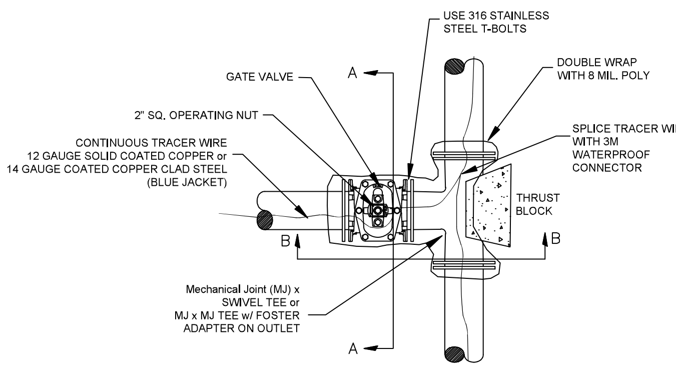
DRAWING TITLE

PIPING DETAILS 1

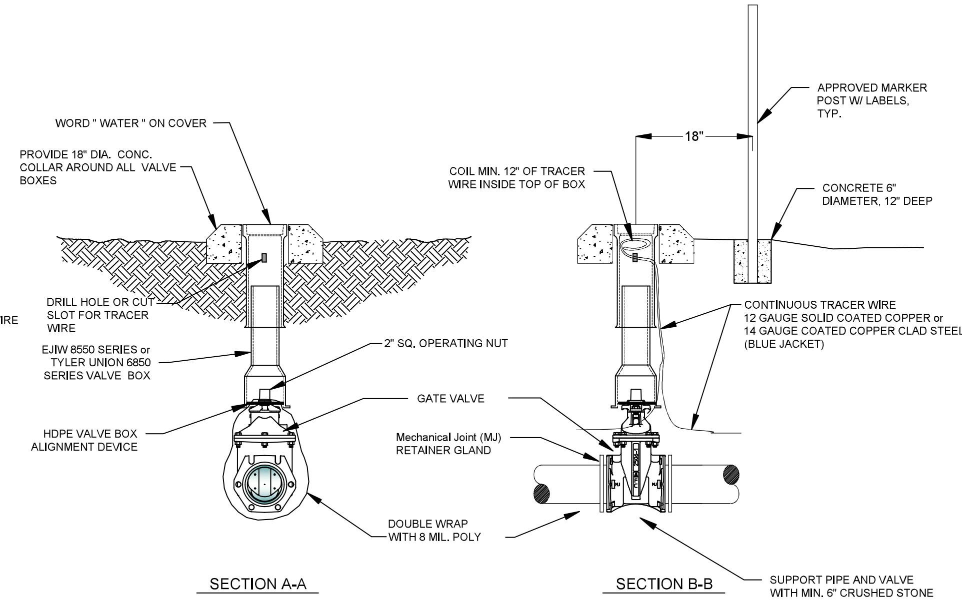
DATE 06,20,2025	ARCHIVE NO.
DESIGNER K.HOD	DRAWING SHEET NO. C110
DRAWN H.HOD	
CHECKED K.HOD	
PROJECT NO. 12445123C0075	SHEET 11 OF 15

7/16/25 09:29 KHODG C:\USERS\KHODG\ONEDRIVE\DOCUMENTS\HODGES ENGINEERING\USFS\WEDINGTON\DWG\CD\DISTRIBUTION\LAKE-WEDINGTON-WATERLINE-DISTRIBUTION-CD-7-14-25 FINAL.DWG;

PERMITTED VALVES (ALL MANUFACTURERS NEED TO BE NOTIFIED THAT BUY AMERICAN IS REQUIRED BY PROJECT)
1. AMERICAN FLOW CONTROL SERIES 2500
2. MUELLER SERIES 2360
3. AMERICAN AVK COMPANY SERIES 25 OR SERIES 45



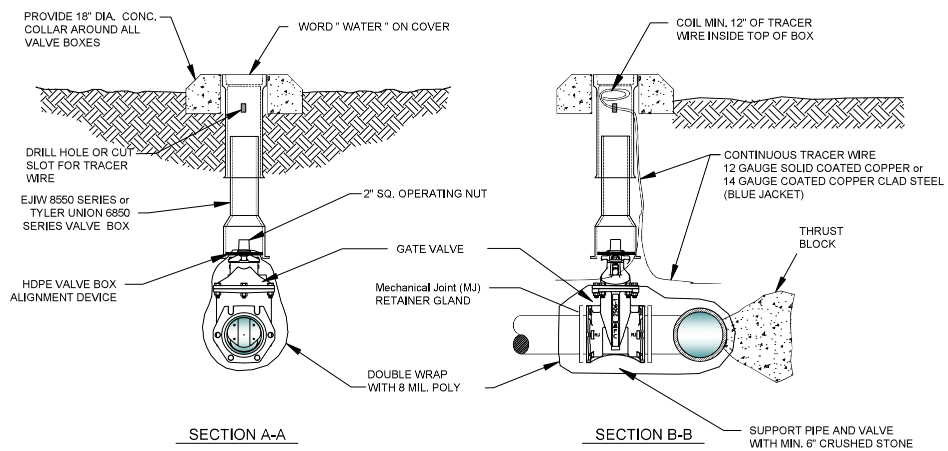
PLAN



SECTION A-A

SECTION B-B

GATE VALVE AT PIPE W/ MARKER DETAIL
NTS



SECTION A-A

SECTION B-B

GATE VALVE DETAIL AT TEE
NTS

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United States Department of Agriculture
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Region 8
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STAMPS, LOGOS, AND SEALS

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NO.	REVISION / ISSUE	DATE
4		
3		
2		
1	USFS Comments	7/14/25

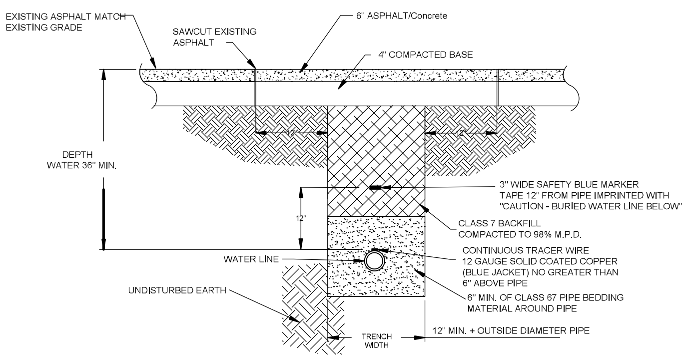
PROJECT NAME

LAKE WEDINGTON WATER DISTRIBUTION

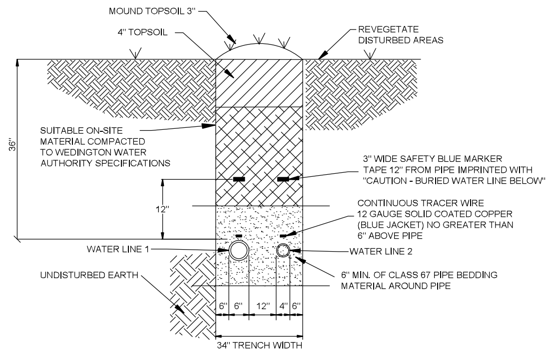
DRAWING TITLE

PIPING DETAILS 2

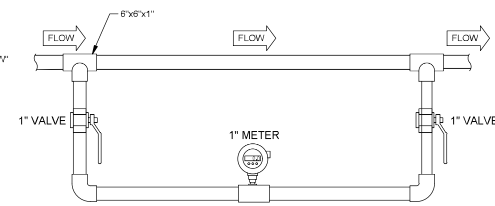
DATE 06,20,2025	ARCHIVE NO.
DESIGNER K.HOD	DRAWING SHEET NO. C111
DRAWN H.HOD	
CHECKED K.HOD	
PROJECT NO. 12445123C0075	SHEET 12 OF 15



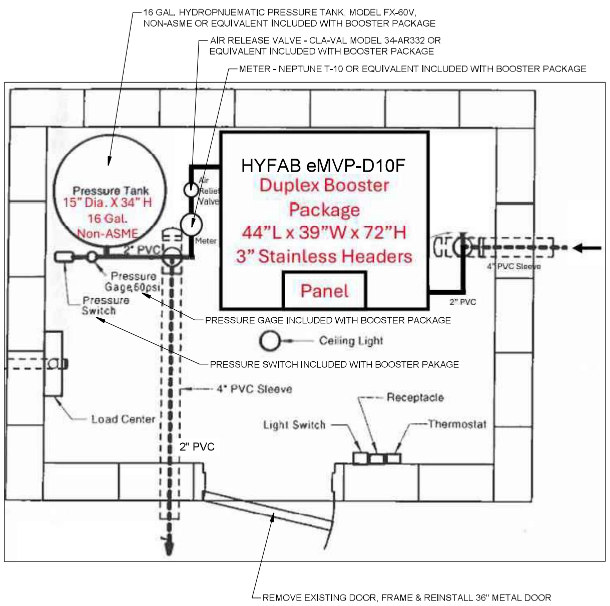
WATERLINE UNDER ASPHALT/CONCRETE DETAIL
NTS



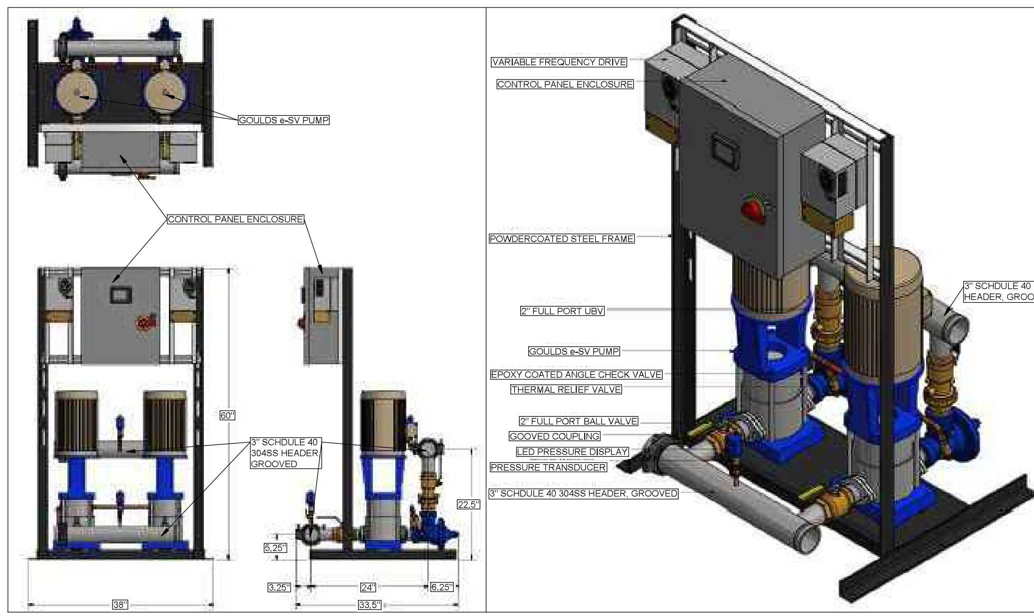
COMBINED TRENCH DETAIL
NTS



METER WITH BYPASS VALVE DETAIL
NTS



BOOSTER PUMP DETAIL PLAN VIEW
NTS



HYFAB eMVP-D10F (OR EQUIVALENT) DUPLEX BOOSTER PACKAGE DETAIL
NTS

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NO.	REVISION / ISSUE	DATE
4		
3		
2		
1	USFS Comments	7/14/25

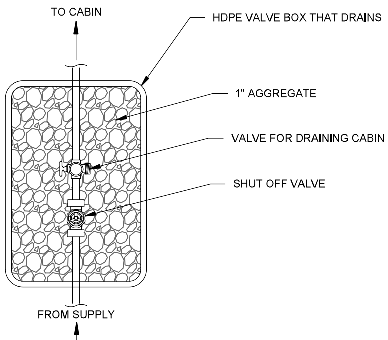
PROJECT NAME

LAKE WEDINGTON WATER DISTRIBUTION

DRAWING TITLE

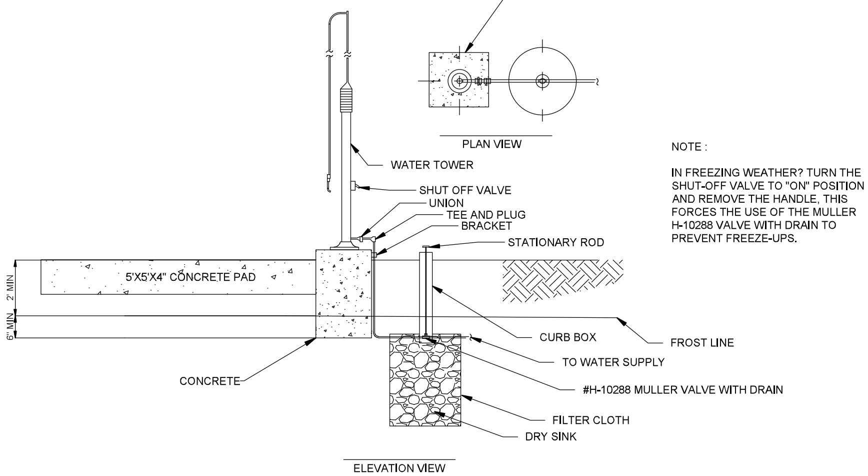
PIPING DETAILS 3

DATE 06.20.2025	ARCHIVE NO.
DESIGNER K.HOD	DRAWING SHEET NO. C112
DRAWN H.HOD	
CHECKED K.HOD	
PROJECT NO. 12445123C0075	SHEET 13 OF 15



CABIN SHUT OFF AND DRAIN DETAIL
NTS

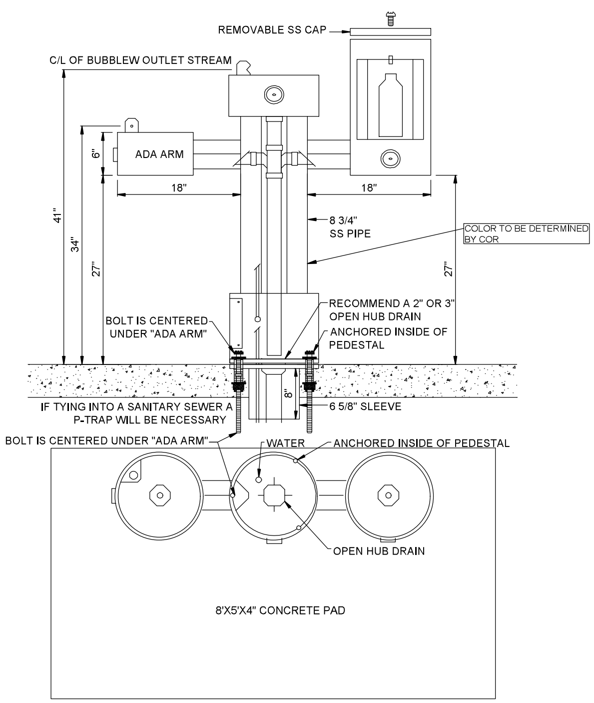
Romort Frost Free Hydrant or Equivalent



NOTE :

IN FREEZING WEATHER? TURN THE SHUT-OFF VALVE TO "ON" POSITION AND REMOVE THE HANDLE, THIS FORCES THE USE OF THE MULLER H-10288 VALVE WITH DRAIN TO PREVENT FREEZE-UPS.

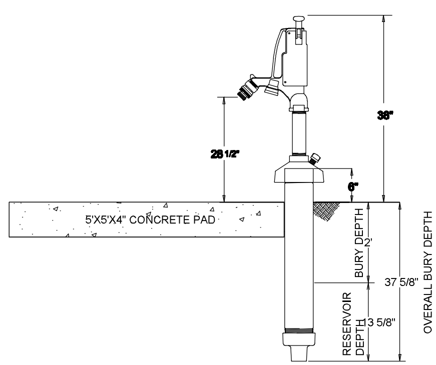
ROMORT FROST FREE HYDRANT INSTALLATION AT DUMP STATION
NTS



Most Dependable Fountains 2440 SMSS, Equivalent, or as Approved by USFS

DRINKING FOUNTAIN DETAIL
NTS

Woodford Model S4H or Equivalent



Rough-In Dimensions

FREEZE PROOF WATER HYDRANT DETAIL
NTS

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NO.	REVISION / ISSUE	DATE
4		
3		
2		7/14/25
1	USFS Comments	

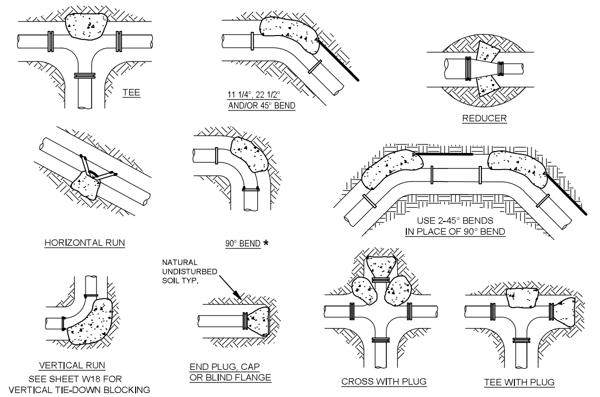
PROJECT NAME

LAKE WEDINGTON WATER DISTRIBUTION

DRAWING TITLE

PIPING DETAILS 4

DATE 06,20,2025	ARCHIVE NO.
DESIGNER K.HOD	DRAWING SHEET NO. C113
DRAWN H.HOD	
CHECKED K.HOD	
PROJECT NO. 12445123C0075	SHEET 14 OF 15



THRUST BLOCK SCHEDULE

(2000 PSF SOIL, 200 PSI WATER PRESSURE, 100 PSI WATER HAMMER)

BEARING AREA OF THRUST BLOCKS IN SQ. FT. (HORIZONTAL BENDS)						VOLUME OF THRUST BLOCK IN CU. FEET (VERTICAL BENDS)						
FITTING SIZE	PLUG TEE OR CAP	BEND ANGLE 90° *	45°	22 1/2°	11 1/4°	FITTING SIZE	BEND ANGLE 90° *	45°	22 1/2°	11 1/4°	ROD DIAM.	EMBED-MENT
2, 3, & 4	1.9	2.7	1.4	0.7	0.4	2, 3, & 4	35.5	19.2	9.8	4.9	.50"	30"
6	5.6	7.9	4.3	2.2	1.1	6	105.8	57.2	29.2	14.7	.75"	30"
8	9.6	13.6	7.4	3.8	1.9	8	181.9	98.5	50.2	25.2	.75"	30"
12	20.5	29.0	15.7	8.0	4.0	12	387.1	209.5	106.8	53.7	1.0"	30"
16	35.7	50.4	27.3	13.9	7.0	16	672.6	364.0	185.6	93.2	1.0"	36"
18	44.8	63.4	34.3	17.5	18.8	18	844.7	457.2	233.1	117.1	1.0"	36"
24	78.4	110.9	60.0	30.6	15.4	24	1478.7	800.3	408.0	205.0	1.5"	36"

THRUST BLOCK NOTES:

1. CONCRETE FOR THRUST BLOCKS - CLASS A CONCRETE SHALL DEVELOP NOT LESS THAN 3000 P.S.I. COMPRESSIVE STRENGTH AT 28 DAYS AND BE PLACED AGAINST UNDISTURBED SOIL.
2. ALL BENDS, BOTH HORIZONTAL AND VERTICAL, SHALL BE BACKED WITH CONCRETE. VERTICAL BENDS SHALL BE PLACED ON CONCRETE PADS WHERE BENDS TURN UP, OR LOADED WHERE BENDS TURN DOWN.
3. DOUBLE WRAP PIPE JOINTS IN 8 MIL "POLYETHLENE" BEFORE PLACING CONCRETE.
4. BEARING AREA SHOWN IN TABLE, IS BASED UPON A 2000 LB/SF. SOIL BEARING, AND UPON A PIPELINE PRESSURE OF 200 psi. PLUS 100 psi. WATER HAMMER. AREAS SHOWN SHALL BE ADJUSTED, SHOULD FIELD CONDITIONS VARY.
5. MJ RESTRAINTS ARE REQUIRED FOR ALL FITTINGS.
6. USE LONG-RADIUS FITTINGS WHEREVER POSSIBLE.
7. ALL BOLTS FOR FITTINGS SHALL BE 316 STAINLESS STEEL.
8. ALL DUCTILE IRON FITTINGS SHALL BE FUSION-BONDED EPOXY COATED INSIDE AND OUTSIDE IN ACCORDANCE WITH ANSI/AWWA C116/A21.16.
9. UNIT WEIGHT OF CONCRETE FOR VERTICAL THRUST BLOCKS IS 150 LBS/CU. FT.

THRUST BLOCKING DETAIL
NTS

THRUST BLOCK LIST

CONSTRUCT THRUST BLOCKS POURED IN PLACE

LINE 1		LINE 2		LINE 3		LINE 4		LINE 5		LINE 6	
STATION	QUANTITY	STATION	QUANTITY	STATION	QUANTITY	STATION	QUANTITY	STATION	QUANTITY		
7+71	1	0+69	2	4+30	1	0+50	1	1+10	1	0+18	2
12+67	1	2+48	1	6+20	1	1+00	1	1+75	1	0+50	1
18+67	1	2+64	1	12+50	1	5+39	1	1+88	1	2+57	1
35+00	1	3+33	1	13+25	1	5+56	1	2+24	2	3+00	1
36+75	1	5+70	1	13+42	1	6+20	1	2+33	1	4+11	2
45+04	1	10+25	1	13+90	1	6+70	1	2+39	1	4+50	1
46+54	1	10+49	1	14+50	1	7+10	1	Total	7	4+83	1
47+07	2	11+70	1	14+65	1	7+58	1			5+97	2
50+75	1	13+14	1	Total	8	9+12	1			Total	11
52+05	1	14+33	1			11+92	2				
53+45	1	15+44	1			Total	11				
55+00	1	17+25	2								
57+75	1	Total	14								
62+05	1										
64+62	1										
72+50	1										
73+50	1										
74+06	1										
75+09	2										
79+03	2										
81+50	1										
82+41	1										
82+53	1										
Total	26										

DISTRIBUTION SYSTEM TOTAL	77

United States Department of Agriculture
Forest Service

Region 8
Southern Region

STAMPS, LOGOS, AND SEALS

Hodges Engineering

CERTIFICATE OF AUTHORIZATION HODGES ENGINEERING No. 3352 ARKANSAS

STATE OF ARKANSAS LICENSED PROFESSIONAL ENGINEER No. 8289 KEVIN E. HODGES

NO.	REVISION / ISSUE	DATE
4		
3		
2		
1	USFS Comments	7/14/25

PROJECT NAME

LAKE WEDINGTON WATER DISTRIBUTION

DRAWING TITLE

THRUST BLOCK DETAILS

DATE 06.20.2025	ARCHIVE NO.
DESIGNER K.HOD	DRAWING SHEET NO. C114
DRAWN H.HOD	
CHECKED K.HOD	
PROJECT NO. 12445123C0075	SHEET 15 OF 15

7/16/25 09:29 KHODG C:\USERS\KHODG\ONEDRIVE\DOCUMENTS\HODGES ENGINEERING\USFS\WEDINGTON\DWG\CD\DISTRIBUTION\LAKE-WEDINGTON-WATERLINE-DISTRIBUTION-CD- 7-14-25 FINAL.DWG;

WEDINGTON RECREATION AREA
WATER DISTRIBUTION SCHEDULE COVER SHEET

TABLE OF CONTENTS

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Engineering

WEDINGTON RECREATION AREA
WATER DISTRIBUTION LINE SCHEDULE

DISTRIBUTION TOTAL LINE SCHEDULE							
	LINE LENGTHS (LF)						
DIA (in)	**LINE 1**	**LINE 2**	**LINE 3**	**LINE 4**	**LINE 5**	**LINE 6**	**TOTAL**
6	5293	0	0	0	0	0	5293
4	2908	1725	0	670	0	0	5303
3	0	0	0	0	0	597	597
2	431	0	1465	522	239	220	2877
1-1/2	0	580	0	0	0	0	580
1	771	0	0	85	385	45	1286

LINE 1			
Designation	**STA**	**Dia. (in)**	**Length (LF)**
LINE 1	0+52 TO 53+45	6	5293
BRANCH TO WIGGINS HOUSE	47+07	1	50
BRANCH TO FROST FREE HYD AT PAVILLION	50+75	1	315
BRANCH TO FOUNTAIN	52+05	1	21
LINE 1	53+45 TO 82+53	4	2908
BRANCH TO LODGE	55+00	2	221
BRANCH TO TOILET	57+75	1	20
BRANCH TO FROST FREE HYD AT BOAT RAMP	73+50	1	165
BRANCH TO DUMP STATION	74+06	2	210
BRANCH TO FROST FREE HYD TO BATH HOUSE	75+09	1	90
BRANCH TO FROST FREE HYD TO CAMP HOST SITE	79+03	1	60
BRANCH TO FROST FREE HYD	81+50	1	50

LINE 2			
Designation	**STA**	**Dia. (in)**	**Length (LF)**
LINE 2	0+00 TO 17+25	4	1725
BRANCH TO CABIN #7	2+48	1-1/2	205
BRANCH TO CABIN #6	3+33	1-1/2	75
BRANCH TO CABIN #5	10+49	1-1/2	120
BRANCH TO CABIN #4	13+14	1-1/2	40
BRANCH TO CABIN #2	15+44	1-1/2	50
BRANCH TO CABIN #1	17+25	1-1/2	90

LINE 3			
Designation	**STA**	**Dia. (in)**	**Length (LF)**
LINE 3	0+00 TO 14+65	2	1465

LINE 4			
Designation	**STA**	**Dia. (in)**	**Length (LF)**
LINE 4	0+00 TO 6+70	4	670
BRANCH TO WORK CENTER	0+50	1	15
BRANCH TO ADMIN OFFICE	1+00	1	15
BRANCH TO LOWER CABIN	5+39	1	25
LINE 4	6+70 TO 11+92	2	522
LINE 4	11+92 TO 12+02	1	10
BRANCH TO HOUSE KEEPING CABIN	9+12	1	20

LINE 5			
Designation	**STA**	**Dia. (in)**	**Length (LF)**
LINE 5	0+00 TO 2+39	2	239
BRANCH TO LOWER CABIN	1+75	1	65
BRANCH TO FROST FREE HYD	2+24	1	160
BRANCH TO FROST FREE HYD	2+24	1	35

LINE 6			
Designation	**STA**	**Dia. (in)**	**Length (LF)**
LINE 6	0+00 TO 5+97	3	597
2" SERVICE LINE TO BLOW OFF	0+18	2	80
1" SERVICE LINE TO FOUNTAIN	0+18	1	45
BRANCH TO BATH HOUSE	0+50	2	40
BRANCH TO DEMO HYD/ADD BLOW OFF	4+11	2	100

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WEDINGTON RECREATION AREA
WATER DISTRIBUTION FITTING SCHEDULE

DISTRIBUTION TOTAL LINE SCHEDULE							
	LINE LENGTHS (LF)						
DIA (in)	LINE 1	LINE 2	LINE 3	LINE 4	LINE 5	LINE 6	TOTAL
6	5293						5293
4	2908	1725		670			5303
3						597	597
2	431		1465	522	239	220	2877
1-1/2		580					580
1	771			85	385	45	1286

CONNECTIONS							
TEE	LINE 1	LINE 2	LINE 3	LINE 4	LINE 5	LINE 6	TOTAL
6"X6"X4"	2						2
6"X6"X3"	1						1
6"X6"X2"	2						2
6"X6"X1"	3						3
4"x4"x2"	3			2			5
4"x4"x1"	5			3			8
3"x3"x2"		9				5	14
3"x3"x1"						1	1
2"x2"x2"				1	2	1	4
2"x2"x1"			2	2	3		7
1"x1"x1"	3						3

REDUCER	LINE 1	LINE 2	LINE 3	LINE 4	LINE 5	LINE 6	TOTAL
6" to 4"							
6" to 3"	1						1
6" to 2"							
4" to 2"				1			1
4" to 1"	1						1
3" to 2"							
3" to 1-1/2"		1					1
3" to 1"						1	1
2" to 1-1/2"		5					5
2" to 1"				1	1	1	3

VALVE	LINE 1	LINE 2	LINE 3	LINE 4	LINE 5	LINE 6	TOTAL
6"	8						8
4"	4			1			5
3"	1						1
2"	5	10	1	3	2	6	27
1-1/2"	1						1
1"	11		2	6	4	2	25

TOTAL FITTING SCHEDULE							
	LINE 1	LINE 2	LINE 3	LINE 4	LINE 5	LINE 6	TOTAL
TEE	19	9	2	8	5	7	50
BLOW OFF	2	4	1	2	2	4	15
FROST FREE HYD	6		2		2	1	11
VALVE	18	10	3	7	6	8	52
REDUCER	2	8		3	1	2	16
BYPASS METER	3						3
FOUNTAIN	1					1	2
THRUST BLOCK	26	14	8	11	7	11	77
3 WAY VALVE	4			1			5
AIR RELEASE VALVE	2		1				3
TANK							
BOOSTER STATION		1					1
ROMORT WT KIT	2						2

Hodges Engineering

WEDINGTON RECREATION AREA
WATER DISTRIBUTION LINE 1 FITTINGS

LINE 1

STATION	ELEMENTS					Tee Size	Reducer	Valve Size(s)	DESCRIPTION
7+71	TEE	THRUST BLOCK	3 WAY VALVE			6"X6"X3"		6"	BEGIN LINE 2 : 4"
								6"	
								4"	
11+00	VALVE							6"	
12+67	THRUST BLOCK								
18+67	THRUST BLOCK								
29+25	BYPASS METER								
35+00	TEE	THRUST BLOCK	3 WAY VALVE			6"X6"X2"		6"	BRANCH TO LINE 3 : 2"
								6"	
								2"	
36+75	THRUST BLOCK								
45+04	TEE	THRUST BLOCK	3 WAY VALVE			6"X6"X4"		6"	BEGIN LINE 4 : 4"
								6"	
								4"	
46+54	TEE	THRUST BLOCK	VALVE	BLOW OFF		6"X6"X2"		2"	
47+07	TEE	THRUST BLOCK	VALVE			6"X6"X1"		1"	BRANCH TO WIGGINS HOUSE
	TEE	THRUST BLOCK	VALVE	FROST FREE HYD		1"x1"x1"		1"	
47+25	BYPASS METER								
50+75	TEE	THRUST BLOCK	VALVE	FROST FREE HYD		6"X6"X1"		1"	FROST FREE HYD AT PAVILLION
52+05	TEE	THRUST BLOCK	FOUNTAIN	VALVE		6"X6"X1"		1"	
53+45	TEE	THRUST BLOCK	3 WAY VALVE	REDUCER		6"X6"X4"	6" to 3"	6"	BEGIN LINE 6 : 3"
								4"	
								3"	
55+00	TEE	THRUST BLOCK	VALVE			4"x4"x2"		2"	BRANCH TO LODGE
57+25	AIR RELEASE VALVE								
57+75	TEE	THRUST BLOCK	VALVE			4"x4"x1"		1"	BRANCH TO TOILET
58+50	VALVE	BYPASS METER						4"	
62+05	THRUST BLOCK								
64+62	THRUST BLOCK								
72+50	THRUST BLOCK								
72+78	AIR RELEASE VALVE								
73+50	TEE	THRUST BLOCK	VALVE	FROST FREE HYD		4"x4"x1"		1"	FROST FREE HYD AT BOAT RAMP
74+06	TEE	THRUST BLOCK	VALVE			4"x4"x2"		2"	BRANCH TO DUMP STATION
	ROMORT WT KIT	ROMORT WT KIT							
75+09	TEE	THRUST BLOCK	VALVE			4"x4"x1"		1"	BRANCH TO FROST FREE HYD, TO BATH HOUSE
	TEE	THRUST BLOCK	VALVE	FROST FREE HYD		1"x1"x1"		1"	
79+03	TEE	THRUST BLOCK	VALVE			4"x4"x1"		1"	BRANCH TO FROST FREE HYD, TO CAMP HOST SITE
	TEE	THRUST BLOCK	VALVE	FROST FREE HYD		1"x1"x1"		1-1/2"	
81+50	TEE	THRUST BLOCK	VALVE	FROST FREE HYD		4"x4"x1"		1"	BRANCH TO FROST FREE HYD
82+41	TEE	THRUST BLOCK	VALVE	BLOW OFF		4"x4"x2"		2"	
82+53	VALVE	REDUCER	THRUST BLOCK				4" to 1"	1"	END LINE AT BATH HOUSE

LINE TOTALS	
TEE	19
BLOW OFF	2
FROST FREE HYD	6
VALVE	18
REDUCER	2
BYPASS METER	3
FOUNTAIN	1
THRUST BLOCK	26
3 WAY VALVE	4
AIR RELEASE VALVE	2
TANK	0
BOOSTER STATION	0
ROMORT WT KIT	2

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WEDINGTON RECREATION AREA
WATER DISTRIBUTION LINE 2 FITTINGS

	LINE 2									
STATION	ELEMENTS				Tee Size	Reducer	Valve Size(s)	DESCRIPTION	LINE TOTALS	
0+69	BOOSTER STATION	THRUST BLOCK	THRUST BLOCK					2 THRUST BLOCKS (LINE IN & LINE OUT)	TEE	9
2+48	TEE	THRUST BLOCK	VALVE	REDUCER	3"x3"x2"	2" to 1-1/2"	2"	BRANCH TO CABIN #7 1-1/2"	BLOW OFF	4
2+64	TEE	THRUST BLOCK	VALVE	BLOW OFF	3"x3"x2"		2"		FROST FREE HYD	0
3+33	TEE	THRUST BLOCK	VALVE	REDUCER	3"x3"x2"	2" to 1-1/2"	2"	BRANCH TO CABIN #6 1-1/2"	VALVE	10
5+70	THRUST BLOCK								REDUCER	8
10+25	TEE	THRUST BLOCK	VALVE	BLOW OFF	3"x3"x2"		2"	DEMO HYD/ADD BLOW OFF	BYPASS METER	0
10+49	TEE	THRUST BLOCK	VALVE	REDUCER	3"x3"x2"	2" to 1-1/2"	2"	BRANCH TO CABIN #5 1-1/2"	FOUNTAIN	0
11+70	THRUST BLOCK								THRUST BLOCK	14
13+14	TEE	THRUST BLOCK	VALVE	REDUCER	3"x3"x2"	2" to 1-1/2"	2"	BRANCH TO CABIN #4 1-1/2"	3 WAY VALVE	0
14+33	TEE	THRUST BLOCK	VALVE	BLOW OFF	3"x3"x2"		2"	DEMO HYD/ADD BLOW OFF	AIR RELEASE VALVE	0
15+44	TEE	THRUST BLOCK	VALVE	REDUCER	3"x3"x2"	2" to 1-1/2"	2"	BRANCH TO CABIN #2 1-1/2"	TANK	0
17+25	TEE	THRUST BLOCK	VALVE	BLOW OFF	3"x3"x2"		2"	BRANCH TO CABIN #1 1-1/2"/ DEMO HYD/ADD BLOW OFF	BOOSTER STATION	1
	VALVE	REDUCER	THRUST BLOCK			3" to 1-1/2"	2"	END LINE AT CABIN #1		

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WEDINGTON RECREATION AREA
WATER DISTRIBUTION LINE 3 FITTINGS

LINE 3

STATION	ELEMENTS					Tee Size	Reducer	Valve Size(s)	DESCRIPTION	LINE TOTALS	
4+13	AIR RELEASE VALVE									TEE	2
4+30		THRUST BLOCK								BLOW OFF	1
6+20		THRUST BLOCK								FROST FREE HYD	2
12+50		THRUST BLOCK								VALVE	3
13+25	TEE	THRUST BLOCK	VALVE	FROST FREE HYD		2"x2"x1"		1"		REDUCER	0
13+42		THRUST BLOCK								BYPASS METER	0
13+90		THRUST BLOCK								FOUNTAIN	0
14+50	TEE	THRUST BLOCK	VALVE	FROST FREE HYD		2"x2"x1"		1"		THRUST BLOCK	8
14+65		THRUST BLOCK	VALVE	BLOW OFF				2"	END LINE AT BLOW OFF	3 WAY VALVE	0
										AIR RELEASE VALVE	1
										TANK	0
										BOOSTER STATION	0

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WEDINGTON RECREATION AREA
WATER DISTRIBUTION LINE 4 FITTINGS

	LINE 4							
STATION	ELEMENTS				Tee Size	Reducer	Valve Size(s)	DESCRIPTION
0+50	TEE	THRUST BLOCK	VALVE		4"x4"x1"		1"	BRANCH TO WORK CENTER
1+00	TEE	THRUST BLOCK	VALVE		4"x4"x1"		1"	BRANCH TO ADMIN OFFICE
5+39	TEE	THRUST BLOCK	VALVE		4"x4"x1"		1"	BRANCH TO LOWER CABIN
5+56	THRUST BLOCK							
6+20	TEE	THRUST BLOCK	VALVE	BLOW OFF	4"x4"x2"		2"	DEMO HYD / ADD BLOW OFF
6+70	TEE	THRUST BLOCK	3 WAY VALVE	REDUCER	4"x4"x2"	4" to 2"	4"	BEGIN LINE 5, 2" DIA HDPE LINE AFTER TEE
							2"	
							2"	
7+10	TEE	THRUST BLOCK	VALVE	REDUCER	2"x2"x1"		1"	BRANCH TO LOWER CABIN
7+58	THRUST BLOCK							
9+12	TEE	THRUST BLOCK	VALVE		2"x2"x1"		1"	BRANCH TO LOWER CABIN
11+92	TEE	THRUST BLOCK	BLOW OFF		2"x2"x2"			10' OF 1" LINE TO ENTRY STATION
	REDUCER	THRUST BLOCK	VALVE			2" to 1"	1"	

LINE TOTALS	
TEE	8
BLOW OFF	2
FROST FREE HYD	0
VALVE	7
REDUCER	3
BYPASS METER	0
FOUNTAIN	0
THRUST BLOCK	11
3 WAY VALVE	1
AIR RELEASE VALVE	0
TANK	0
BOOSTER STATION	0

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WEDINGTON RECREATION AREA											
	LINE 5										
STATION	ELEMENTS				Tee Size	Reducer	Valve Size(s)	DESCRIPTION	LINE TOTALS	
1+10	TEE	THRUST BLOCK	VALVE	BLOW OFF	2"x2"x2"		2"		TEE	5
1+75	TEE	THRUST BLOCK	VALVE		2"x2"x1"		1"	BRANCH TO LOWER CABIN	BLOW OFF	2
1+88	THRUST BLOCK								FROST FREE HYD	2
2+24	TEE	THRUST BLOCK	VALVE	FROST FREE HYD	2"x2"x1"		1"	FROST FREE HYD FOR TRAILER HOMES	VALVE	6
	TEE	THRUST BLOCK	VALVE	FROST FREE HYD	2"x2"x1"		1"		REDUCER	1
2+33	TEE	THRUST BLOCK	VALVE	BLOW OFF	2"x2"x2"		2"		BYPASS METER	0
2+39	REDUCER	VALVE	THRUST BLOCK			2" to 1"	1"	BRANCH TO LOWER CABIN	FOUNTAIN	0
									THRUST BLOCK	7
									3 WAY VALVE	0
									AIR RELEASE VALVE	0
									TANK	0
									BOOSTER STATION	0

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WEDINGTON RECREATION AREA
WATER DISTRIBUTION LINE 6 FITTINGS

	LINE 6									
STATION	ELEMENTS				Tee Size	Reducer	Valve Size(s)	DESCRIPTION	LINE TOTALS	
	TEE	THRUST BLOCK	VALVE		3"x3"x2"		2"	2" SERVICE LINE TO BLOW OFF THEN 1" LINE TO DRINKING FOUNTAIN	TEE	7
	TEE	THRUST BLOCK	VALVE	BLOW OFF	2"x2"x2"		2"		BLOW OFF	4
0+18	REDUCER	FOUNTAIN				2" to 1"			FROST FREE HYD	1
0+50	TEE	THRUST BLOCK	VALVE		3"x3"x2"		2"	BRANCH TO BATH HOUSE	VALVE	8
2+57	TEE	THRUST BLOCK	VALVE	BLOW OFF	3"x3"x2"		2"	DEMO HYD/ADD BLOW OFF	REDUCER	2
3+00	THRUST BLOCK								BYPASS METER	0
4+11	TEE	THRUST BLOCK			3"x3"x2"				FOUNTAIN	1
	THRUST BLOCK	VALVE	BLOW OFF				2"	DEMO HYD/ADD BLOW OFF	THRUST BLOCK	11
4+50	THRUST BLOCK								3 WAY VALVE	0
4+83	TEE	THRUST BLOCK	VALVE	FROST FREE HYD	3"x3"x1"		1"		AIR RELEASE VALVE	0
	TEE	THRUST BLOCK	VALVE	BLOW OFF	3"x3"x2"		2"		TANK	0
5+97	REDUCER	THRUST BLOCK	VALVE			3" to 1"	1"	1" LINE TO BOAT HOUSE	BOOSTER STATION	0

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"General Decision Number: AR20260058 01/02/2026

Superseded General Decision Number: AR20250058

State: Arkansas

Construction Type: Heavy
HEAVY CONSTRUCTION PROJECTS (Including Water and Sewer Lines)

County: Washington County in Arkansas.

Modification Number	Publication Date
0	01/02/2026

SUAR2015-055 01/09/2017

	Rates	Fringes
CARPENTER, Includes Form Work	$ 17.50	0.00
CEMENT MASON/CONCRETE FINISHER	$ 14.36	0.00
ELECTRICIAN	$ 22.88	7.19
LABORER: Common or General	$ 12.36	0.93
LABORER: Pipelayer	$ 14.14	0.00
OPERATOR: Backhoe/Excavator/Trackhoe	$ 16.26	0.00
OPERATOR: Bulldozer	$ 17.00	1.92
OPERATOR: Crane	$ 24.21	6.79
OPERATOR: Loader	$ 15.45	0.00
PAINTER (Brush and Roller)	$ 18.00	0.00
TRUCK DRIVER: Dump Truck	$ 14.02	2.18

WELDERS - Receive rate prescribed for craft performing operation to which welding is incidental.

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Note: Executive Order (EO) 13706, Establishing Paid Sick Leave for Federal Contractors applies to all contracts subject to the Davis-Bacon Act for which the contract is awarded (and any solicitation was issued) on or after January 1, 2017. If this contract is covered by the EO, the contractor must provide employees with 1 hour of paid sick leave for every 30 hours they work, up to 56 hours of paid sick leave each year. Employees must be permitted to use paid sick leave for their own illness, injury or other health-related needs, including preventive care; to assist a family member (or person who is like family to the employee) who is ill, injured, or has other health-related needs, including preventive care; or for reasons resulting from, or to assist a family member (or person who is like family to the employee) who is a victim of, domestic violence, sexual assault, or stalking. Additional information on contractor requirements and worker protections under the EO is available at https://www.dol.gov/agencies/whd/government-contracts.

Note: Executive Order 13658 generally applies to contracts subject to the Davis-Bacon Act that were awarded on or between January 1, 2015 and January 29, 2022, and that have not been renewed or extended on or after January 30, 2022. Executive Order 13658 does not apply to contracts subject only to the Davis-Bacon Related Acts regardless of when they were awarded. If a contract is subject to Executive Order 13658, the contractor must pay all covered workers at least $13.30 per hour (or the applicable wage rate listed on this wage determination, if it is higher) for all hours spent performing on the contract in 2025. The applicable Executive Order minimum wage rate will be adjusted annually. Additional information on contractor requirements and worker protections under Executive Order 13658 is available at www.dol.gov/whd/govcontracts.

Unlisted classifications needed for work not included within the scope of the classifications listed may be added after award only as provided in the labor standards contract clauses (29CFR 5.5 (a) (1) (iii)).

The body of each wage determination lists the classifications and wage rates that have been found to be prevailing for the type(s) of construction and geographic area covered by the wage determination. The classifications are listed in alphabetical order under rate identifiers indicating whether the particular rate is a union rate (current union negotiated rate), a survey rate, a weighted union average rate, a state adopted rate, or a supplemental classification rate.

Union Rate Identifiers

A four-letter identifier beginning with characters other than ""SU"", ""UAVG"", ?SA?, or ?SC? denotes that a union rate was prevailing for that classification in the survey. Example: PLUM0198-005 07/01/2024. PLUM is an identifier of the union whose collectively bargained rate prevailed in the survey for this classification, which in this example would be Plumbers. 0198 indicates the local union number or district council number where applicable, i.e., Plumbers Local 0198. The next number, 005 in the example, is an internal number used in processing the wage determination. The date, 07/01/2024 in the example, is the effective date of the most current negotiated rate.

Union prevailing wage rates are updated to reflect all changes over time that are reported to WHD in the rates in the collective bargaining agreement (CBA) governing the classification.

Union Average Rate Identifiers

The UAVG identifier indicates that no single rate prevailed for those classifications, but that 100% of the data reported for the classifications reflected union rates. EXAMPLE: UAVG-OH-0010 01/01/2024. UAVG indicates that the rate is a weighted union average rate. OH indicates the State of Ohio. The next number, 0010 in the example, is an internal number used in producing the wage determination. The date, 01/01/2024 in the example, indicates the date the wage determination was updated to reflect the most current union average rate.

A UAVG rate will be updated once a year, usually in January, to reflect a weighted average of the current rates in the collective bargaining agreements on which the rate is based.

Survey Rate Identifiers

The ""SU"" identifier indicates that either a single non-union rate prevailed (as defined in 29 CFR 1.2) for this classification in the survey or that the rate was derived by computing a weighted average rate based on all the rates reported in the survey for that classification. As a weighted average rate includes all rates reported in the survey, it may include both union and non-union rates. Example: SUFL2022-007 6/27/2024. SU indicates the rate is a single non-union prevailing rate or a weighted average of survey data for that classification. FL indicates the State of Florida. 2022 is the year of the survey on which these classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. The date, 6/27/2024 in the example, indicates the survey completion date for the classifications and rates under that identifier.

?SU? wage rates typically remain in effect until a new survey is conducted. However, the Wage and Hour Division (WHD) has the discretion to update such rates under 29 CFR 1.6(c)(1).

State Adopted Rate Identifiers

The ""SA"" identifier indicates that the classifications and prevailing wage rates set by a state (or local) government were adopted under 29 C.F.R 1.3(g)-(h). Example: SAME2023-007 01/03/2024. SA reflects that the rates are state adopted. ME refers to the State of Maine. 2023 is the year during which the state completed the survey on which the listed classifications and rates are based. The next number, 007 in the example, is an internal number used in producing the wage determination. The date, 01/03/2024 in the example, reflects the date on which the classifications and rates under the ?SA? identifier took effect under state law in the state from which the rates were adopted.

WAGE DETERMINATION APPEALS PROCESS

1) Has there been an initial decision in the matter? This can be:

a) a survey underlying a wage determination
b) an existing published wage determination
c) an initial WHD letter setting forth a position on a wage determination matter
d) an initial conformance (additional classification and rate) determination

On survey related matters, initial contact, including requests for summaries of surveys, should be directed to the WHD Branch of Wage Surveys. Requests can be submitted via email to davisbaconinfo@dol.gov or by mail to:

Branch of Wage Surveys
Wage and Hour Division
U.S. Department of Labor

200 Constitution Avenue, N.W.
Washington, DC 20210

Regarding any other wage determination matter such as conformance decisions, requests for initial decisions should be directed to the WHD Branch of Construction Wage Determinations. Requests can be submitted via email to BCWD-Office@dol.gov or by mail to:

Branch of Construction Wage Determinations
Wage and Hour Division
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

2) If an initial decision has been issued, then any interested party (those affected by the action) that disagrees with the decision can request review and reconsideration from the Wage and Hour Administrator (See 29 CFR Part 1.8 and 29 CFR Part 7). Requests for review and reconsideration can be submitted via email to dba.reconsideration@dol.gov or by mail to:

Wage and Hour Administrator
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210

The request should be accompanied by a full statement of the interested party's position and any information (wage payment data, project description, area practice material, etc.) that the requestor considers relevant to the issue.

3) If the decision of the Administrator is not favorable, an interested party may appeal directly to the Administrative Review Board (formerly the Wage Appeals Board). Write to:

Administrative Review Board
U.S. Department of Labor
200 Constitution Avenue, N.W.
Washington, DC 20210.

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END OF GENERAL DECISION
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